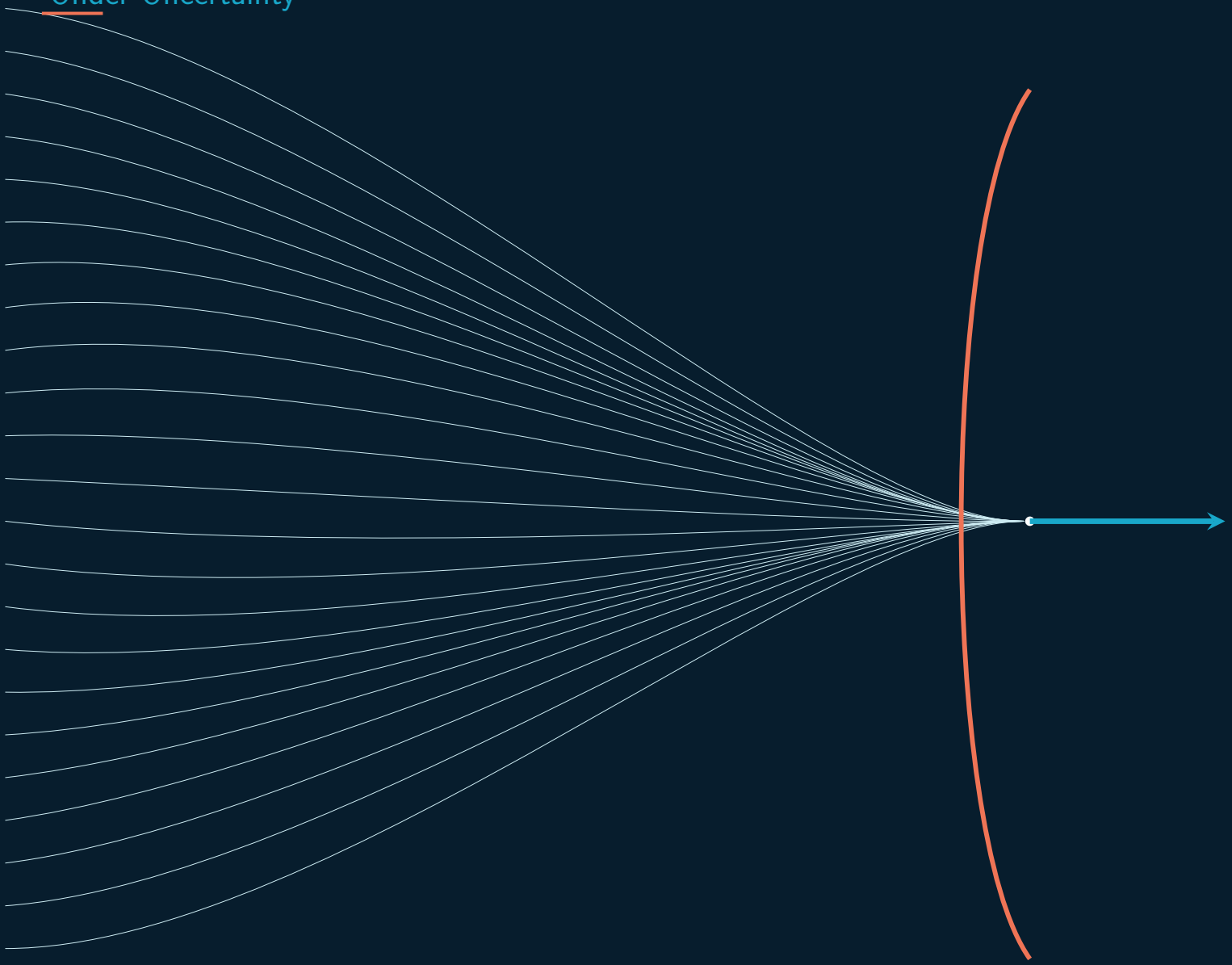
FORECASTING FOR DECISIONS

From Prediction to Action
Under Uncertainty



Adrian Antico

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CHAPTER 1

A Forecast Is Not the Decision

Depends on. None; this is the opening argument.

Why does anyone forecast? Not because the future is interesting. An act has to be chosen before its consequence is observed, and some of those acts change the system that will be observed.

A warehouse orders this week because replenishment arrives next week. Waiting for next week's stock-out is not a plan; it is the thing the order was meant to prevent. A clinic staffs tonight because the waiting room is a stock of people already in the building, not a rumor about tomorrow's calendar. A claims shop assigns reviewers because OPEN files already exist, aging while they wait. In each case the decision is *now*, the numbers that would make the decision easy are *not yet*, and waiting for those numbers is itself a decision — usually a bad one.

One small visitor shop will recur through the book. Ten people, `v01` through `v10`, have already arrived when its first forecast is issued. Seven are still OPEN. The shop also expects more people to arrive, but those future people do not yet have identities. Its future population is therefore built from two different objects: people known at the origin and unlabeled arrival mass forecast beyond it. Chapters 2 and 16 will make that boundary visible and carry both populations through the same process.

That is the whole subject of the book, stated without machinery:

> A forecast is a statement about the future, issued from what can be known now, for an act that cannot wait. The act is not the statement.

Chapter 2 will make that sentence technically tight: origins, information sets, leakage, kinds of uncertainty, and the process language in which many of the quantities live. What matters here is *why the sentence is worth a book*, and what happens if you treat the statement as if it were the act.

1.1 The number is not the endpoint

Most of the world's forecasting software stops at a number or a band. That is a legitimate product. It is not this book's endpoint. Three facts keep pushing past the printed number.

First, the act may change the thing being forecast. Spend, staffing, outreach, and a hiring freeze are not spectators. If a control u enters the system, tomorrow's Y is not the same object as yesterday's extrapolation of Y. A completion forecast that is right this month, while this month's freeze empties next month's OPEN, did not include the act. Treating the forecast as a recommendation is then a category error, not a rounding error.

Second, the relevant object may be stock, not flow. Forty files closed this month can look stable while OPEN explodes, or while OPEN is already empty and the shop is about to go idle. Completions are a flow through a window. The decision often cares about what remains. Chapter 2 will draw that distinction as occupancy reconstructed at a cut; here it is enough that a dashboard KPI and a staffing remainder are not the same sentence.

Third, the relevant functional may not be the mean. A ward that fails if occupancy exceeds a floor, a cash desk that cares about the mean, an insurer that cares about $P(Y > \tau)$: these are different questions about the same future. A forecast that wins on RMSE can starve the floor. Granger, and later Elliott and Timmermann, already said that a point forecast is a functional of a predictive law under a loss. This book takes that seriously *after* the law has been pushed through process state, allocation or control, and a declared preference — and it treats refusal as a first-class output. A mathematically local-optimal schedule is not a recommendation. A recommendation is not an authorized action. Those sentences are later chapters. They are why this one refuses to end on "better models."

Not every act needs a forecast, and not every forecast should move an act. A reorder rule that already fires at a calibrated quantile may not need a process engine. A stress scenario may be propagated without anyone optimizing. Two forecasts that differ in RMSE by eight percent, and issue the same act, are not decision-relevant no matter how sophisticated the runner-up. Accuracy that does not change α has no value of information; Chapter 19 will name that fact. The opening version is simpler: if the act does not move, stop.

1.2 Stopping early is a result

The book is not a command to use every layer on every problem. It is a sequence of questions, and an honest answer to any of them can be: you do not need what follows.

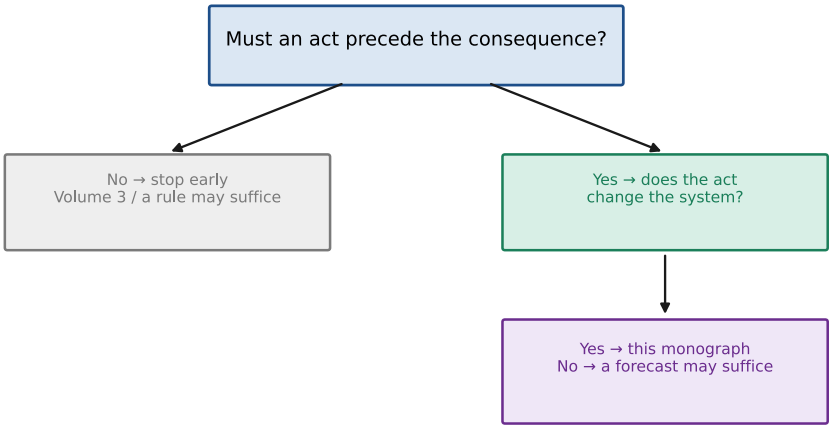


Figure 1.1: An act that must precede its consequence is the only reason to forecast; an act that also changes the system is the reason this monograph exists. Either "no" is a stop, not a failure.

Read the fork from the top. Must an act be chosen before the consequence is seen? If not, you do not need this book's machinery. You may still be curious about the future; curiosity is not a decision problem. If yes, a second question: does the act change the system that will be observed? If not, a forecast — perhaps an ordinary time-series forecast of the kind Volume 3 of this series teaches in a laboratory — may suffice. A well-calibrated replenishment rule, a seasonal naive that already implies "do nothing," a stress scenario that nobody is about to optimize: each of those can be a terminal result. If yes, you are in this monograph. Spend will change arrivals, staffing will change queue age, a freeze will change next month's OPEN, and the statement about the future has to know that.

A mathematical frontier, later, may itself be the terminal result when no risk preference is supplied. A process forecast of occupancy may be enough when nobody is asking where to put the next hour of overtime. **Stopping early is a result.** The later chapters exist so that, when you do not stop, you can say which object you are holding.

Volume 3 remains the laboratory of ordinary forecasting: baselines, rolling origins, ETS against ARIMA, leakage as a crime against an origin (Hyndman and Athanasopoulos 2021; Tashman 2000). This monograph does not replace it. If your problem is "which univariate method wins on a rolling origin," go there first.

1.3 What the next chapters are for

Chapter 2 makes the opening sentence tight: origin, leakage, and why two shops with the same completions are not the same decision. Chapters 3–4 separate intervening from watching, and a decimal from an identified lever. Chapters 5–8 stay in a world where tomorrow does not remember the stock: maps, an efficient mix, a parent that only sees totals, several objectives. When the stock *does* carry, Chapter 7 shows the parent's totals were the wrong object. Chapters 9–16 keep that stock as a process. Chapters 16–19 push the same occupancy from an expectation to paired worlds and a floor. Chapters 19 and 23 ask which kind of not-knowing you have, and which preference — if any — you will declare. Chapter 23 spends a Monday morning using all of that on one roster.

None of the ingredients is new. Diebold, Hyndman and Athanasopoulos, Granger, Pearl, Manski, Gneiting: the laboratory, the decision-theoretic point, the intervention calculus, the identified set, the proper score, all exist (Diebold 2007; Hyndman and Athanasopoulos 2021; Granger and Newbold 1986; Pearl 2009; Manski 2003; Gneiting and Raftery 2007). What is unusual is the insistence that purpose, information, identification, composition, validation, preference, and refusal stay visible when they are composed.

1.4 Exercises

C1-E1. Conceptual. Name an act in your domain that must be chosen before the consequence is observed. What, exactly, must be known about the future?

C1-E2. Method choice. A reorder rule already fires at a calibrated quantile. Do you need a process engine? If you cannot name a stock that the rule ignores, the honest answer may be: stop early.

C1-E3. Assumption criticism. "We have a funnel, therefore we optimize the funnel." Which distinction did that sentence collapse?

C1-E4. Decision reasoning. Two forecasts differ in RMSE by 8% and both already imply the same act. Is further sophistication decision-relevant?

C1-E5. Failure diagnosis. A dashboard titled "forecast" uses month-end inventory known only at month-end. What did it use that this morning did not have?

C1-E6. Decision reasoning. You are asked for "the recommended policy" and given only a point forecast. What is missing?

1.5 Transition

Chapter 2 is the intellectual contract: what a forecast is allowed to see, what kind of uncertainty a band is pretending to be, and why a business is a process whose stock remains after the KPI has been totaled.

CHAPTER 2

Representing a Business as a Process

Depends on. Chapter 1.

Two shops closed 40 files in June. A dashboard that ranks shops by completions calls them equal. A hiring freeze is then "the same act in both places." It is not.

Shop A started June with 42 OPEN, received 3 arrivals, closed 40, and ends at 5 OPEN, all young. A freeze is survivable. Next month's completions will drop because there is almost nothing left to complete, and that is the point of the freeze: demand arrived, was cleared, and stopped.

Shop B started June with 180 OPEN, received 50 arrivals, closed 40, and ends at 190 OPEN, many past a service-age that predicts dropout rather than completion. Completions are identical. The freeze is a different act. OPEN will keep aging; next month's "stable completions" will be a harvest of a decaying pile, not a capacity match. Overtime versus a freeze is a decision about *what remains*, not about last month's flow.

A period KPI is a flow through a window. A decision often cares about stock that remains. Forecasting only the terminal KPI discards the inventory that makes the *next* forecast depend on *this* action. The rest of this chapter exists so that "40 completions" cannot impersonate occupancy.

Occupancy is reconstructed from events at a *cut time*, not from a label attached to a cohort at the horizon. Using "current stage at day 84" as occupancy at a day-28 origin is illegal: the horizon label is not in the information that existed at the cut. The honest sentence is that occupancy is not available, not that a number can be interpolated from a later screenshot.

2.1 A forecast is issued at an origin

A *prediction* is any mapping from features to a target. Time may be absent. A *forecast* is a prediction issued at an *origin*, about a *horizon*, from an *information*

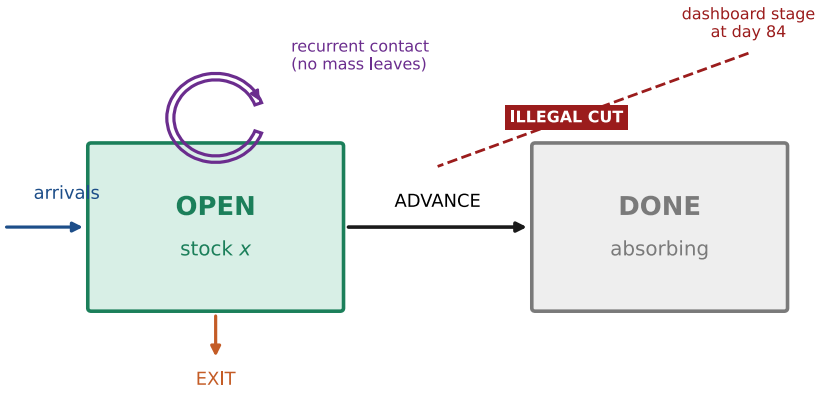


Figure 2.1: Occupancy is the area of OPEN. Completions are the ADVANCE flow through a window. Recurrent contact loops without removing mass. A dashboard stage at day 84 offered as day-28 occupancy is an illegal cut.

set that must not contain the future (Diebold 2007; Clements and Hendry 1998; West and Harrison 1997; Hyndman and Athanasopoulos 2021). The origin is not "the last row of the spreadsheet." It is a time at which some things are known and others are not.

Let $\mathcal{I}_t$ be the information available at origin t. A point forecast of Y_{t+h} is a functional $\hat{y}_{t+h|t} = g_h(\mathcal{I}_t)$. A probabilistic forecast is a predictive law $P(Y_{t+h} \in \cdot \mid \mathcal{I}_t)$. If g_h uses a variable that is not known at t, the object is not a forecast. It is leakage with a plot (Tashman 2000; Hyndman and Athanasopoulos 2021).

2.1.1 Ten people at the cut

Return to the visitor shop introduced in Chapter 1. During week 0, ten observed people arrived: `v01` through `v10`. By the start of week 1, `v01` and `v02` have completed, `v03` has exited, and `v04` through `v10` are still OPEN. All ten identities are in the origin information set because all ten people have already appeared. Seven contribute to origin occupancy because seven remain in process.

The shop forecasts expected arrivals of 8, 6, and 6 people in weeks 1, 2, and 3. Those numbers are **future arrival mass**. At the origin there is no `visitor_id` for any of those people. Assigning names would pretend to know who will arrive, not merely how many are expected.

Three objects must remain distinct:

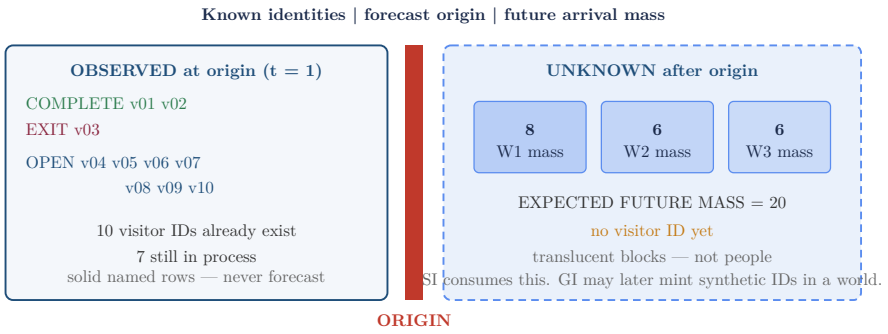


Figure 2.2: Ten observed visitor identities lie to the left of the forecast origin. Expected future arrival mass lies to the right without visitor IDs.

object	what exists at the origin?	identity
observed entity	a real person already observed	real `visitor_id`
future arrival mass	an expected count in a future period	none
synthetic future entity	a simulated person created later inside one realized world	world-specific synthetic ID

The third object does not exist yet in this chapter. Chapter 17 will create it only after a future world is drawn. The information boundary is therefore memorable before it is mathematical: **we know these people exist; future people do not have identities yet.**

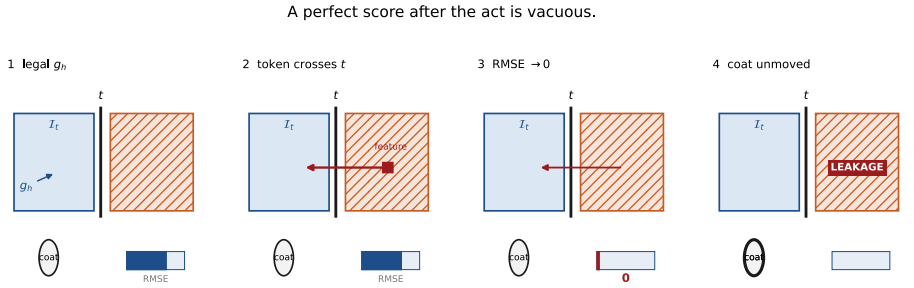


Figure 2.3: Four panels: a legal forecast from $\mathcal{I}_t$; a feature token crossing the origin from the hatched future; RMSE falling to zero; the coat decision unmoved, stamped LEAKAGE.

Panel 1 is a legal $g_h(\mathcal{I}_t)$. Panel 2 admits tomorrow's observed temperature as if it were known this morning. Panel 3 is a perfect score. Panel 4 is the act that had to be taken at t: the coat does not move. Lowest RMSE produced a vacuous decision.

The same crime in quieter clothes: a month-end inventory label used as if it were known at month-start; a campaign's realized spend used as a "forecast" of that campaign. *Forecastability* is a property of the process relative to $\mathcal{I}_t$, not of the model class. A random walk is forecastable — the last value is hard to beat. White noise is not. Baselines exist to embarrass you (Hyndman and Athanasopoulos 2021; Diebold 2007; Tashman 2000). If a sophisticated

Almost every useful forecast is *conditional*. "If the published tariff path holds, expected filings are …" is a forecast. "Filings will be …" is a slogan. Conditioning on an observed covariate, intervening on a lever, and declaring a scenario are three different sentences; Chapter 3 separates them. Here it is enough that a known-future covariate is *read*, not *set*.

2.2 Two intervals of the same width can license opposite acts

A hospital is deciding whether to buy overflow insurance that pays if midnight occupancy exceeds 40. Two forecasts both issue a 90% interval of width 8, centered near 36.

Forecast A is Poisson occupancy given *known* arrival and completion rates. Almost all of the width is noise given those rates. Buying insurance against that tail is a coherent act; the premium can be compared with $P(Y > 40 \mid \theta, \mathcal{J}_t)$.

Forecast B has the same width, but it comes from not knowing whether the arrival rate is 12 or 28. Once the rate is fixed, entity noise is tiny. You are insuring a parameter you have not identified, dressed as a tail of noise. Two scores can be equal and the decisions opposite (Der Kiureghian and Ditlevsen 2009).

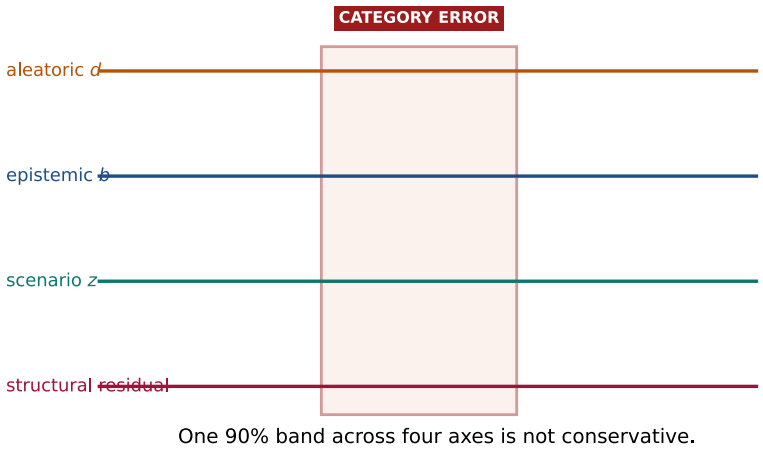


Figure 2.4: Four uncertainty axes as separate number lines. One 90% band drawn across all four is a category error, not conservatism.

Noise given a model, ignorance of a parameter, a declared other world, and a mechanism that might be wrong are not one ribbon. Identification — the *set*

of parameters compatible with the assumptions — is Chapter 4's object, not a tighter interval. Chapter 19 keeps those axes from being added in a pipeline. This chapter only forbids treating a width as a provenance.

RMSE is a scoring rule for the mean under quadratic loss, and only for that functional (Gneiting and Raftery 2007). A staffing decision that fails if occupancy exceeds a floor cares about a tail, not RMSE. If no act changes, further accuracy has no value for the decision (Chapter 19). A last-value forecast that already implies "do nothing" can be enough.

2.3 Occupancy in one page

A cell of occupancy is mass in a process state, with clocks the future still depends on: calendar time, how long this unit has been OPEN, when it arrived, how long since last contact. Omit a clock only when the future is independent of it given what you kept.

Conservation is ordinary bookkeeping (McKendrick 1926; Andersen et al. 1993): arrivals in, completions and exits out, remainder still in the building. Recurrent contact can emit an outcome *without* removing stock — a reminder on an OPEN file is not a second occupancy. Competing completion and exit probabilities cannot silently sum above one.

A Kalman filter treats Y as a noisy view of a persistent state (Kalman 1960; Durbin and Koopman 2012). A static map $Y = f(u)$ is a response. A process $x_{t+1} = F_t(x_t, u_t, \theta)$ is structure. Using a response map when stock carries is the failure Chapter 7 will compute. Chapter 9 owns the engine that advances age. This chapter only insists that Shop A and Shop B are different objects, and that the engine is not a substitute for the cut at the origin.

When no defensible stock exists at the origin, do not invent one from a later dashboard. An ordinary time-series forecast of the KPI may still be honest — and Chapter 1 already permitted stopping there.

2.4 When the naive forecast is better

A short, noisy series with a stable mean: the historical mean versus a ten-lag neural net. Sixteen draws of $N(0, 1)$. The net can interpolate the sixteen points; out of origin it is fitting noise. A seasonal series whose forcing is *not* in $\mathcal{I}_t$: seasonal naive versus a regression on next month's unobserved weather. The regression is leakage or a disguised scenario.

Sophistication is not automatically a better forecast. If every plausible forecast issues the same act, extra complexity has no value for that act.

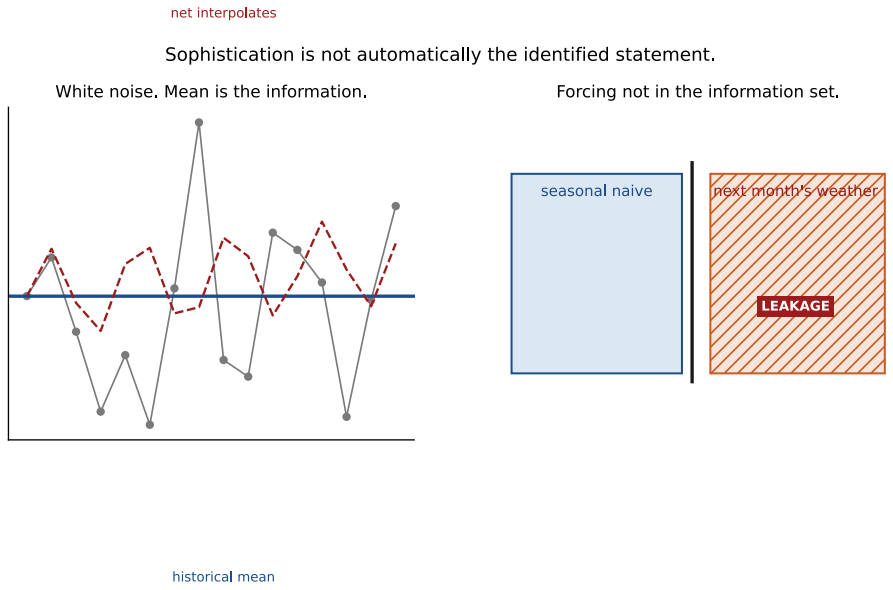


Figure 2.5: Naive mean versus an interpolating net on sixteen noise draws; seasonal naive versus a regression that uses a future forcing.

2.5 Exercises

C2-E1. Conceptual. Shop A ends June at 5 OPEN; Shop B at 190. Both closed 40. Which freeze is the staffing decision, and why?

C2-E2. Failure diagnosis. A table of "current stage at day 84" is offered as day-28 occupancy. Why is that illegal?

C2-E3. Mathematical. A model uses tomorrow's promotion flag to "forecast" tomorrow's sales. What information did it use that $\mathcal{J}_t$ did not contain?

C2-E4. Interpretation. Insurance pays if $Y > \tau$. Forecast A wins RMSE; forecast B is better calibrated in the tail. Which do you issue, and what must you know to answer?

C2-E5. Method selection. When is a seasonal naive better than a process model? Name the assumptions.

C2-E6. Assumption criticism. A stakeholder wants "the 90% band" covering parameter uncertainty, micro noise, and a recession scenario. Write the refusal in one sentence.

C2-E7. Decision reasoning. Two forecasts differ in RMSE by 8% and both already imply "do nothing." Is further sophistication decision-relevant?

2.6 Transition

Chapter 3 asks what happens when the statement is no longer about a future we will watch, but about a future we will *set*.

CHAPTER 3

Intervention Is Not Prediction

Depends on. Chapters 1–2.

Chapter 2 gave occupancy, information sets, and a vocabulary of forecast objects. The next question is not "which engine?" It is: if I *do something*, is the statement I am making still a forecast of the same kind?

People collapse three sentences that look like cousins.

"Given that spend was high, completions were high." That is *conditioning*. A covariate took a value, and Y is described in the slice of the world where it did.

"If we set spend high, completions will be high." That is *intervention*. A lever the decision maker can actually move is set, and the claim is about what happens *because* it was set.

"If we imagine a world in which spend is high, the fitted process says completions are high." That is a *scenario*. The mechanism F is held fixed; a future path z is declared; the process is asked what it would do. Nobody has claimed that z is identified as the effect of a human act.

Those are not synonyms. Mixing them into one "impact estimate" is not conservative. It is a category error. Potential-outcomes calculus (Imbens and Rubin 2015) and the do-calculus (Pearl 2009) already named the first two; scenario analysis is older still. What this chapter forbids is using the first sentence as if it were the second, so that later chapters can refuse a fake lever rather than decorate it.

Weather versus cloud-seeding is the clean pair. You can wait and observe tomorrow's weather; that is a known-future covariate, a conditional forecast. You can seed clouds; that is setting a lever, and it needs a lever identified as weather-modification, not a regression of rainfall on yesterday's seeding in seasons when storms were already coming. The sentences are not interchangeable because one of them is an act.

3.1 Three arrows into the same Y

Let Y be a future observable, X a covariate, u a lever the decision maker can set, and z a declared future path. Write $\mathcal{I}_t$ for the information available at the

origin, as in Chapter 2.

A *conditional forecast* is $P(Y_{t+h} \mid X_{t+h}, \mathcal{I}_t)$ when X_{t+h} is in $\mathcal{I}_t$, or is a known-future role (a published tariff, a contracted weather path). Honest language sounds like: "Given the published tariff, expected filings are …" The covariate is being *read*, not *set*.

An *intervention* is $P(Y_{t+h} \mid \mathrm{do}(u), \mathcal{I}_t)$. Honest language sounds like: "If we staff +4 FTE, occupancy by Friday is …" The lever is being *set*. The data that identified the map, if any, have to speak to setting, not merely to co-occurrence. Chapter 4 will ask when that identification is a singleton, a set, or `PRIOR_ONLY`.

A *scenario* is $P(Y_{t+h} \mid z, \mathcal{I}_t)$ under a declared z, with the same fitted F. Honest language sounds like: "If supply is cut 20%, the process says …" Replay is not automatically an identified counterfactual. Chapter 11 will own scenario as an object; here it is enough that declaring z does not mint a causal effect.

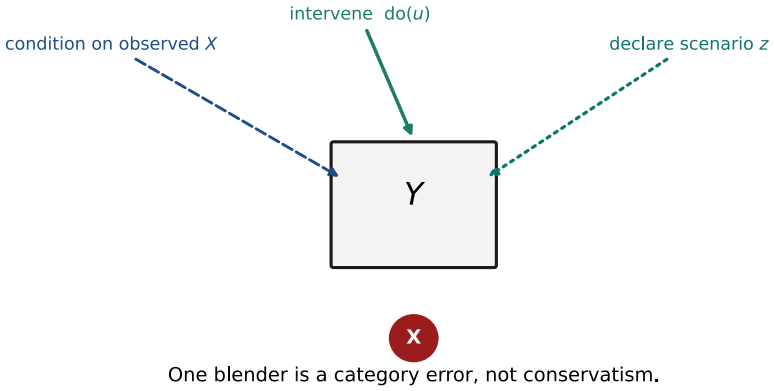


Figure 3.1: Conditioning reads an observed X; intervention sets a lever u; a scenario declares a path z. Blending the three arrows into one "impact" is not conservatism.

The figure's three arrows land on the same Y. A blender joining them is marked as a category error because each arrow authorizes a different sentence. Conditioning on an observed covariate does not authorize setting it. Setting a lever does not authorize treating a residual spike as "the COVID scenario." Declaring a scenario does not identify a causal effect.

A small picture of confounding makes the first collapse concrete. Suppose spend U and completions Y both rise in busy weeks because organic demand S is high then. Then $P(Y \mid U = u)$ describes a slice in which S is also high. $P(Y \mid \mathrm{do}(u))$ describes a world in which u is set and S is whatever S is. One graph is enough: $S \to U$, $S \to Y$, $U \to Y$. Conditioning on U mixes the path through S; intervening on U does not.

Twenty weeks, ten quiet and ten busy. Completions are constant within each cell:

season	spend	completions	weeks
quiet	10	20	8
quiet	20	24	2
busy	10	40	2
busy	20	50	8

The conditional means weight weeks by how spend *actually occurred*:

$$E[Y \mid U = 20] = \frac{2 \cdot 24 + 8 \cdot 50}{10} = 44.8, \qquad E[Y \mid U = 10] = \frac{8 \cdot 20 + 2 \cdot 40}{10} = 24.$$

The naive gap is 20.8. High spend is mostly busy weeks. The intervention sets spend and leaves season alone:

$$E[Y \mid \mathrm{do}(20)] = \tfrac{1}{2} \cdot 24 + \tfrac{1}{2} \cdot 50 = 37, \qquad E[Y \mid \mathrm{do}(10)] = \tfrac{1}{2} \cdot 20 + \tfrac{1}{2} \cdot 40 = 30.$$

The interventional gap is 7. Staffing on the regression treats extra spend as if it bought 20.8 completions; setting spend in a typical week buys 7. Same table, two laws, different act.

The same collapse, in quieter clothes: using a campaign's realized spend as a "forecast" of that campaign (that is Chapter 2's leakage, not an intervention at all); treating a GLM of historical arrivals as a causal forcing of next month's A_t; reading a residual as if it were a declared z.

3.1.1 Two levers in the visitor shop

The visitor shop makes another distinction unavoidable. Overtime applied to `v04` through `v10` is a **stock intervention**: it changes progression among people already OPEN. An acquisition action is an **arrival intervention**: if identified, it changes future arrival mass, not the seven known people. Neither action can give future people identities at the origin. Chapter 16 will put the two levers on the cohort triangle; for now, keep their targets separate.

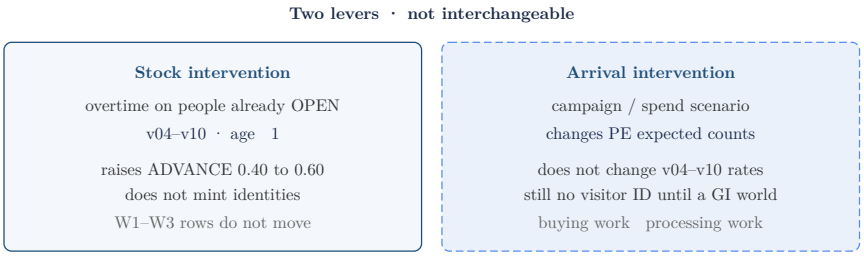


Figure 3.2: A stock intervention acts on observed OPEN visitors, while an arrival intervention changes unlabeled future arrival mass.

3.2 A map of co-occurrence is not a lever

Chapter 5 will treat maps $u \mapsto (R, Y)$ as allocation objects. In Chapter 2's vocabulary those maps are *response-driven forecasts*. They become interventions only under identification (Chapter 4). Using a regression of historical completions on historical spend as if it were a staffing lever is the collapse this chapter names, so that Chapter 5 can refuse it rather than print a frontier on top of it.

The same collapse, in process language, is treating a GLM of *arrivals* as a causal forcing of A_t. A 10% spend increase is not a 10% arrival increase. June: spend 100, arrivals 1000. July: spend 110, arrivals 1100. The two points lie on a line of slope 10 arrivals per unit spend, elasticity one. July is also the month organic demand rises, and the extra ten units of spend were bought on a channel already saturating. The conditional statement — given that spend was high, arrivals were high — is true of the slice. The intervention "set spend to 110 in a June-like month" is not identified by the pair. Exposure, saturation, timing, and the OPEN already in the building all sit in between. Later, in a qualified microsimulation of an existing pipeline, on the order of 95% of near-horizon harvest comes from stock that is already there at the origin (Chapter 17). That number is a fixture, not a field claim, but it is the right kind of insult: most of what will complete shortly is not waiting on this week's spend. Percentage spend as percentage exposure is an identity the process does not owe you.

A predictive response may still support *conditional planning*. "Given that the published tariff path holds, expected filings are …" is a forecast. It does not become "raise the tariff" by being stored in a revision. Rewrite the sentence, or refuse the lever.

3.3 How a lever enters a process

An intervention does not enter a process as a generic shock. It enters through a *type*. The types are older than this book: they are the arrows on a compartment diagram. Naming them on a process, and refusing to clip competing probabilities in silence, is governance over established mass balance, not a new physics.

Arrival forcing moves age-0 mass A_t. It does not, by itself, change the ADVANCE rates of units already in OPEN. Buying more work is not the same act as processing the work you have.

Transition or ADVANCE forcing moves competing exit probabilities of stock. It does not create new arrivals. Overtime that accelerates review is not a marketing campaign.

Recurrent forcing emits events on survivors without removing stock — a contact on an OPEN file, a rework loop, a follow-up visit. Occupancy does not change except insofar as those events later alter competing risks. Recurrence is not a second occupancy; Chapter 2 already forbade counting it as one.

Capacity or resource forcing changes constraints or rates that depend on a budget. It does not move "the KPI" directly. Adding four FTE changes how fast OPEN can ADVANCE. The completions are a consequence, not the lever.

Forcing, state, and measured response need not peak together. That is the distributed-lag / impulse-response fact (and the thermal-lag intuition of Chapter 2): outdoor temperature, wall heat, and indoor comfort are not simultaneous; media spend, latent awareness, and conversions need not be either. The metaphor does not identify causality. It only forbids assuming that peaks coincide.

The typed operators themselves — exposure, timing, magnitude, temporal kernels — live in Chapter 10. What is needed here is only that a lever has a type, so that feeding a GLM arrival prediction in as A_t can be refused as a type error rather than debated as a tuning choice. Reconstructing an arrival rate from a control path is a mapping contract, not a physical law and not an identification result.

A pure extrapolative forecast of Y (Volume 3) is legitimate when u is not being set, or when the decision-material effect of the lever is below the tolerance that Chapter 21 will call ε. Conditioning on a known-future X — weather, a published tariff — is a forecast, not an intervention. A scenario on a declared z, without claiming an intervention, is Chapter 11. The uncertainty named here is *category* uncertainty: which of the three objects is being issued. Mixing the three into one band is the crime Chapter 19 will spend a chapter on; the blender in Figure 3.1 is already illegal.

Language is earned separately for each object: conditional, interventional, or scenario. A causal recommendation is not earned from a predictive association.

When the sentence is the wrong object, refuse the lever and, if the sentence can be rewritten as a conditional forecast, issue that instead. An irreversible plant closure justified from a predictive elasticity is not a close call. The map is not identified as an intervention, and the act cannot be taken back.

3.4 Exercises

C3-E1. Conceptual. Rewrite "TV causes search" as (a) a conditional association, (b) an intervention, (c) a scenario. Which sentence do you actually have evidence for?

C3-E2. Mathematical. From the twenty-week table, compute $E[Y \mid U = 20] - E[Y \mid U = 10]$ and $E[Y \mid \text{do}(20)] - E[Y \mid \text{do}(10)]$. Why do they differ? One graph.

C3-E3. Failure diagnosis. A model uses next month's observed promotion flag to "forecast" next month's sales. Name the object (hint: Chapter 2 leakage).

C3-E4. Method choice. You can wait and observe weather, or you can seed clouds. Which object of this chapter is each?

C3-E5. Assumption criticism. "We will feed the GLM arrival prediction in as A_t." What would that require of the map, and what should you issue if it is not supplied?

C3-E6. Interpretation. Reconstructing an arrival rate from a control path produces a typed forcing. In one sentence, say what that object is **not**.

C3-E7. Decision reasoning. An irreversible plant closure is justified from a predictive elasticity. What, if anything, may you recommend?

3.5 Transition

Chapter 4 asks the next question: when is a map *identified* as an intervention, when is it `PRIOR_ONLY`, and why drawing it a million times does not promote its authority. Chapter 10 then builds the operator chain for readers who have a defensible lever.

CHAPTER 4

Identification, Support, and Authority

Depends on. Chapters 2–3.

Chapter 3 distinguished conditioning, intervention, and scenario. The reader now has a map, or a coefficient, or a posterior draw. The question is no longer which sentence was uttered. It is: *what set of worlds is that number compatible with — and which acts does that set authorize?*

Availability of a decimal is not identification. A tight interval is not support. A million simulations are not an experiment.

4.1 Identification is a set

Partial identification, in Manski's sense (Manski 2003), is the observation that assumptions plus data imply a *set* of values for an estimand. A singleton is a special case, earned by assumptions, not by software. Write Θ_I for the identified set of a response or a forcing under a named assumption class. If Θ_I is not a singleton, a point $\hat{\theta}$ is a *selection*, not a discovery. The honest issued object is the set, or a labeled selection from it.

That sounds abstract until you put a number on it. Suppose the data, under the maintained assumptions, are compatible with an arrival elasticity $\beta \in [-1, 4]$. A vendor curve, or a regularizing prior, or an optimizer's starting value, will still produce a point — say $\hat{\beta} = 0.8$. The point can be stored, plotted, and pushed through a process. It does not become the set. The set is $[-1, 4]$. Everything that follows from 0.8 as if it were known is a selection effect, and it should be labeled as one.

`PRIOR_ONLY` is this book's name for the case in which the lever's effect on the process is not identified from the data under the maintained assumptions. A prior, an analogy, or a vendor curve may still produce a number. The number does not become an experiment by being stored in a revision. Simulation does not promote `PRIOR_ONLY`. Validation of a shortlist (Chapter 18) evaluates consequences *under the declared object*. It does not upgrade the object.

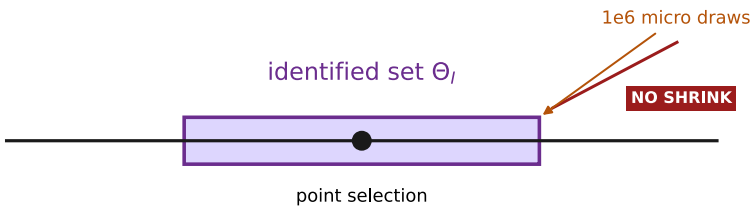


Figure 4.1: The identified set Θ_I is a region on a line. A point inside it is a selection. A million micro draws at that point bounce off the set: simulation does not shrink Θ_I, and does not promote `PRIOR_ONLY`.

Read the figure as a refusal of a very common move. The thick interval is Θ_I. The dot in the middle is a point selection. A million entity draws at that point explore d, not Θ_I.

Put an act on that geometry. The process maps $\Theta_I = [-1, 4]$ to expected harvest in $[22, 58]$. A selected $\widehat{\theta} = 0.8$ maps to harvest 41. A million draws at that point produce a Monte Carlo interval of width 0.3 around 41; the identified harvest set remains thirty-six units wide.

The board will close the plant if expected harvest is below 30, and keep it open otherwise.

parameter location	expected harvest	act
low end of Θ_I	22	close
selected point	41	stay
high end of Θ_I	58	stay

The set does not settle the act. Some legally compatible worlds close the plant; some do not. The tight Monte Carlo ribbon around 41 says stay more loudly while never visiting 22. Authority did not move. If every point in Θ_I issued the same act, the selection would not matter for this decision. Here it is the whole decision.

A Bayesian posterior $P(\theta \mid \text{data}, \text{prior})$ is a different object again. It mixes a prior into a law. It is not a synonym for Θ_I. When the likelihood does not inform a coordinate, the posterior on that coordinate *is the prior*. That is not a philosophical complaint; it is what Bayes' rule says. If $L(\beta \mid \text{data})$ is flat

on $[-1, 4]$ and the prior is $N(0, 0.1^2)$, the posterior is, to any resolution that matters, $N(0, 0.1^2)$ truncated to the interval. The data contributed the interval; the spike at zero is the prior. Calling that coordinate "estimated" is a lie. Calling the posterior interval "the identified set" is a second lie: the identified set was $[-1, 4]$, and the posterior interval is a probability measure the analyst brought to it.

Identification uncertainty is not axis b and not axis d. Mixing it into a residual band is Chapter 19's crime, previewed here. Draws d at a selected $\theta \in \Theta_I$ do not explore Θ_I. An unconditional mixture over an unidentified set is not a qualification path. For `PRIOR_ONLY` coordinates the honest object is the set, and "unconditional reliability" is `UNAVAILABLE`.

4.2 Support is identification's cousin on the control domain

A map $Y = f(u)$ is a statement on a *domain*. Outside that domain it is a different statement. Chapter 5 will refuse silent extrapolation of an allocation map; Chapter 6 will call a target resource outside the constructed trusted set infeasible. The same discipline applies to a response surface before anyone allocates.

Suppose $f(u)$ is identified — even as a singleton — on $u \in [0.5, 2]$. An optimizer asks for $u = 8$. A Hill curve, or any other parametric shape, will happily emit a number at 8. The identified estimand will not. Evaluating f past the last observed dose is not "being conservative." It is changing the estimand. Modest extrapolation can be a declared policy. Unsupported evaluation should fail closed.

The same distinction appears as support classes: interpolation, modest extrapolation, unsupported. If the new control is unsupported, the honest output is not a number. It is an unavailable identified evaluation, or a closed refusal (Chapter 10). A number without a class is how an optimizer launders an unsupported point into a schedule.

The same geometry will reappear as trusted-boundary traversal on an allocation frontier. The idea is not package-specific. A statement $Y = f(u)$ always has a domain; the software either names the domain or silently leaves it.

4.3 Authority is a claim class, not a capability

Capability is what the code can produce. Authority is which *sentence* the evidence class permits. The code can always produce a schedule. The evidence class does not always permit "do this."

A forecast of completions next month, given current OPEN, needs an information set $\mathcal{I}_t$. If the cut is missing, the object is Chapter 2's leakage or Chapter 2's `UNAVAILABLE` occupancy — not a quieter occupancy number. A sentence that begins "if we *observe* higher spend" needs a conditional association, and it does not license "if we *set*." Chapter 3 already forbade the collapse; significance does not un-forbid it. A regression coefficient on spend can be "significant" in a confounded series while the identified set for do(spend) is still huge. A p-value is not Θ_I.

The sentence that begins "if we *set* spend" needs an identified response, an experiment, or a design. If what you have is `PRIOR_ONLY`, do not recommend an irreversible act. A reversible weekly bid and an irreversible plant closure can share the same unidentified elasticity and still not share an authority. The weekly bid can be informed, cautiously, by a labeled prior, because the act can be taken back. The closure cannot. Same map, different sentence, different permission.

Even a well-identified map does not mint the later sentences. "This schedule is locally optimal on the stated nonlinear program" needs solver evidence (Chapter 15); it is not a decision rank. "This schedule is best among the shortlist under micro realization" needs the validation of Chapter 18; it is not a global optimum. "Do this" needs preference, a business tolerance ε, authority, and a human remainder (Chapters 21 and 26). If preference is missing, say so. Do not emit a quieter interval.

Reporting the *declared* authority of a candidate does not mint a better one. Allocation evidence, when it arrives in Chapter 5, is a decision object, not a causal claim. Placing a map on a frontier does not acquire causality for it.

4.4 What to issue instead of a fake point

Several honest objects are available once Θ_I is named.

Bound analysis, in Manski's style (Manski 2003), issues Θ_I itself. That is the most direct sentence: "under these assumptions the elasticity lies in $[-1, 4]$."

A Bayesian analysis issues a posterior, and says when the posterior is prior-dominated. The unidentified-coordinate case above is the extreme: say that the posterior *is* the prior, and do not call the coordinate estimated.

A point selection with sensitivity issues $\hat{\theta}$ *and* how the preferred act α^* moves as θ ranges over Θ_I. If every $\theta \in \Theta_I$ issues the same act, the selection does not matter for this decision (value of information is then zero; Chapter 19). If some θ close the plant and some do not, the selection is the whole decision, and it should not hide inside a point.

The remaining honest move is to refuse the lever: `DO_NOT_USE_THIS_MODEL` as intervention, while still forecasting Y extrapolatively if that is the act. A shop

that cannot identify how spend creates arrivals can still forecast completions from current OPEN. Those are different sentences, and only one of them was asked for.

A named estimand, a named assumption class, a domain for u: missing any one of them is not a reason to emit a tighter number. It is a reason to emit a set, a status, or a refusal.

4.5 Exercises

C4-E1. Conceptual. Why is a singleton Θ_I an assumption-laden claim rather than a software feature?

C4-E2. Mathematical. Data compatible with $\beta \in [-1, 4]$. A prior $N(0, 0.1^2)$ yields a posterior near 0. What did the data contribute?

C4-E3. Failure diagnosis. Validation returns a crisp $P(A > B)$ for two `PRIOR_ONLY` schedules. What did the simulation promote, and what must it not promote?

C4-E4. Method choice. Close the plant if harvest is below 30. Using 22 / 41 / 58, does Θ_I settle the act? A reversible weekly bid shares the same unidentified elasticity. Which act, if either, may a labeled prior inform?

C4-E5. Public API. Evaluate a response at an unsupported u. Report the support class. Do not treat the number as identified.

C4-E6. Decision reasoning. You are asked to recommend plant closure from a predictive elasticity. What, if anything, may you authorize?

C4-E7. Assumption criticism. "The Bayesian interval is the identified set." Write the refusal in one sentence.

4.6 Transition

Chapter 5 asks the opposite of Chapter 2's persistence. If, at this grain, retained state does not change the act, allocation among static maps remains. Those maps inherit this chapter's support and authority. They do not acquire causality by being placed on a frontier.

CHAPTER 5

When Tomorrow Does Not Remember: Allocation as a Frontier

Depends on. Chapters 2–4.

A forecast of next month's visits does not say which clinic should lose an hour. A blended "ROI" does not say it either: it has already averaged away the slope that would have told you. Once Chapter 2's stock and Chapter 4's identification have been faced honestly, the remaining problem is smaller, and sharper.

> If, at the grain of the act, retained process state does not change the preferred allocation, what is left to optimize?

What is left is a static assignment of a scarce resource among units that do not remember one another. That is a different epistemic object from a schedule through F_t. Using a process optimizer because "we have a funnel" is how sophistication arrives before it has been earned. If memory *is* material at this grain, the honest move is Chapter 9, not a more elaborate static map. Chapter 7 will say when the parent's accounting has already become a lie. Here we stay in the world where tomorrow does not remember, and we refuse to pretend that a forecast of Y is already a rule for placing R.

The sequence of methods is not a catalog of packages. A naive forecast of Y cannot say *where* to put R. Trusted maps $u \mapsto (R, Y)$ per unit can, but they do not yet construct the efficient *set* (Chapter 6). That set, at one grain, does not yet speak for an organization that decides at many grains (Chapter 7). Composition of those grains is exact only when the cut is Markov-sufficient *and* the decision impact of dropping occupancy is not larger than the user's ε (Chapters 7 and 9). Vector outcomes and parameter draws (Chapter 8) still leave preference undeclared (Chapter 21). Each layer exists because the previous one failed a question the act actually asks.

5.1 Control is not the budget, and the budget is not the outcome

People collapse three objects into one "ROI."

A **control** u is a lever someone can set: shift length, a bid, hours scheduled, a targeting knob. A **resource** R is what the constraint sees: payroll, media cost, bed-days. An **outcome** Y is what the objective sees: completed visits, harvest, claims closed. Sometimes $u = R$: spend is the lever, and the budget is denominated in the same units. Often not. Target ROAS is not spend. Staff hours on the floor are not completed visits. Shift *length* is not payroll once overtime premia exist.

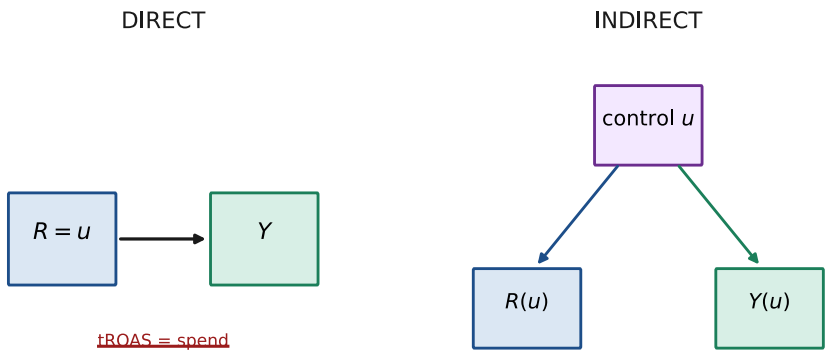


Figure 5.1: Control, resource, and outcome are three objects. Direct maps identify resource with the lever; indirect maps split a lever into a cost the budget sees and an outcome the objective sees. Equating a ratio such as tROAS with spend is a category error.

Call a unit **direct** when $Y_i = f_i(R_i)$ and $R_i = u_i$. Call it **indirect** when the lever produces a pair $(R_i(u_i), Y_i(u_i))$. Direct is a special case, not a moral improvement. The invariant is that the three objects remain distinct even when two of them happen to share a name.

A clinic makes the split numerical. Let u be hours of presence, the lever a scheduler actually sets. Completions Y depend on presence: an overtime hour is still one hour of a clinician in the building. Payroll R does not. With a $1.5\times$ premium after eight hours,

$$R(u) = \begin{cases} u & u \le 8, \\ 8 + 1.5(u - 8) & u > 8. \end{cases}$$

The pair $(R(u), Y(u))$ is an indirect map. A ten-hour shift makes the split numerical: presence is $u = 10$, payroll is $R = 8 + 1.5 \times 2 = 11$. Completions are a function of the ten hours a clinician was in the building, $Y(10)$, not of eleven hours of budget. Anyone who inverts payroll and reads a direct map at $R = 11$ has invented an eleventh hour of presence. Treating completions as $Y = f(R)$ asserts, more generally, that an overtime hour "buys" as many visits as 1.5 straight-time hours of budget. The constraint sees 1.5; the waiting room sees 1. Identifying $Y(R)$ directly skips the labor identity. That skip is often wrong, and it is wrong *before* any optimizer runs.

The same collapse in a marketing costume is the sentence "tROAS is the spend." It is not. The ratio is an outcome-per-resource *summary*. The lever, if there is one, is a bid or a budget cap; the resource is cost; the outcome is a conversion or a harvest. Collapsing them hides the constraint.

Maps of this kind are **response-driven forecasts** in Chapter 2's taxonomy. They become interventions only under identification (Chapters 3–4). A regression of historical completions on historical spend is not a staffing lever. Frontier optimization consumes maps; it does not mint causal authority. The maps arrive from ResponseIntelligence, or from another identified response. The frontier layer does not refit them. A silent restart that refits the maps is a different statement pretending to be the same object.

5.2 Three clinics, and why a blended return is already a lie

The laboratory that Chapters 6–8 will keep using is three independent sites of a claims clinic. Resource is paid hours. Outcome is completed visits. The maps are direct, $u = R$, and Michaelis–Menten in shape,

$$Y = \frac{VR}{K + R}, \qquad R > 0.$$

V is a ceiling, K a half-saturation. The form is a teaching map, not a marketing law, and not a claim that visits are enzymatic. It is concave and increasing on $R > 0$: the second derivative is negative, so each additional hour buys fewer visits than the hour before. Trusted support is the compact interval [0.5, 7.5] on each site. Outside that interval the map is a different statement.

site	role	V	K	$Y(0.5)$	$Y(7.5)$
A	urgent	10	1.0	3.33	8.82
B	routine	8	3.0	1.14	5.71
C	elective	12	6.0	0.92	6.67

At half an hour, A already produces 3.33 visits and C only 0.92: the small-K site is steep near the origin. At 7.5 hours, A has almost reached its ceiling of 10, while C, slow to start, has climbed to 6.67 on the way to 12. The derivatives make the rank reversal explicit. Differentiating $Y = VR/(K + R)$ gives $Y' = VK/(K + R)^2$, so at the trusted lower bound

$$Y'_A(0.5) = 4.44, \qquad Y'_B(0.5) = 1.96, \qquad Y'_C(0.5) = 1.70,$$

and at the trusted upper bound

$$Y'_A(7.5) = 0.138, \qquad Y'_B(7.5) = 0.218, \qquad Y'_C(7.5) = 0.395.$$

Near the origin A is more than twice as steep as C. At the high bound C is almost three times as steep as A. Two units with different (V, K) trade rank as the total budget moves. At a low budget the next hour belongs to A. At a high budget C's unused ceiling starts to matter. Nothing in that sentence requires a process engine. It does require that the maps remain separate objects, not a blended return that has already mixed their slopes. A last-quarter average that happened to weight the three sites equally would report a single "visits per hour" and would be unable to see that the next hour and the last hour belong to different sites.

A blended ROI is the average of those slopes, weighted by whatever hours happened to be worked last quarter. It cannot tell you where the *next* hour should go, because the next hour is an increment on a particular map, at a particular occupancy of that map. Chapter 6 will construct the portfolio states those increments generate. This chapter only insists that the increments exist, are trusted on a named domain, and are not secretly functions of one another.

5.3 Separability is an assumption, not a default

Write n units, a single resource, a single outcome. If

$$Y = \sum_{i=1}^{n} Y_i(u_i), \qquad \sum_{i=1}^{n} R_i(u_i) = \bar{R},$$

and the maps do not depend on one another's controls, the portfolio is **separable**. Concave increasing $Y_i(R_i)$ and one resource is the setting in which ranking by marginal sacrifice is optimal on the constructed trusted set — a standard diminishing-returns knapsack (Fox 1966), in the lineage of Koopmans's activity analysis (Koopmans 1951), not a system invention. The next hour of overtime should go to the clinic where an extra hour still buys visits. Once that clinic saturates, the ranking changes. Chapter 6 proves the greedy claim and walks

the three-site arithmetic. The claim is a theorem *on this class*. It is not a personality trait of the software.

Complements replace the sum. If television makes search more productive, the outcome is not $Y_1(R_1)+Y_2(R_2)$; the cross partial is positive, and the stationarity condition of the separable problem is the wrong equation. If two channels share one impression pool, they are two straws in one well: spending on one moves the map of the other.

A teaching well of 100 impressions. Channel A at 60 already extracts $Y_A = 48$. Channel B, treated as independent, is scored as if 100 remain and $Y_B = 70$ looks available. After A has drunk, 40 remain; B's honest yield is 28. The separable construction over-states the portfolio by 42 visits. That is not a small coupling residual. It is the wrong object.

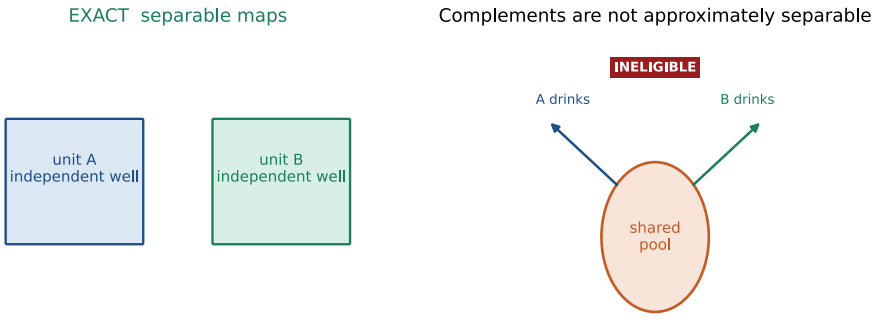


Figure 5.2: Left: two independent wells. Right: two straws in one well.

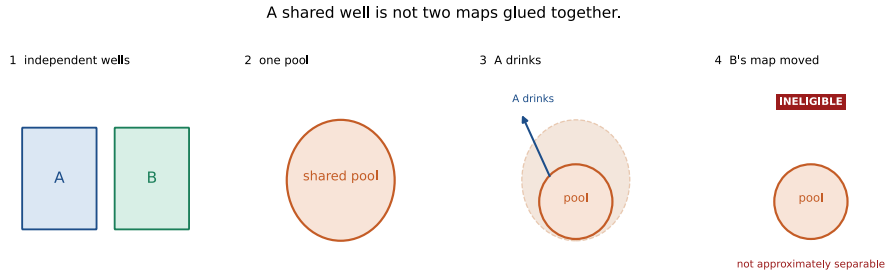


Figure 5.3: Four stills: independent wells; one pool; A drinks; B's remaining map has moved.

Product complements and shared saturation are not "approximately separable if we squint." They are a different mathematical object. The separable construction over-stated the portfolio by 42 visits because it scored B as if A had not already drunk.

Static coupling of maps is not process carryover x_t. Carryover is Chapter 7. Coupling is the question whether Y adds across units *now*. The construction reports a class, it does not negotiate. Separable or additive maps are `EXACT`. A declared weak interaction is `APPROXIMATE`. Product complements and shared saturation are `INELIGIBLE`. That last status is not a curve with an apology. Patching two channels that share a pool as separable "to get a number" is the vice Chapter 2 already named: sophistication without identification. The failure object is the lesson.

The maps are statements on a domain. The construction consumes them; it does not fit them. A laboratory fixture is not a business recommendation. On the three-site laboratory the maps are independent, so the coupling class is exact. Asking the same construction to treat two channels that share a pool as independent does not produce a slightly worse frontier. It is ineligible: the object is wrong.

5.4 A map is a statement on a domain

A dose–response evaluated past the last dose is not conservative. It is a different estimand. Support restriction is ordinary scientific hygiene; what the system adds is that a restart must not silently refit, and that the failure is labeled rather than returned as a curve.

Trusted support here is a compact control interval, an observed grid, or a trusted neighborhood. The three clinics are trusted on [0.5, 7.5]. If someone asks for control 0, the honest output is unsupported — not $f(0)$. Chapter 6 will add a second brick: a target total resource that lies below the constructed minimum of the *portfolio* is infeasible, even if each unit's map could be written down at that point. Interpolation inside trusted support is construction. Evaluation outside it is extrapolation of a statement that was never made.

Until a stochastic campaign (Chapter 8), a frontier may honestly report that uncertainty is `UNAVAILABLE`. That is not $\sigma = 0$. It is the absence of an uncertainty object. Treating missing uncertainty as a degenerate distribution is how a point map impersonates a law.

Validation follows the same hygiene. Out-of-support application is not a forecast. Restart must replay the same maps; it must not refit them. Coupling class must match the physics. The maps come from an identified response; frontier optimization does not grow them.

5.5 Three frontiers, and why "globally optimal" is usually unearned

Even after a successful build, three different objects still answer to the word "frontier." They have three owners, and "optimal" means three different things.

The object of this chapter and the next is a mathematical set $\mathcal{F}$ of raw trusted states, constructed among the units that were actually supplied. Envelope-scoped reoptimization among those units is not a process global, and it is not a decision. ScientificIntelligence, in Chapters 14–16, generates candidate *schedules* through a process operator F_t, locally, on a stated nonlinear program, with an evaluation limit. The decision-layer overlay, in Chapters 15 and 23, ranks a supplied set after a human has named materiality and risk. Those are not the same sentence. Collapsing them is how a successful allocation is reported as "the globally optimal spend of the industry."

The governing refusal is `no_unearned_global_optimum`. Envelope-scoped `GLOBAL_REOPTIMIZATION` among available units is a label on an envelope: which units may move, which are frozen, which are merely context. It is not a solver claim about the true process, and it is not a recommendation. Chapter 7 will make the envelope visible as a DAG. Chapter 8 will refuse to invent the preference that would turn an undominated set into a single point. This chapter only forbids the noun from drifting.

A successful build therefore still does not authorize the sentence "this is globally optimal." It authorizes a narrower one: among these trusted maps, under this coupling class, inside this support, these portfolio states are undominated for a single resource and a single outcome. That sentence is already worth having. It is not worth inflating.

5.6 Refusals that look like analysis

Two channels share one impression pool. The construction must refuse. A consultant who labels them separable in order to ship a chart has changed the problem. The chart can be accurate as a description of last quarter's blended return and still be the wrong object for next quarter's split.

Rank the three clinics by the historical RMSE of a volume forecast and give the next hour to the "most accurate" site. Accuracy of Y is not marginal productivity of R. Site A can have RMSE 3 and a remaining slope of 0.14; site B RMSE 8 and a remaining slope of 0.40. If the act is allocation and the maps are visible, the next hour goes to B: it buys almost three times as many visits, noisily. If the maps are not visible, the honest status is `INSUFFICIENT_INFORMATION`, not a ranking by forecast score. A more accurate forecast of a flat map is a more accurate forecast of a unit that should not receive the next hour.

A map offered as a causal claim with `PRIOR_ONLY` authority, attached to an irreversible act, is Chapter 4's refusal, not this chapter's optimizer. Product complements, shared saturation, decision-material OPEN carrying across units, and empty trusted support are this chapter's. OPEN that carries from this week's "unit" into next week's is not a coupling class. It is a process, and it is Chapter 7. Do not compress it as a courtesy to a calendar widget.

The reader should refuse the machinery, not "try it anyway and add a fudge." Chapters 6–8 assume the refusal has already happened. What they will construct is an efficient set of raw trusted states, then a DAG of those sets, then a vector and a campaign. None of that construction refits the maps, and none of it becomes a process engine by being nested.

5.7 Exercises

C5-E1. Mathematical. Write $Y(u)$ and $R(u)$ for overtime versus completions with a $1.5\times$ overtime premium after eight hours. Direct or indirect?

C5-E2. Failure diagnosis. Two channels share one impression pool. What coupling class is that? If someone still asks for a frontier, what status is owed: a curve, or `INELIGIBLE`?

C5-E3. Assumption criticism. A frontier builds successfully. Why is the sentence "the frontier is globally optimal" still forbidden?

C5-E4. Interpretation. A frontier object has `uncertainty$availability = UNAVAILABLE`. Interpret that in one sentence that does not say "zero risk."

C5-E5. Method selection. OPEN carries from this week's unit into next week's unit. Is this chapter's independence assumption true? Which chapter owns the problem?

C5-E6. Decision reasoning. Unit A has RMSE 3 and a flat remaining $Y(R)$. Unit B has RMSE 8 and a steep remaining slope. Budget is scarce. Which unit gets the next dollar if the act is allocation? If you cannot see the maps, what status is honest?

C5-E7. Method selection. The three clinics are independent maps. Two advertising channels share one impression pool. Which of those constructions is eligible as a separable frontier, and what do you issue for the other?

CHAPTER 6

Raw Efficient States, Not a Fitted Curve

Depends on. Chapter 5.

A trusted map $Y_i(R_i)$ per unit still does not answer the question the act asks. The act asks which *portfolio* states are efficient: which combinations of unit-level controls, at a stated total resource, are undominated inside trusted support. People skip that combinatorial object and fit a Hill curve through imagined points. The curve is a summary of a set that does not yet exist. This chapter constructs the set. The curve, if any, comes last, and it is labeled derived.

6.1 Efficiency is a subset of what you actually built

Let $\mathcal{Z}$ be a set of feasible portfolio states $z = (R, Y)$, a single resource to minimize and a single outcome to maximize. A state $z \in \mathcal{Z}$ is **efficient** when no $z' \in \mathcal{Z}$ has $R' \leq R$ and $Y' \geq Y$ with at least one inequality strict. Write $\mathcal{F}$ for that subset. Never write F: that is the process operator of Chapter 2. Never write f: that is a fitted curve through $\mathcal{F}$, and it is not $\mathcal{F}$.

This is ordinary two-dimensional Pareto efficiency (Miettinen 1999). The system-specific claim is narrower: **raw trusted states are authoritative**; f is compression. A fitted Hill curve can be evaluated at any R, including values the construction never visited and values outside support. That evaluation looks like a forecast. It is an extrapolation of a summary, twice removed from the maps. Compression error of f against $\mathcal{F}$ can be tiny while the act that needs a tail of a later process is unrelated. Forecast accuracy of the derived curve is not decision quality.

The set $\mathcal{F}$ is a polyline of raw states. Occupancy, when it appears in later chapters, is a box, not a point on that polyline. A Hill curve drawn as if it *were* $\mathcal{F}$ is a picture of the wrong object.

6.2 Frontier construction as a complete method

Reuse the verified laboratory: three independent clinic maps, trusted hours [0.5, 7.5], start at the rich corner, target $\bar{R} = 12$, mix (3.0, 3.5, 5.5), $Y = 17.547$.

A trusted map per site still does not answer the act. The act asks which *portfolio* states are efficient at a stated total resource, and then which of those states, if any, a declared preference will pick. People skip the construction and fit a curve through imagined points. This sequence refuses that skip. The reader watches $\mathcal{F}$ being built, then watches preference fail to appear by itself.

Never write F for this set: that is the process operator of Chapter 2. Never write f for this set: that is a fitted curve through $\mathcal{F}$, and it is not $\mathcal{F}$.

6.2.1 Unit maps, trusted support

Three independent sites of a claims clinic. Resource is paid hours. Outcome is completed visits. Direct maps, $u = R$, Michaelis–Menten in shape, teaching maps not a marketing law:

$$Y = \frac{VR}{K + R}.$$

site	role	V	K	trusted interval
A	urgent	10	1.0	[0.5, 7.5]
B	routine	8	3.0	[0.5, 7.5]
C	elective	12	6.0	[0.5, 7.5]

At the lower bound A is already steep ($Y_A'(0.5) = 4.44$); C is not ($Y_C'(0.5) = 1.70$). At the upper bound the rank has reversed ($Y_A'(7.5) = 0.138$, $Y_C'(7.5) = 0.395$). A blended last-quarter ROI cannot see that reversal. Chapter 5 already forbade the blend. This sequence uses the three maps as separate objects.

Trusted support is the compact interval on each site, hence a box in the portfolio. The rich corner (7.5, 7.5, 7.5) has total resource $R = 22.5$ and harvest

$$Y_A = 8.824, \quad Y_B = 5.714, \quad Y_C = 6.667, \quad Y = 21.204.$$

The constructed minimum, every site at 0.5, is $R = 1.5$. Target $R = 0.1$ is not a point to the left of a curve. It is a brick: **infeasible**. Evaluating a fitted Hill curve at 0.01 is decoration.

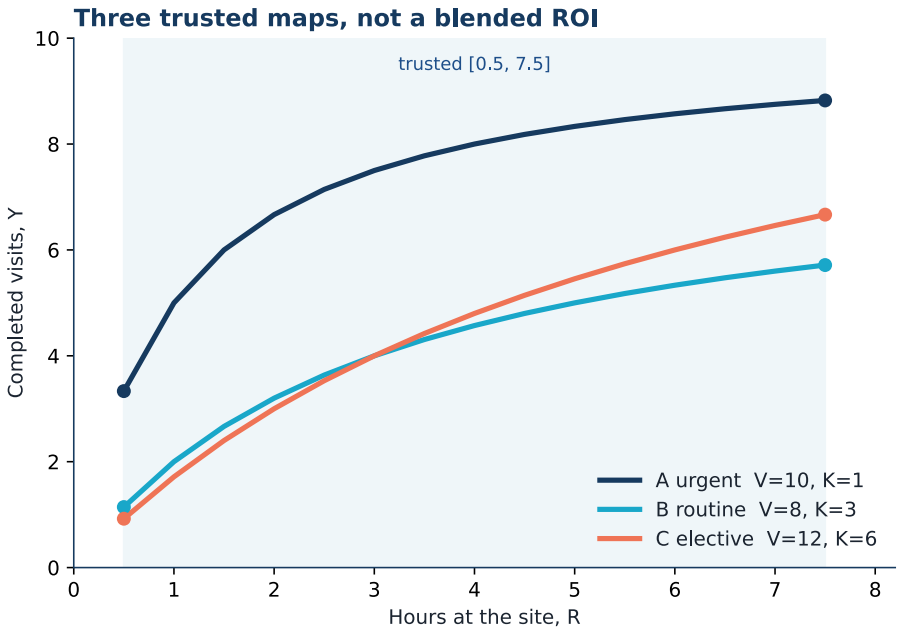


Figure 6.1: Three trusted maps, not a blended ROI

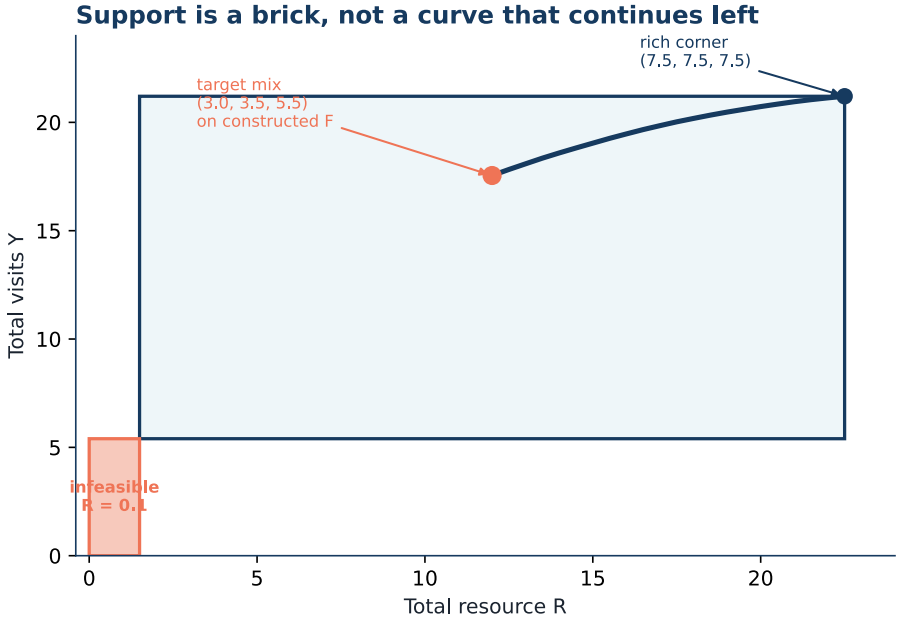


Figure 6.2: Support is a brick, not a curve that continues left

6.2.2 Marginal sacrifice, one legal step

Remove-resource traversal starts rich and gives hours back until the total hits the target. The next decrement of size $\delta = 0.5$ comes from the site whose **loss of visits per hour given back** is smallest,

$$s_i(R_i) = \frac{Y_i(R_i) - Y_i(R_i - \delta)}{\delta},$$

provided the step remains inside [0.5, 7.5]. That is discrete marginal sacrifice. It is the slope of the chord we are about to walk, not the slope of a global curve we have not built.

At the rich corner:

$$\begin{aligned} s_A(7.5) &= \frac{10 \cdot 7.5/8.5 - 10 \cdot 7.0/8.0}{0.5} = \frac{8.8235 - 8.7500}{0.5} = 0.147, \\ s_B(7.5) &= 0.229, \\ s_C(7.5) &= 0.410. \end{aligned}$$

A is cheapest. Giving back half an hour costs 0.074 visits. The first token that moves is A's, from 7.5 to 7.0. Total resource becomes 22.0. Harvest becomes 21.131. **That pair** $(R, Y) = (22.0, 21.131)$ **is now a vertex of** $\mathcal{F}$**.** The set did not exist at the target. It is being constructed.

6.2.3 Watch the set grow

Twenty-one steps of 0.5 land exactly on $\bar{R} = 12$. After every step the three chords are recomputed and the cheapest legal site is cut. Re-ranking is not a defect. It *is* diminishing returns.

Read the six frames left to right, top then bottom.

- **Step 0.** One point: the rich corner. $\mathcal{F}$ has a vertex and no path.
- **Step 4.** Four cuts, all from A. Mix (5.5, 7.5, 7.5), $R = 20.5$, $Y = 20.842$. A's sacrifice has steepened from 0.147 to 0.256, which is now *above* B's still-flat 0.229. The next cheapest site is B. The polyline exists.
- **Step 8.** A and B have started to trade the badge. Mix (5.0, 6.0, 7.5). C has not moved. Its chord is still 0.410.
- **Step 13.** Mix (4.0, 4.5, 7.5), $R = 16.0$, $Y = 19.467$. Now $s_A = 0.444$, $s_B = 0.457$, $s_C = 0.410$. **C enters.** The high-ceiling site, ignored at the rich corner because it was still productive, is the first to give hours back once A and B have been driven onto their steep shoulders.
- **Step 17.** Mix (3.5, 4.0, 6.5), $R = 14.0$. All three sites are in the walk.

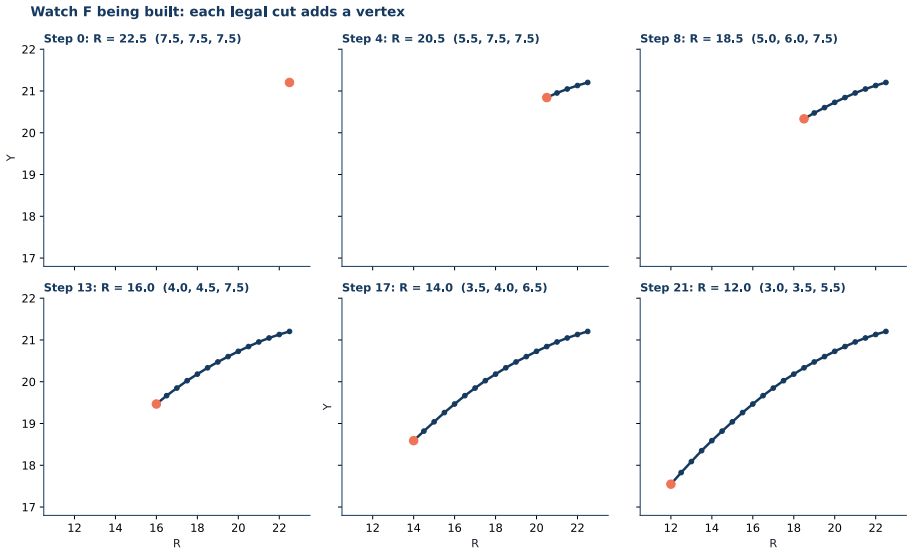


Figure 6.3: Watch F being built: each legal cut adds a vertex

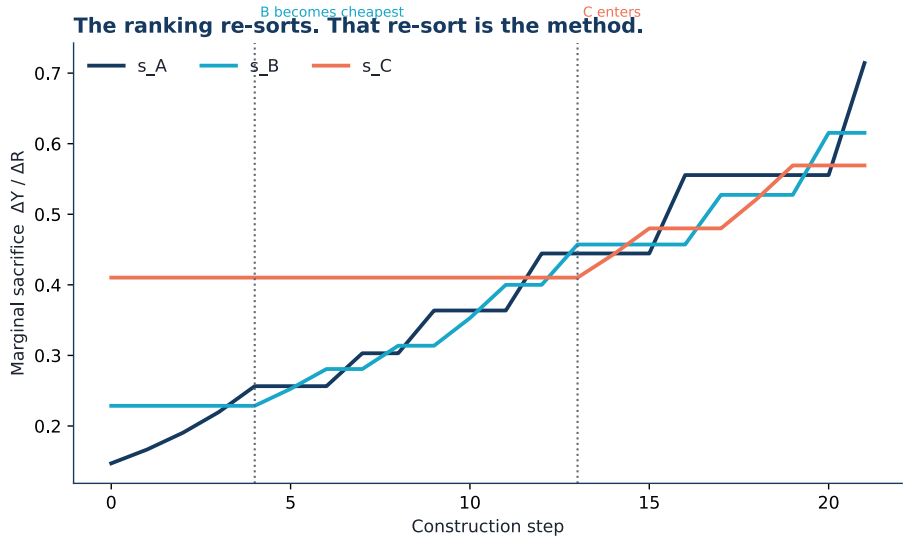


Figure 6.4: The ranking re-sorts. That re-sort is the method.

- **Step 21.** Mix (3.0, 3.5, 5.5), $R = 12.0$, $Y = 17.547$. The target. Twenty-one vertices sit on $\mathcal{F}$. No curve has been fitted.

The full walk is in `arithmetic/frontier_walk.csv`. Four load-bearing rows:

after cut	mix (R_A, R_B, R_C)	R	cheapest next
start	(7.5, 7.5, 7.5)	22.5	$s_A = 0.147$
A has steepened	(5.5, 7.5, 7.5)	20.5	$s_B = 0.229$
C enters	(4.0, 4.5, 7.5)	16.0	$s_C = 0.410$
target	(3.0, 3.5, 5.5)	**12.0**	—

On this laboratory the grid hits 12 exactly. If a later laboratory would overshoot, the last state is the convex combination along *that unit's* trusted segment. Interpolation is construction, not evaluation of a Hill curve. The Hill curve has not been fitted yet. When it is, it is labeled **derived**. Compression error is how well that curve tracks the vertices already built. It is not occupancy next quarter, and it is not the allocation.

6.2.4 Traceback is the allocation

Harvest at the target is KIT-06 arithmetic, restated so the sites stay named:

$$Y_A = 10 \cdot 3/4 = 7.500, \quad Y_B = 8 \cdot 3.5/6.5 = 4.308, \quad Y_C = 12 \cdot 5.5/11.5 = 5.739,$$

total $Y = 17.547$. Urgent care still has more hours than a "cut the expensive site first" rule would have left it, because A was cheap to cut only *while* it sat on its ceiling. Elective C gave back two hours only after A and B had steepened past 0.410.

A number $f(12)$ does not name the sites. The mix does. That is why traceback is not a reporting convenience. It is the act.

6.2.5 Four allocations at the same budget, only one on constructed $\mathcal{F}$

Fix $R = 12$ and consider three rivals the construction did *not* emit. They are legal portfolios. They are not vertices of $\mathcal{F}$.

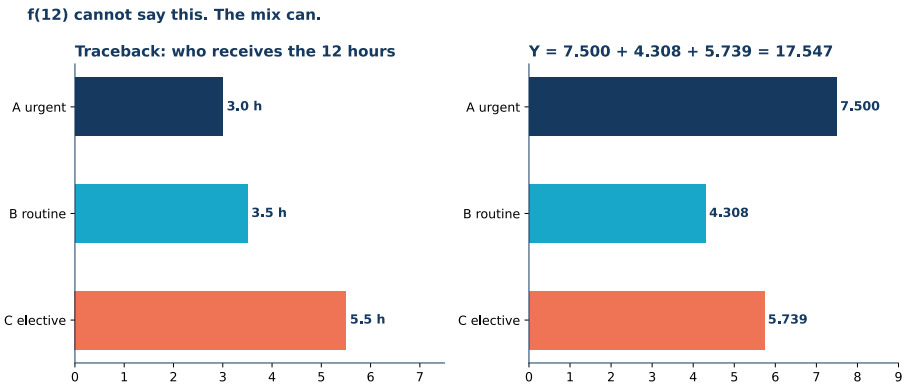


Figure 6.5: Traceback: who receives the 12 hours

name	mix	Y	wait incidents $8 - R_A$	on constructed $\mathcal{F}$?
F-mix	(3.0, 3.5, 5.5)	17.547	5.0	yes
Equal split	(4.0, 4.0, 4.0)	17.371	4.0	no
A-heavy	(5.5, 4.0, 2.5)	16.562	2.5	no
C-heavy	(2.0, 2.5, 7.5)	16.970	6.0	no

Incidents are a declared second map, $I = 8 - R_A$: wait events fall as urgent hours rise. It is an input, not a discovery, and not a fitted residual.

Equal split harvests 17.371, which is 0.176 below the constructed mix at the same R. On the single outcome Y, Equal is dominated by F-mix. So are A-heavy and C-heavy. **Constructed $\mathcal{F}$ at $R = 12$ contains one point.** The rivals are feasible. They are not efficient in Y.

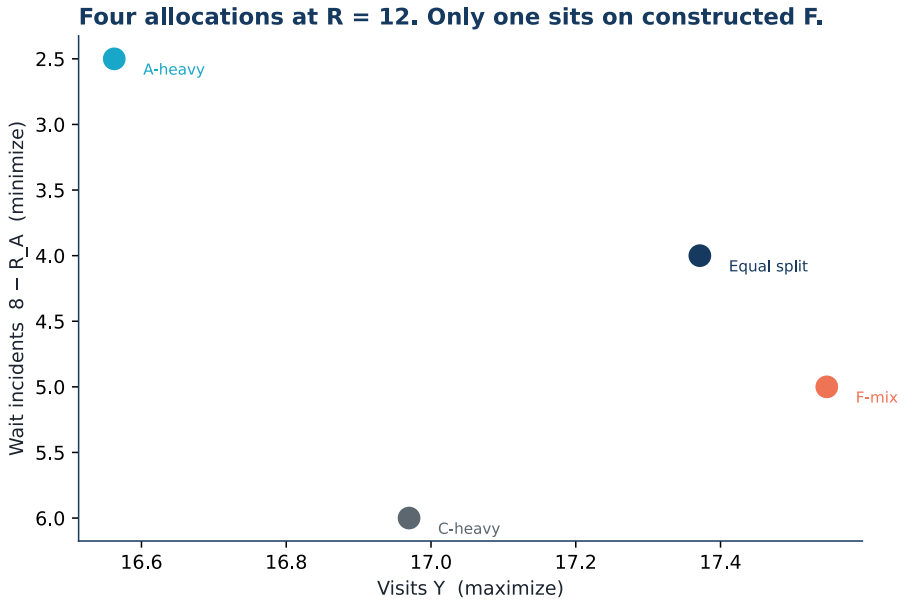


Figure 6.6: Four allocations at R = 12. Only one sits on constructed F.

C-heavy has worse Y *and* worse incidents than F-mix. C-heavy is dominated. Remove it.

F-mix versus Equal: F-mix harvests 0.176 more and has one more incident. Incomparable.

F-mix versus A-heavy: F-mix harvests 0.985 more and has 2.5 more incidents. Incomparable.

Equal versus A-heavy: Equal harvests more and has more incidents. Incomparable.

The mathematical frontier of the *supplied* set is {F-mix, Equal, A-heavy}. It is not a global optimum of the industry. `globally_optimal` is hard-false. This is Chapter 8's object, sitting on Chapter 6's construction. It is still not Chapter 15's process-candidate pool, and still not Chapter 21's decision overlay.

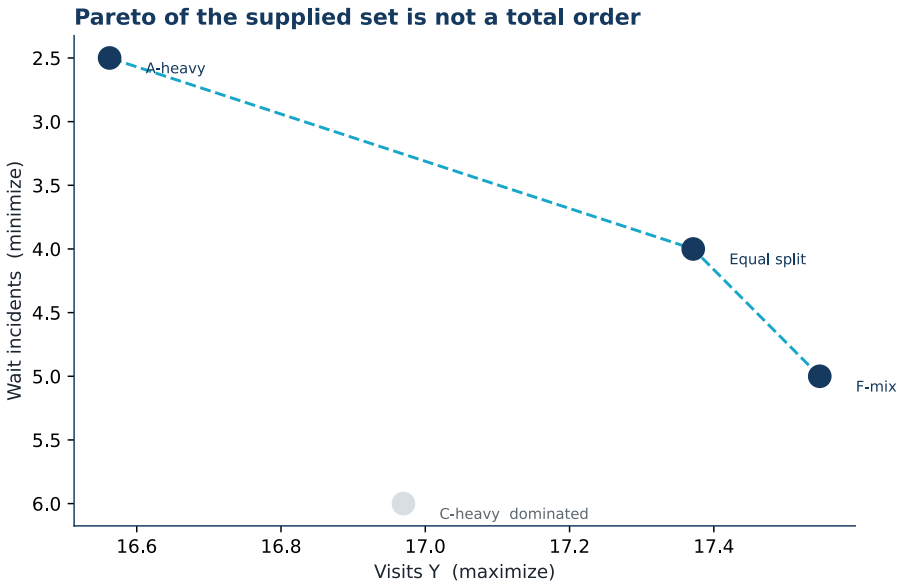


Figure 6.7: Pareto of the supplied set is not a total order

A hidden weight that scores $Y - 0.2 \times I$ picks F-mix ($17.547 - 1.0 = 16.547$ versus Equal $17.371 - 0.8 = 16.571$ — Equal, actually, at 0.2). At weight 0.1 it picks F-mix (17.047 versus 16.971). The weight is the ethics. The arithmetic is the ethics with a decimal. Missing weights fail closed.

6.2.7 ε, then risk, then a decision or a stop

The organization writes: a third of a visit is noise, $\varepsilon_Y = 0.30$. Distance from Equal to constructed F-mix is $0.176 < 0.30$. Distance from A-heavy is $0.985 >$

0.30. The ε-near set of constructed $\mathcal{F}$, in the Y sense, is

$$\mathcal{N}_\varepsilon = \{\text{F-mix, Equal}\}.$$

A-heavy is on the two-objective mathematical frontier of the supplied set and **outside** the near set of the constructed mix. That is the side door Chapter 8 already marked. Risk without ε can pick A-heavy as "safer" (incidents 2.5) while sacrificing almost a full visit the organization already called material. Overlay order forbids the skip: $\mathcal{F}$, then near, then risk *inside* the near set.

Inside $\mathcal{N}_\varepsilon$:

- Risk-neutral, maximizing Y, picks **F-mix** (17.547).
- Incident-minimizing picks **Equal** (incidents 4 versus 5).
- If nobody will say which person they are, the honest output is `RISK_PREFERENCE_UNRESOLVED`. It is not a quieter ranking.

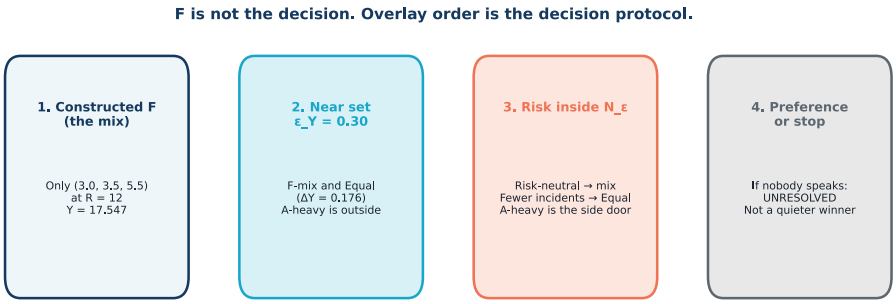


Figure 6.8: F is not the decision. Overlay order is the decision protocol.

The machine constructed $\mathcal{F}$, named the mix, filtered domination, applied a declared ε, and stopped when preference was missing. That stop is the result. Inventing expected-value because the ranking table looks empty is a preference, not a default.

Chapter 15's five hospital policies are the same overlay on a *process* candidate pool. Do not collapse the two constructions. Chapter 21's Sure/Gamble table is the same overlay with a CVaR sentence. This sequence does not reprint it. It only shows that the clinic mix, once constructed, still needs a human remainder.

6.2.8 What the reader can now reconstruct

Given concave separable maps, trusted bounds, a step δ, and a target $\bar{R}$, the reader can:

1. start at a trusted corner;
2. rank current chords, cut the cheapest legal unit, re-rank;
3. write down each (R, Y) vertex as it is born;
4. stop at the target, or interpolate along the last unit's segment, or brick an infeasible resource;
5. read the mix off the last vertex (traceback), and refuse to substitute $f(\bar{R})$;
6. place rival portfolios next to that mix and say which are not on constructed $\mathcal{F}$;
7. Pareto-filter a supplied set on a second declared outcome;
8. apply ε, then a named risk mode inside the near set, or stop.

Greedy is a theorem on this class (Chapter 6's swap argument remains). It is a heuristic the moment a map is convex or a setup lump appears. It is the wrong language the moment OPEN carries from one "unit" into the next: that is a process, and the triangle of Anchor 1, not this polyline.

6.3 Why greedy is a theorem on this class, and where it stops

The portfolio problem is

$$\max_{R_1,\dots,R_n} \sum_{i=1}^{n} Y_i(R_i) \quad \text{subject to} \quad \sum_{i=1}^{n} R_i = \bar{R}, \quad R_i \in [a_i, b_i].$$

Assume each Y_i is concave and increasing, the maps are separable, and there is one resource. At an interior optimum the Karush–Kuhn–Tucker condition equalizes derivatives,

$$Y_i'(R_i) = \lambda$$

for every unit not sitting on a bound. Units at the lower bound have $Y_i' \geq \lambda$ (further cuts would be too expensive). Units at the upper bound have $Y_i' \leq \lambda$ (they have already been left rich because they were cheap to leave rich). Ranking by discrete sacrifice $s_i = \Delta Y_i / \Delta R_i$ is the grid analogue: always reduce the unit whose current chord is smallest.

A swap argument makes the same claim without multipliers. Suppose, at some constructed state, a feasible step cuts unit j while unit i still has room to be cut and $s_i < s_j$. Replacing j's cut with i's cut loses s_i visits per hour instead of s_j. Total resource is unchanged and total visits rise. The alleged construction was not efficient. Therefore every step of the algorithm, including the steps taken after a re-sort, must cut a currently cheapest legal unit. Concavity is what makes the argument global rather than myopic: as a unit is cut, its remaining slope

rises, so a unit that was correctly cheap stays correctly compared to units not yet cut, and the only thing that changes the ranking is this honest steepening. Re-ranking is not a defect of greedy. It is greedy tracking Y'.

The argument fails as soon as a map is not concave. Increasing returns, setup costs, and S-shaped productivity all break the invariant that a currently cheapest cut remains correctly compared to units not yet cut. A two-unit counterexample is enough. Let unit 1 produce nothing until five hours have been committed and then produce ten visits per additional hour, and let unit 2 produce one visit per hour from the first hour. Budget six. Add-resource greedy, ranking by the next small chord, gives every hour to unit 2, because unit 1's next chord is zero until the lump is paid: harvest 6. The optimum pays the lump, assigns five hours to unit 1 and one hour to unit 2, and harvests 11. Lumps beat drops. Greedy remove-resource on the same maps is then a heuristic and must be labeled as such. It is not a theorem about "how we allocate," and the software does not become a proof by running.

The selected point is the mix (3.0, 3.5, 5.5) and the harvest 17.547. The fitted curve, if any, is a summary of the polyline already built. Compression error is how well that curve tracks the vertices — not how well occupancy will hold a floor next quarter. Do not treat a number read off the curve at $R = 12$ as the selected state. That number does not name the sites.

6.4 Where the construction should refuse

A fleet of depots, each with a trusted maintenance map on [0.5, 7.5] hours, and a target budget of 0.1: the constructed minimum is again 1.5 if there are three depots, and the honest output is infeasible. The consultant who fits a Hill curve through last year's (R, Y) pairs and reads it at 0.01 has not produced a plan. They have produced a picture of a set that was never constructed, evaluated at a resource the trusted bounds forbid. Smoothness is not support.

Two raw states at the same R, with ΔY equal to 0.01% of scale, and maps that differ at $p < 0.01$: statistically different, and without a declared ε (Chapter 21) you do not have a preference. The status is `INSUFFICIENT_INFORMATION` for action, not "pick the higher Y."

A fitted cubic through $\mathcal{F}$ with tiny compression error, attached to an act that is "will visits fall below a contractual floor this quarter?": the curve can be an excellent summary of the polyline and a useless summary of the tail. Compression error can be excellent while decision quality is unrelated.

Target resource outside the trusted set; maps no longer trusted; coupling `INELIGIBLE` (Chapter 5); a request for "the globally optimal spend of the industry": refuse. Envelope-scoped optimality among the supplied units is the most the sentence may say, and even that sentence is about $\mathcal{F}$, not about $f(\bar{R})$.

If the maps are convex, or have a setup lump, the language of this chapter must change from theorem to heuristic before any number is issued. If OPEN carries from one "unit" into the next, the language must change from frontier to process (Chapter 7) before any compression is issued. In both cases the failure is the question, not the ranking of chords. The chords, on the three-site laboratory, did their job: they told you to cut A first, to re-sort onto B, to bring C in only after A and B had steepened, to land on (3.0, 3.5, 5.5) at $R = 12$, and to brick $R = 0.1$. That is already a complete allocation answer in a world that does not remember. It becomes a lie only when the world does.

6.5 Exercises

C6-E1. Mathematical. Under separability and diminishing returns, show that swapping a higher-sacrifice remove-resource step for a lower-sacrifice one cannot improve Y at fixed R. Where does the swap fail if a map is convex?

C6-E2. Failure diagnosis. A stakeholder wants $R = 0.1$ on the three-site laboratory. The constructed minimum is 1.5. What do you owe them?

C6-E3. Interpretation. The fitted curve at $R = 12$ reads 17.55. Why is that not the selected allocation?

C6-E4. Assumption criticism. The mix at $R = 12$ is (3.0, 3.5, 5.5). Why is "globally optimal hours of the industry" still forbidden?

C6-E5. Decision reasoning. Two raw states, same R, harvests differing by 0.01% of scale. The maps differ statistically. Without a declared business tolerance, can you pick one?

C6-E6. Method selection. Maps are convex in R (increasing returns). Is greedy a theorem?

C6-E7. Application. From the table, whose hours changed between $R = 16$ and $R = 12$? Why did C move at all?

CHAPTER 7

Hierarchy, and When the Parent's Totals Lie

Depends on. Chapters 5–6.

Chapter 6 built one efficient mix. Organizations decide at many grains. The naive move is to fold weeks into a month because the calendar has those folders. That is not mathematics.

A parent can treat each child as a one-dimensional lever — *how many hours this week gets* — only when two things are true. The child's efficient set is a sufficient statistic for the parent's act. And dropping the stock the child would have left behind does not change the preferred act by more than the organization already called noise. This chapter constructs the honest parent, then computes the lie when the second condition fails.

This is not Dantzig–Wolfe, not Benders, and not hierarchical *forecast* reconciliation (Dantzig and Wolfe 1960; Benders 1962; Wickramasuriya et al. 2019). Those cohere forecasts or decompose linear programs. Here the parent adds independent (R, Y) chips. Occupancy does not travel.

7.1 The Parent Must Receive Enough of the Child

Two operating regions, North and South, each offer three efficient staffing policies. Output is good, cost is constrained, and terminal backlog is inventory that next week inherits.

Unit	Policy	Output	Cost	Terminal backlog
North	N1	6	5	7
North	N2	9	7	4
North	N3	11	10	2
South	S1	5	4	8
South	S2	8	7	5

Unit	Policy	Output	Cost	Terminal backlog
South	S3	10	9	1

The parent needs total output at least 18, total cost at most 17, and at least two units of terminal backlog in each child.

7.1.1 First pass: totals only

If the children pass only output and cost upward, the cheapest feasible combination is N2 + S3:

Pair	Output	Cost	North backlog	South backlog
N2 + S3	19	16	4	**1**
N3 + S2	19	17	2	5

The totals-only parent selects N2 + S3. Its arithmetic is correct and its decision is unusable. South's backlog falls below two. The summary discarded a state that the next act can see.

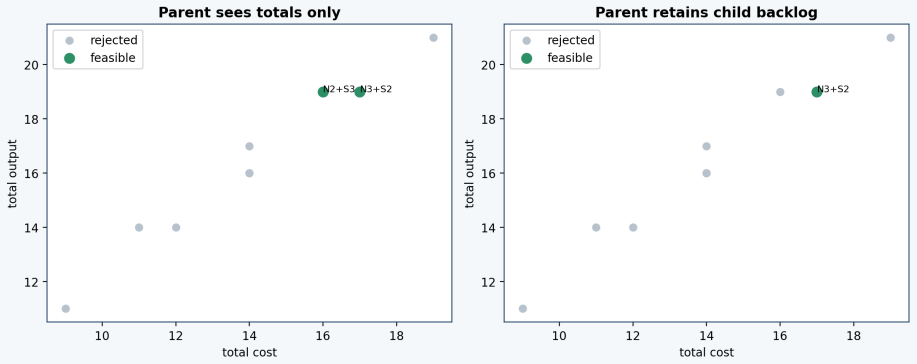


Figure 7.1: The totals-only parent accepts a cheap combination that fails a child inventory floor; the state-aware parent selects the legal alternative.

7.1.2 Second pass: sufficient child information

Pass the child backlog with output and cost. N2 + S3 is now rejected. N3 + S2 is the lowest-cost legal pair:

1. North executes N3: output 11, cost 10, ending backlog 2.

2. South executes S2: output 8, cost 7, ending backlog 5.
3. Parent totals are output 19 and cost 17.
4. Both children preserve the declared floor.

That is traceback. The parent did not merely pick a total of 19. It picked a pair of child policies that can actually be executed.

The summary test is operational: remove one child field and resolve. If the preferred parent act changes materially, the removed field was not safely compressible. A hierarchy is not made correct by having levels. It is made correct by passing sufficient decision state between them.

7.2 Does dropping the stock change the act?

Days-to-mature is the wrong number to route on. A long trickle can sit in the building for a year and never move the preferred roster. A short backlog can move it next Monday.

Let J with the stock kept be compared with J after that stock is emptied, over the controls you would actually consider. The gap is the decision impact of dropping the boundary. On a claims month the process engine, run twice, gives 41.98 with OPEN retained and 39.10 with OPEN emptied. The gap is 2.88. If the organization already called a harvest difference of 0.60 material, do not compress the month as independent weeks. If it called 5.00 noise, the same physics is *not* material at that grain. Same files, different decision. If nobody wrote a tolerance, you may still report the gap. You may not report that state "doesn't matter."

Two tails that both take 400 days to die can split on that test: a trickle whose presence never moves J by more than 0.01, and a backlog that moves it by 2.88. Routing on "400 days of memory" would send both to a process engine, or neither. Routing on the decision gap splits them.

The sibling check is composition error: the absolute difference between independent-chunk J and a process path that *does* age the stock, compared with the same tolerance. A small error does not make the parent a process global. A large one does not make the process schedule "the truth." It says the parent's question was wrong. The decision layer compares numbers the process engine supplied. It does not invent occupancy.

A state-conditioned child — an efficient set *given incoming stock*, then Bellman at the parent (Bellman 1957) — would close the pipe. That object is not a shipped method in this book. The auditor is the method: measure the gap, and if it is material, go to the process chapters (10–16) rather than folding the weeks.

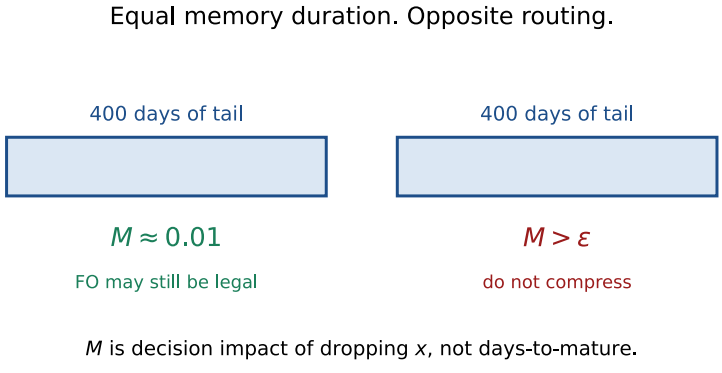


Figure 7.2: Two memory bars of equal length. One barely moves the act; one does. Equal duration, opposite routing.

7.3 Exercises

C7-E1. Application. From the 10-hour parent, whose hours in week 2 are zero? Why was that legal on static maps?

C7-E2. Mathematical. Two weeks, harvest $Y_1(u_1, x_0)$ and $Y_2(u_2, x_1(u_1))$. Why is (R_1, Y_1) not enough for week 2?

C7-E3. Failure diagnosis. Both shops post (80, 40) in week 1. Week 2 harvests 18 versus 8. Solver bug or question bug?

C7-E4. Interpretation. Gap 2.88, no business tolerance written. Can you say the stock is immaterial?

C7-E5. Method selection. A claims tail takes 400 days to die. A manager routes to a process engine "because memory is long." What number would you ask for instead?

C7-E6. Hostile. January maps look "broken" versus December. Correction, regime, or composition? If you cannot see OPEN, what is honest?

CHAPTER 8

Many Objectives, Many Draws, No Hidden Preference

Depends on. Chapters 5–7.

Chapter 6's $\mathcal{F}$ was one resource, one outcome, and no draws from a law of θ. Real acts care about cost *and* incidents *and* wait. Parameter uncertainty b is not entity noise d. People scalarize anyway: a weighted sum, a "risk-adjusted ROI," a tenth-percentile re-rank of the whole set. Each of those encodes a **hidden preference**. This chapter makes preference declared or unresolved. It does not make it disappear by being numerical.

A scalar Y cannot represent a tradeoff. A point map cannot represent $P(\theta \mid \text{data})$. Each addition is a new object, not a new package. Until a campaign is actually run, `uncertainty$availability = UNAVAILABLE` remains honest, and remains not $\sigma = 0$.

8.1 Pareto is not a total order

A point a **dominates** a point b when a is at least as good in every coordinate and strictly better in one. The **mathematical frontier of a supplied set** is the undominated subset of that set. It is not a global optimum of an unenumerated universe, and the construction that reports it sets `globally_optimal = FALSE` always. The claim is `MATHEMATICAL_FRONTIER_AMONG_SUPPLIED_STATES`. This $\mathcal{F}$ is still not the decision frontier of Chapters 15 and 23, and still not ScientificIntelligence's candidate generator through F_t. Three objects, three owners; Chapter 5's distinction is stricter here, not looser.

A point that is a little worse on harvest and much better on incidents is not "worse." It is incomparable until a human names how those coordinates trade. Scalarization — a weighted sum, an ε-constraint, a lexicographic order — is how incomparability becomes a total order (Miettinen 1999). That conversion is a preference, not a scientific discovery. Weighted and lexicographic policies are legal only with declared weights or order. Missing weights fail closed. That is not user-hostility. It is refusing to invent ethics.

Three points make the geometry unforgettable. Harvest is to be maximized, incidents minimized.

$$A = (40,\ 5), \qquad B = (39,\ 1), \qquad C = (38,\ 8).$$

C is dominated: A harvests more and has fewer incidents, and B does too. A and B are incomparable. A harvests one more unit; B has four fewer incidents. A weight vector $(1, 0)$ picks A. A weight vector $(0, 1)$ picks B. A hidden "risk-adjusted ROI" that happens to be harvest minus $0.1\times$ incidents scores A at $40-0.5 = 39.5$ and B at $39-0.1 = 38.9$, so it also picks A — but that 0.1 was a preference. Change the weight to 0.5 and the scores reverse: A at $40-2.5 = 37.5$, B at $39-0.5 = 38.5$. The mathematical frontier is $\{A, B\}$. The decision frontier is not yet defined. A weighted arrow drawn across the figure without numbers is an X, not a ranking. The weight is the ethics; the arithmetic is just the ethics with a decimal.

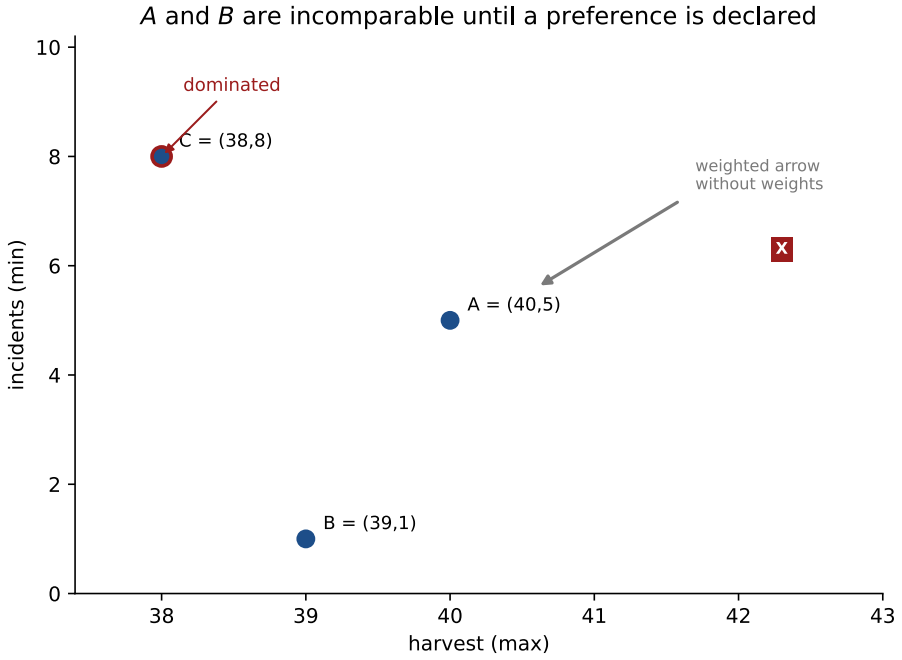


Figure 8.1: Points $A = (40, 5)$, $B = (39, 1)$, $C = (38, 8)$. C is dominated; A and B are incomparable until a preference is declared. A weighted arrow without weights is a refusal, not a ranking.

The same geometry in a depot costume: cost, on-time rate, safety incidents. A plan that spends a little more and almost never injures anyone is not "worse than" a cheaper, more dangerous plan. It is incomparable. Ranking the states

by the standard deviation of a single outcome and calling the minimum “best” is the same hidden preference in a statistical costume, and it is a refusal.

8.2 Same night forecasts, different roster

Vector outcomes add a second jump that scalar Pareto does not catch. Two night clinics, North and South. Each overflows, needing four extra nurses, on one night in five. A float nurse can cover one clinic on a given night.

night	North overflow	South overflow (separate nights)	South overflow (same nights)
1	yes	no	yes
2	no	no	no
3	no	yes	no
4	no	no	no
5	no	no	no

Each clinic still overflows on 1/5 of nights in both calendars. Under the separate-nights calendar, a shared float of four covers every overflow. Under the same-nights calendar, both clinics overflow on night 1; four extra nurses cover one clinic and leave the other uncovered. Covering every overflow now takes eight extra nurses.

Same one-clinic forecasts, different roster. In a long run the independent both-fail chance is $0.20 \times 0.20 = 0.04$; the same-nights chance is 0.20 (Nelsen 2006). The five-night table makes the dependence assumption countable.

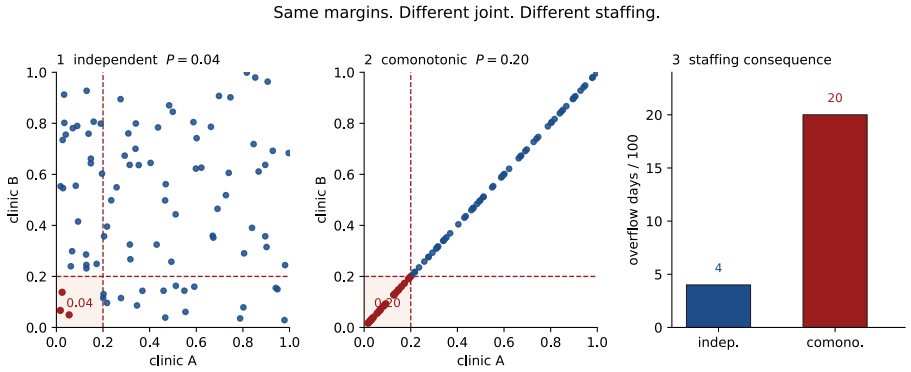


Figure 8.2: Same uniform margins. Left: independent, both-fail mass 0.04. Right: comonotonic, both-fail mass 0.20.

8.3 Do not average the maps first

Parameter draws b index epistemic worlds: draws from a posterior, or from any declared law of θ. They are not micro draws d of entity realization. The honest stochastic allocation is: draw b, then re-solve the same mathematical problem, then report the *distribution of decisions* and the *distribution of outcomes*. Averaging the maps first —

$$\bar{f} = E_b[f_b], \qquad \text{then optimize } \bar{f}$$

— is a different interchange. Jensen's inequality, and the failure of ∇ and E to commute for nonlinear J, are the reason; Chapter 20 returns to the dense form. Sample-average approximation is established stochastic programming (Shapiro et al. 2009; Birge and Louveaux 2011). This book does not claim two-stage recourse or a chance-constrained mixed-integer program as frontier primitives. Chance language is a **user constraint**, $P(J \geq \tau) \geq 1-\alpha$ (Charnes and Cooper 1959), not a default, and exact chance-constrained MIP is not implemented here.

Two equally likely worlds, two clinic sites, ten hours to allocate. In world θ_1, urgent care returns 1.2 visits per hour and routine returns 0.2. In world θ_2, the slopes reverse. The averaged map returns 0.7 on both sites. Greedy on the average is indifferent among splits and, used as a unique rule, splits 5–5. Realized harvest is

$$1.2 \times 5 + 0.2 \times 5 = 7$$

in either world, so expected harvest is 7.

Wipe the split. Re-optimize in each world. In θ_1 every hour goes to urgent care; in θ_2 every hour goes to routine. Harvest is 12 in each world. Expected harvest of the re-optimized policy is 12. The averaged map was a different problem. "Average then optimize" is not conservative. It is a distinct, and here strictly worse, objective: the policy that looked cautious, a five–five split in both worlds, left five visits a world on the table.

The same geometry with saturating maps is quieter and still one-sided. Let θ_1 give site A the map $Y = 20R/(0.5 + R)$ and site B the map $Y = R/(20 + R)$, and let θ_2 reverse them, still equally likely, still ten hours. The average of the two maps is symmetric, so greedy splits 5–5. In θ_1 that split harvests $20{\cdot}5/5.5{+}5/25 = 18.38$; the re-optimized all-A allocation harvests $20{\cdot}10/10.5 = 19.05$. The other world is the mirror. Expected J on the left-hand panel is 18.38; on the right-hand panel, 19.05. Concavity ate most of the gap — the steep site was already near its ceiling at five hours — but the inequality still points the same way. Re-optimizing per b puts almost everything on whichever site is steep in that world. Campaigns re-solve per b. They do not average maps first.

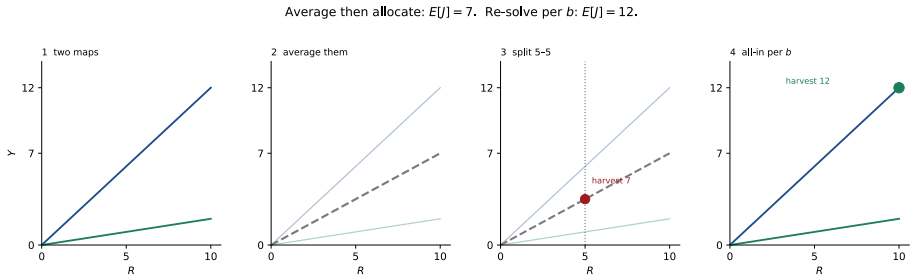


Figure 8.3: Left: average the maps, split 5–5, harvest 7. Right: re-optimize per world, all-in, harvest 12.

The interchange is the same one Chapter 20 writes as $\nabla E_\theta[J]$ versus $E_\theta \nabla J$. Here it is visible without gradients: the operator "take the mean of the maps" does not commute with the operator "assign the next hour to the steepest site." Averaging first produces a fictitious pair of mild clinics that do not exist in either world.

Entity noise d is a different axis. A campaign over b can move the ranking while micro draws at frozen θ do not, or the reverse. One ribbon covering both is illegal. Chapter 19 owns the axes; this chapter only refuses to mix them in the allocation.

8.4 Preference is not a tenth percentile of the whole set

A and B are still incomparable. A weight vector is one way to pick; it is a preference, not a discovery. Another way is to say how close is close enough, then apply a risk functional *inside* that near set. A tenth-percentile re-rank of the *whole* set can pick B even if the organization would have called one harvest unit noise. That skip is a preference smuggled past the tolerance. Chapter 21 computes a downside functional on a two-act table and shows the pick move. This chapter only needs the geometry: incomparable points, then a declared trade, or an honest stop if nobody will declare one.

One outcome, no parameter draws, separable maps: this chapter is unearned. What you refuse to add is a weighted sum "for robustness," a q10 re-rank "for safety," and a ribbon that pretends parameter worlds and entity worlds were one uncertainty (Chapter 19).

Statistical difference is not business materiality. Chapter 21 draws those as two different pairs of points. A p-value does not set the tolerance and does not substitute for a missing risk mode.

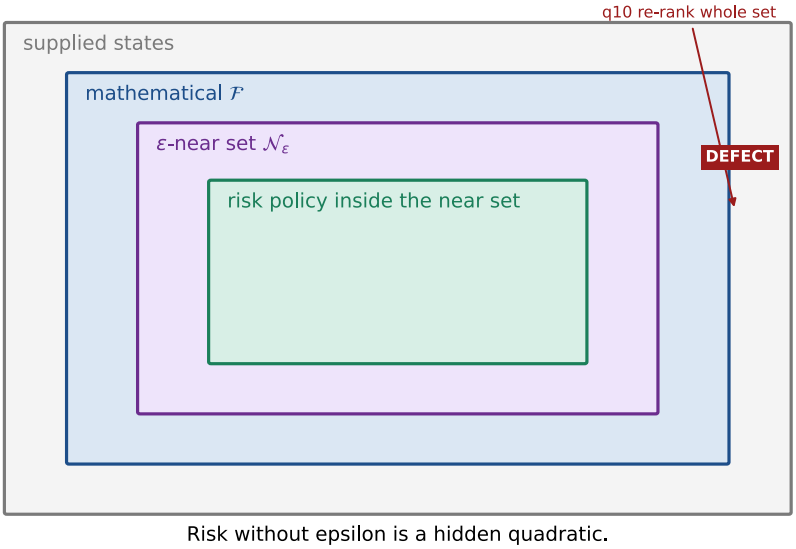


Figure 8.4: Nested sets from supplied states to the mathematical frontier to a near set to a risk slice. A side door that re-ranks the whole set is a defect.

8.5 Exercises

C8-E1. Mathematical. Prove that Pareto efficiency is not a total order. Exhibit three points in $\mathbb{R}^2$, and say which are dominated.

C8-E2. Failure diagnosis. Rank states by the standard deviation of a single outcome and call the minimum "best." Write the refusal.

C8-E3. Interpretation. No business tolerance is supplied. Is the near set a proper subset of the mathematical frontier, or is it the same set?

C8-E4. Assumption criticism. Why is `globally_optimal` hard-false on a mathematical frontier of supplied states?

C8-E5. Failure diagnosis. A weighted policy is requested and no weights are supplied. What status is expected?

C8-E6. Interpretation. Same control u, campaign draws b, micro draws d. Why is one ribbon illegal?

C8-E7. Decision reasoning. No one has said how harvest trades against incidents. Can you pick A or B?

C8-E8. Method selection. You have one outcome, no parameter draws, separable maps. Do you need this chapter?

CHAPTER 9

Forecasting What Persists

Chapter 7 ended with a pipe running beneath two weekly boxes. The boxes counted work completed. The pipe carried unfinished work into the next week. If that pipe matters to the decision, adding weekly forecasts is the wrong calculation.

This chapter begins where that failure becomes visible.

9.1 The visitor shop at the forecast origin

Ten people arrived in week 0. At the start of week 1, the origin used throughout this running case, their observed ledger is:

visitor	observed week-0 result	state at origin
`v01`, `v02`	completed	COMPLETE
`v03`	left without completing	EXIT
`v04`-`v08`	still in process	OPEN
`v09`, `v10`	still in process	OPEN

All ten are observed people. Seven remain exposed to future progression. That seven-person state, not the original ten and not the two completions, is what the next process step receives.

The shop also expects 8, 6, and 6 arrivals in weeks 1, 2, and 3. At the origin these are counts without identities. They join persistent state only when their cohort arrives. The future therefore has two sources:

future population = propagated known people+propagated future arrival mass.

The picture contains the central object of this chapter. The seven OPEN people after week 0 are a starting condition for week 1. Resetting the shop to ten people would resurrect `v01`, `v02`, and `v03`. Forecasting only new arrivals would discard `v04` through `v10`. Both calculations throw away the pipe.

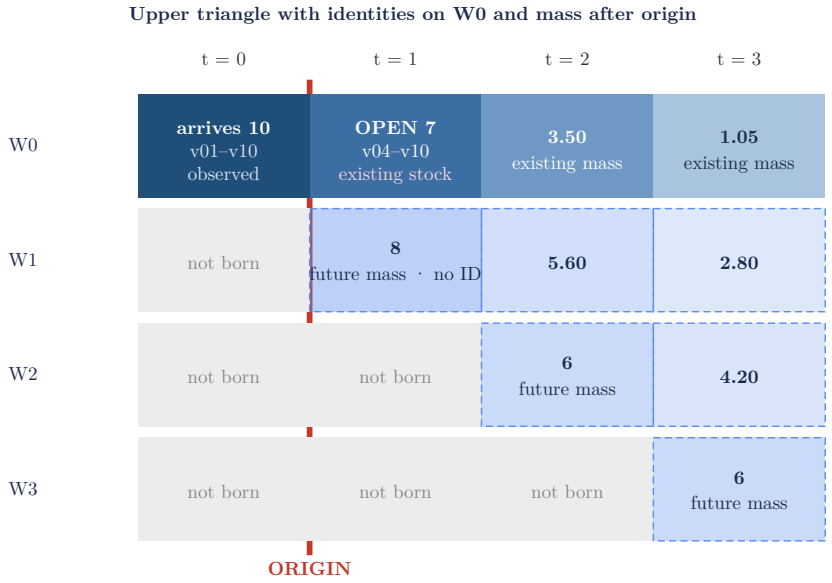


Figure 9.1: The observed W0 identities continue through the origin as existing stock; future cohorts enter as unlabeled mass.

9.2 From the ledger to a state equation

Let O_t be OPEN population at the start of week t, and let p_t be the fraction completing during that week. Then

$$D_t = p_t O_t, \qquad O_{t+1} = O_t - D_t + A_{t+1},$$

where A_{t+1} denotes new arrival mass before the next week begins.

The pair of equations says exactly what the table did:

1. flow D_t is generated from the stock currently at risk;
2. the unresolved stock survives;
3. new arrivals join the surviving stock.

This is a simple state-space recurrence (Durbin and Koopman 2012). More elaborate processes may have several states, competing exits, recurrent events, and clocks such as time since arrival (Andersen et al. 1993). The purpose is unchanged: retain enough of the past to forecast the relevant future.

9.3 When one state is not enough

Suppose two visitor cohorts each have 60 OPEN people.

- Cohort A arrived yesterday.
- Cohort B has remained in process for six weeks.

If resolution probability rises with waiting age, the same total $O_t = 60$ does not imply the same forecast. The state must distinguish age, perhaps as $O_{a,t}$, the number open at age a.

A useful state is not "every fact we can store." It is the smallest summary that makes histories with different futures distinguishable. If previous service attempts change the next hazard, attempt history belongs in the state. If it does not, retaining it adds complexity without predictive value.

This is the practical test:

> Could two histories have the same recorded state but materially different future distributions under the same planned actions?

If yes, the state is too small.

9.4 Conservation is the first diagnostic, not the last

For a closed three-state process OPEN → COMPLETE or EXIT, the ledger should balance:

$$N_{t+1} = N_t + \text{arrivals}_{t+1} - \text{exits}_{t+1}.$$

Competing exits must share the stock they deplete. If OPEN can move to COMPLETE or EXIT during one step, their probabilities cannot independently remove more than 100% of OPEN.

Conservation catches impossible accounting. It does not prove the model is adequate. A model can conserve every claim while using the wrong transition rates, omitting waiting age, or extrapolating to an unsupported claim type.

9.5 The ten-person hand calculation

In week 0, two of the ten observed visitors COMPLETE, one EXITs, and seven remain OPEN:

State	Start	Enter	Leave for COMPLETE	Leave for EXIT	End
observed W0	10	0	2	1	7 OPEN

The row conserves exactly: $2 + 1 + 7 = 10$. At the start of week 1, expected future arrival mass of 8 enters as a new cohort, so calendar OPEN becomes $7 + 8 = 15$. The eight arrivals do not retroactively acquire observed identities.

More importantly, week 1 begins with two distinguishable pieces of state: seven known visitors continuing from W0 and eight units of newly arrived cohort mass. They have different ages and therefore can have different futures. Chapter 16 will propagate every cohort cell by hand.

9.6 When to stop early

State modeling adds no value when the act depends only on a future quantity whose distribution does not depend materially on carried inventory or delayed effects. Tomorrow's independently generated electricity load may need a predictive distribution but no persistent operational stock.

The right question is not "Can we build a state model?" It is "Would retaining state change the forecast or decision enough to matter?"

9.7 Optional public-API lab

The analytical lesson does not depend on a package. For a reproducible process lab, the exported ScientificIntelligence sequence is:

1. declare states and transitions with ‘sci_dynamic_process_spec()‘;
2. declare an age-structured engine with ‘sci_age_structured_engine_spec()‘;
3. fit the supplied transition evidence with ‘sci_fit_age_structured_process()‘;
4. forecast from the fitted state with ‘sci_forecast_age_structured_process()‘.

Save and reload the fitted object before forecasting again to verify that the future can be reproduced without refitting. The code is a lab implementation of the ledger above, not the explanation of why the ledger is needed.

9.8 Failure cases

- **Resetting state at every reporting period.** This resurrects completed or exited visitors and reproduces Chapter 7's missing-pipe error.

- **Keeping only the terminal KPI.** Equal completions can conceal very different remaining visitor populations.
- **Treating conservation as validation.** Balanced mass can still follow wrong dynamics.
- **Adding state indiscriminately.** Unused history can make estimation harder without changing the decision.

9.9 Exercises

C9-E1. At the origin, seven observed visitors remain OPEN and eight new arrivals enter. Compute calendar OPEN. Which part has known identities at the origin? **C9-E2.** OPEN contains 40 people. During one week, 30% complete and 10% exit. Write the mass balance and compute the remaining OPEN stock. **C9-E3.** Two queues each contain 50 jobs, but one has waited one day and the other ten. State the condition under which queue size alone is an adequate state. **C9-E4.** Give a forecasting problem where persistent state is unnecessary, and explain why adding it would not change the act. **C9-E5.** Construct a mass-conserving process model that is nevertheless predictively wrong. Identify the missing or misspecified mechanism.

Transition to Chapter 10. We can now carry unfinished work honestly. The next question is how an action enters that process without confusing a changed prediction with an intervention effect.

CHAPTER 10

When an Action Enters the Forecast

A hospital's completion forecast rises after a new laboratory result arrives. The same forecast also rises after the hospital adds an evening shift.

The two lines may look similar. They answer different questions.

- New information changes what we believe will happen.
- An intervention changes the process we are asking about.

Confusing them turns a predictive revision into an effect claim.

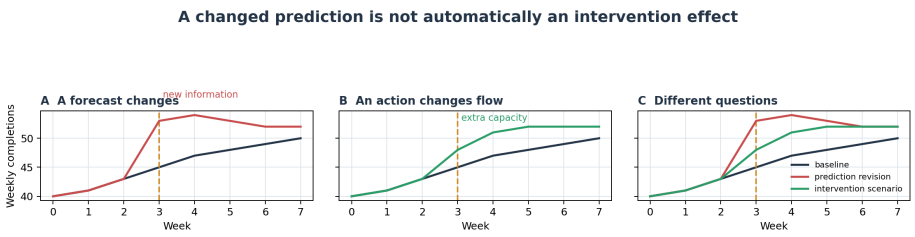


Figure 10.1: Three panels contrast a prediction revision, an intervention scenario, and the two trajectories together.

10.1 One staffing pulse

Return to the claims desk. Baseline completions over four weeks are

$$(20,\ 16,\ 12.8,\ 10.2).$$

Management can add one evening shift in Week 1. The shift can be described in three layers:

1. **Action:** eight additional reviewer-hours are scheduled.
2. **Delivery:** all eight hours are actually worked.

3. **Mechanism:** those hours increase the Week 1 resolution fraction from 0.20 to 0.25.

With 100 open files, Week 1 completions become 25 rather than 20. But the later weeks also change because only 75 files remain:

Week	Baseline completions	Shift scenario completions	Difference
1	20.0	25.0	+5.0
2	16.0	15.0	-1.0
3	12.8	12.0	-0.8
4	10.2	9.6	-0.6
Total	**59.0**	**61.6**	**+2.6**

The action produces five extra immediate completions but only 2.6 extra completions over four weeks. Some of Week 1's gain is work pulled forward from later weeks.

That interpretation is impossible if we compare only Week 1 totals.

10.2 Timing is part of the intervention

An action may keep affecting the process after the scheduled hour. Advertising can create carryover; maintenance can reduce failures for months; a treatment can have delayed effects. A distributed-lag representation writes delivered exposure u_t as

$$m_t = \sum_{k=0}^{K} w_k u_{t-k},$$

where w_k describes timing (Almon 1965).

The weights describe a response operator. They do not identify a causal effect by themselves (Pearl 2009). If the weights are prior assumptions, every conclusion remains conditional on those assumptions.

For a geometric carryover with decay 0.65 and a ten-unit pulse, the active exposure is

Week after pulse	Active exposure
0	10.00
1	6.50
2	4.23
3	2.75

A reader can now see what “decay 0.65” means. It is not a mysterious tuning parameter: 65% of the previous active exposure survives one more period.

10.3 Prediction, scenario, and intervention

Use three different sentences.

Prediction: Given information available now, completions next week are expected to be 45.

Scenario: If eight extra hours are delivered and the declared response model is correct, completions are projected to be 49.

Intervention effect: Delivering eight extra hours causes four additional completions.

The third sentence requires identification that the first two do not. A scenario may be useful without earning causal language. Calling it a scenario is not a weakness; it is a precise statement of what was computed.

10.4 Support and extrapolation

Suppose historical shifts ranged from zero to ten extra hours. A scenario at eight lies within observed support. A scenario at forty asks the response model to extrapolate far beyond the evidence.

A forecast engine can still calculate a number at forty. The analytical result should make the extrapolation visible, because numerical availability is not evidential support.

For irreversible or expensive actions, unsupported extrapolation should block a recommendation even if the scenario itself can be displayed.

10.5 Adaptation is not an extra intervention

A state-space forecast may assimilate a surprise into its latent level. If the same residual is also added as an external correction, one observation has been counted twice.

Ask what each mechanism says:

- **state update:** the underlying level moved;
- **temporary correction:** a one-off miss should be offset;
- **new response slope:** the action-outcome relationship changed;
- **regime challenger:** a different dynamic law may now be better.

These are competing explanations, not ingredients to blend automatically.

10.6 Optional public-API lab

The temporal-response lab uses exported calls:

- ‘sci_temporal_response_spec()‘ to declare the carryover family and normalization;
- ‘sci_fit_temporal_response()‘ to store the declared or estimated timing law;
- ‘sci_execute_temporal_response()‘ to apply it to a schedule;
- ‘sci_temporal_response_jacobian()‘ to inspect how each scheduled action contributes to later active exposure.

The important choice is visible in the resulting impulse table before any process forecast is run.

10.7 Failure cases

- A prediction revision is described as a treatment effect.
- Scheduled action is assumed to equal delivered exposure.
- The same forecast error is assimilated twice.
- A response learned on small actions is silently extrapolated to a large action.
- Two mechanisms with the same immediate outcome are treated as equivalent even though they leave different state behind.

10.8 Exercises

C10-E1. For $p^* = \text{logit}^{-1}(\text{logit}(0.30) + r)$, compute p^* and dp^*/dr at $r = 0.40$. **C10-E2.** Starting with 100 open files, compare one week with 10 new arrivals against one week with a five-point increase in the resolution probability. Construct parameters that give equal immediate completions, then compare ending stock. **C10-E3.** Compute four periods of active exposure for a pulse of 12 and geometric decay 0.5. Explain what the calculation does and does not identify. **C10-E4.** Describe a concrete case in which a latent-state update and an external error correction would double count the same observation. **C10-E5.** Rewrite the sentence "the campaign caused 200 sales" as an honest scenario claim when the response is prior-only.

Transition to Chapter 11. An action and its timing law are now explicit. We can compare complete baseline and candidate paths through the same fitted process.

CHAPTER 11

Following a Scenario Through Time

A manager has one block of overtime. It can be used early, when the pipeline is large, or late, near the reporting deadline. The budget is identical. The consequences are not.

This chapter follows both schedules through the same process before introducing summary windows or software contracts.

11.1 The early-versus-late calculation

Let open inventory evolve as

$$x_{t+1} = 0.9x_t - u_t,$$

where u_t is extra work completed during period t. Begin with $x_0 = 100$. Compare two schedules:

- **Early:** $u = (10, 0, 0, 0)$
- **Late:** $u = (0, 0, 0, 10)$

Period	Early inventory	Late inventory
0	100.00	100.00
1	80.00	90.00
2	72.00	81.00
3	64.80	72.90
4	58.32	55.61

Both spend ten units. Late action leaves less inventory at the end of period 4, but early action reduces inventory sooner. Which is better depends on what the decision values: near-term relief, terminal backlog, cumulative waiting, or some combination.

The visual uses the same recurrence and schedules as the table. Timing changes the state inherited by every later period.

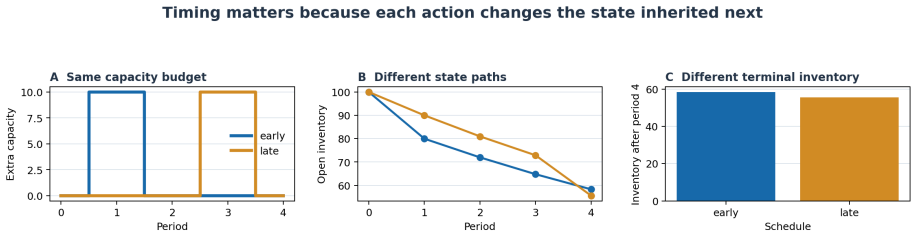


Figure 11.1: The same capacity budget, placed early or late, creates different state paths and terminal inventory.

11.2 A scenario is a paired replay

A defensible comparison holds fixed:

- the fitted process and origin state;
- future arrivals and context;
- the consequence horizon;
- every assumption except the candidate action path.

Then it replays baseline and candidate schedules separately.

This pairing matters. If the candidate is evaluated with a different arrival forecast, a different origin, or a refitted process, the difference cannot be attributed to the schedule alone.

The recurrence

$$x_{t+1} = F_t(x_t, u_t; \theta)$$

is a conditional state-space projection (Durbin and Koopman 2012). Replaying a controlled input path does not by itself make the comparison causal. When controls are selected sequentially rather than merely replayed, this same state recursion becomes the substrate of dynamic programming (Bellman 1957).

11.3 Three horizons, three questions

The action window and consequence window should not be collapsed.

Window	Question
Decision	When may the schedule change?

Window	Question
Objective	Which periods count directly in the chosen score?
Consequence	How long must the process run to reveal material effects?

Suppose overtime can change only in Weeks 1–4, management scores completions in Weeks 1–4, but unresolved files influence Weeks 5–8. The consequence simulation must run through Week 8 even though the optimizer cannot act there.

Stopping at Week 4 would value harvesting and ignore the inventory left behind.

11.4 Creation versus harvesting

Two schedules can have the same near-term output and opposite long-term consequences.

- An acquisition action creates future pipeline.
- A conversion action harvests existing pipeline.

Moving resources from creation to harvesting may improve the reporting-period KPI while leaving less work available next quarter. A scenario report should retain both output flows and terminal state.

For an additive discounted objective, split the difference at objective horizon H:

$$\Delta V = \sum_{t \le H} d_t \Delta y_t + \sum_{H < t \le T} d_t \Delta y_t + d_T \Delta V_T(x_T).$$

The first term is the scored window. The second and third prevent the score from hiding post-window consequences.

11.5 Summary evidence versus full trajectories

A summary is sufficient when it retains every quantity needed for the declared decision:

- objective totals by horizon;
- constraint margins;

- selected terminal-state inventories;
- support and failure diagnostics.

Full trajectories are needed when the reader must inspect timing, diagnose a constraint failure, validate a new functional, or explain why two schedules differ.

Summary mode is an efficient representation of a known question. It is not a different model and should not be treated as permission to discard unknown future questions.

11.6 Optional public-API lab

Use ‘sci_process_summary_spec()‘ to declare windows and retained state summaries, then call ‘sci_forecast_age_structured_process()‘ twice from the same fitted object: once with baseline forcing and once with candidate forcing. A no-refit replay should reproduce both paths after serialization.

Inspect the trajectories before relying only on summary fields.

11.7 Failure cases

- Candidate execution mutates the baseline state.
- The objective window is mistaken for the consequence horizon.
- A prior-only response is called causal because propagation is exact.
- Equal budget is assumed to imply equal future state.
- A terminal KPI is retained while the pipeline that generates later outcomes is discarded.

11.8 Exercises

C11-E1. Using the recurrence above, verify all five inventory values for the early and late schedules. **C11-E2.** Compute cumulative inventory $\sum_{t=1}^{4} x_t$ for both schedules. Which schedule reduces waiting exposure more? **C11-E3.** Give a decision for which the objective window can end at Week 4 but the consequence window must extend to Week 12. **C11-E4.** Explain why exact replay of a prior-only response does not identify a causal effect. **C11-E5.** List the minimum summary quantities needed to compare two schedules subject to a terminal-inventory floor.

Transition to Chapter 12. Paired replay can compare a few chosen schedules. Searching many schedules requires knowing how one small action changes the whole future path.

CHAPTER 12

How One Action Changes the Future

A hospital can add one nurse-hour today. The immediate effect is easy to count. The difficult question is how that hour changes tomorrow's queue, and therefore every later queue.

12.1 A four-day calculation

Let unfinished cases evolve as

$$x_{t+1} = 0.8x_t + 20 - 2u_t,$$

where u_t is extra nurse-hours and each hour clears two cases. Start at $x_0 = 50$. With no extra hours, the path is

$$50,\ 60,\ 68,\ 74.4,\ 79.52.$$

Add one hour on Day 0 and nothing later. The path becomes

$$50,\ 58,\ 66.4,\ 73.12,\ 78.50.$$

The first difference is -2 cases. Persistence carries 80% of that difference forward: $-1.6, -1.28, -1.024$. A local action has become a trajectory.

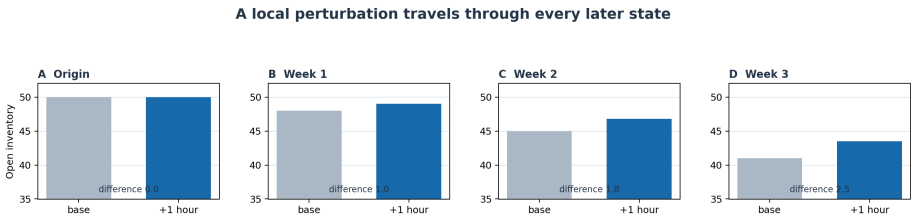


Figure 12.1: Four panels show a local perturbation entering a state, propagating forward, and producing a horizon-wide response.

12.2 From differences to derivatives

For a small perturbation h, a finite-difference estimate is

$$\frac{x_t(u_0 + h) - x_t(u_0)}{h}.$$

Here the system is linear, so $h = 1$ already gives the exact derivatives

Time	$\partial x_t / \partial u_0$
1	-2.000
2	-1.600
3	-1.280
4	-1.024

For a general recurrence $x_{t+1} = f_t(x_t, u_t)$, define $S_t = \partial x_t / \partial u_0$. The chain rule gives

$$S_{t+1} = \frac{\partial f_t}{\partial x_t} S_t + \frac{\partial f_t}{\partial u_0}.$$

The second term is nonzero only when the chosen control enters that transition. The first term carries its consequence onward.

12.3 Local is not propagated

The local derivative $\partial f_0 / \partial u_0 = -2$ answers how Day 1 changes. It does not answer how a four-day objective changes. If

$$J = \sum_{t=1}^{4} x_t,$$

then

$$\frac{\partial J}{\partial u_0} = -2 - 1.6 - 1.28 - 1.024 = -5.904.$$

Ignoring propagation understates the effect by almost a factor of three.

12.4 One Derivative Becomes Two at a Boundary

Let cleared backlog after a control be

$$f(u) = \max(0, 2 - 2u).$$

At (u=0.5), the process is smooth. At (u=1), the queue has just cleared and the `max` creates a kink.

Location	(h)	Forward	Backward	Centered
Smooth, (u=0.5)	1.00	-1.00	-2.00	-1.50
Smooth, (u=0.5)	0.10	-2.00	-2.00	-2.00
Smooth, (u=0.5)	0.01	-2.00	-2.00	-2.00
Kink, (u=1.0)	1.00	0.00	-2.00	-1.00
Kink, (u=1.0)	0.10	0.00	-2.00	-1.00
Kink, (u=1.0)	0.01	0.00	-2.00	-1.00

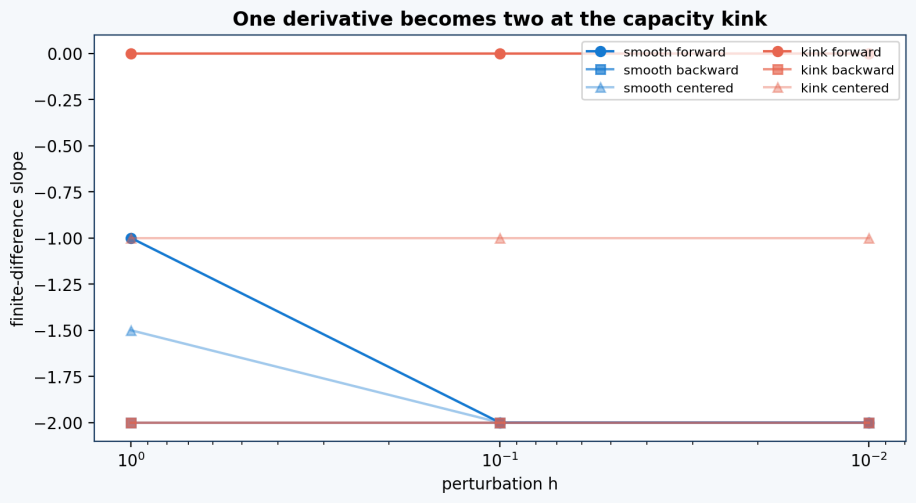


Figure 12.2: Forward and backward differences agree in the smooth region and remain different at the capacity kink.

At the kink, the centered value (-1) is not a derivative hiding between two noisy estimates. It averages two different directional statements. More control cannot clear a negative queue, so the right derivative is 0. Less control rebuilds backlog at two cases per unit, so the left derivative is -2. Report the direction the contemplated act will move.

12.5 When the calculation can mislead

At a capacity boundary, the derivative may be one-sided or undefined. For example, replacing the transition by $\max(0, 0.8x_t + 20 - 2u_t)$ creates a kink when the queue clears. Compare forward and backward perturbations, report the direction, and do not call a derivative stable when the two sides disagree.

12.6 Optional software lab

The public ScientificIntelligence sensitivity interface can propagate a control perturbation through a fitted process. The conceptual contract is the same as the table above: a control, a state path, a horizon, and a declared objective. Inspect the returned path, not only its terminal derivative.

12.7 Exercises

C12-E1. Recompute the perturbed queue path when persistence is 0.6. What is $\partial x_4/\partial u_0$?

C12-E2. For persistence 0.8, compute $\partial J/\partial u_0$ when $J = x_2 + x_3 + x_4$.

C12-E3. Compare finite differences using $h = 1$ and $h = 0.01$ for the linear recurrence. Explain why they agree.

C12-E4. Give an operational example in which a one-sided derivative is more honest than a centered derivative.

C12-E5. A colleague reports only the Day 1 effect of an action. State what additional evidence is needed before using it for a four-day decision.

CHAPTER 13

Many Controls, One Outcome

Forward sensitivity is natural when there is one action. It becomes expensive when a planner controls ten channels over 52 weeks: 520 control directions must be propagated to differentiate one objective.

The adjoint reverses that arithmetic. It asks how one final objective sends value backward through every state and control.

13.1 The Complete Adjoint Tape

The hospital tracks waiting inventory (q_t) and staff fatigue (f_t). Controls are regular staffing (r_t) and overtime (o_t):

$$\begin{bmatrix} q_{t+1} \\ f_{t+1} \end{bmatrix} = \begin{bmatrix} 0.75 & 0 \\ 0 & 0.60 \end{bmatrix} \begin{bmatrix} q_t \\ f_t \end{bmatrix} + \begin{bmatrix} -1.5 & -2.5 \\ 0 & 0.8 \end{bmatrix} \begin{bmatrix} r_t \\ o_t \end{bmatrix} + \begin{bmatrix} 12 \\ 0 \end{bmatrix}.$$

Start at $(40, 0)$ and apply controls $(1, 1), (1, 0.5), (1, 0)$. The objective sums quadratic state and control costs and adds terminal fatigue value.

13.1.1 1. Forward state pass

Time	Waiting	Fatigue	Regular	Overtime	Stage cost
0	40.000	0.000	1.0	1.0	
1	38.000	0.800	1.0	0.5	722.628
2	37.750	0.880	1.0	0.0	712.886
3	38.812	0.528			753.361

The forward tape stores states and the transition derivatives. Nothing runs backward yet.

13.1.2 2. Objective contributions

Each stage contributes

$$\ell_t = \tfrac{1}{2} x_{t+1}^{\mathsf{T}} Q x_{t+1} + \tfrac{1}{2} u_t^{\mathsf{T}} R u_t,$$

with $Q = \mathrm{diag}(1, 0.4)$, $R = \mathrm{diag}(0.2, 0.8)$, and terminal contribution $2f_3$. Total objective is 2189.930965.

13.1.3 3. Backward value pass

Initialize at the terminal state:

$$\lambda_3 = Q x_3 + (0, 2)^{\mathsf{T}} = (38.8125, 2.2112)^{\mathsf{T}}.$$

Then accumulate

$$\lambda_t = Q x_t + A^{\mathsf{T}} \lambda_{t+1}.$$

State time	(lambda_q)	(lambda_f)
3	38.812500	2.211200
2	66.859375	1.678720
1	88.144531	1.327232

The first-period waiting state has the largest backward value because it enters every later waiting state.

13.1.4 4. Control gradients appear

For each period,

$$\nabla_{u_t} J = R u_t + B^{\mathsf{T}} \lambda_{t+1}.$$

Control period	Regular gradient	Overtime gradient
0	-132.016797	-218.499543
1	-100.089062	-165.405462
2	-58.018750	-95.262290

Negative means a small increase reduces this cost. Early controls have larger magnitude because their state consequences persist.

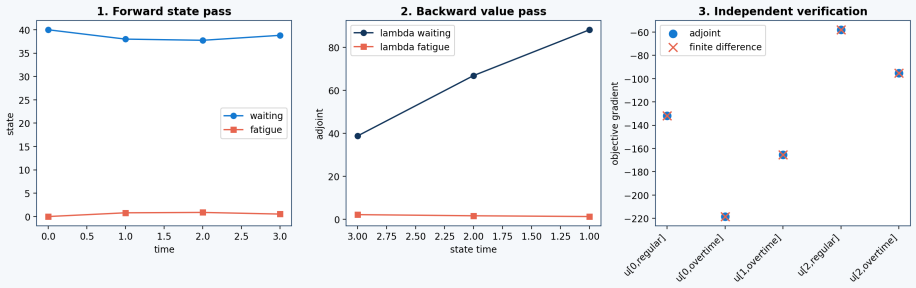


Figure 13.1: The same hospital case runs forward once, accumulates value backward once, and verifies selected gradients independently.

13.1.5 5. Finite-difference verification

Component	Adjoint	Centered difference	Absolute error
r_0	-132.01679688	-132.01679690	2.20×10^{-8}
o_0	-218.49954252	-218.49954253	4.57×10^{-9}
o_1	-165.40546150	-165.40546151	7.36×10^{-9}
r_2	-58.01875000	-58.01875002	1.91×10^{-8}

One reverse pass produced six gradients. Six separate forward perturbations would recover the same vector at roughly six times the directional work. Verification is not optional: the largest observed discrepancy is 2.21×10^{-8}.

13.2 Why reverse mode scales

Suppose there are p controls and one objective. Forward mode carries roughly p sensitivity directions. Reverse mode carries one adjoint vector and then reads all p control gradients. When objectives are few and controls are many, this changes the dominant cost from scaling with controls to scaling with objectives (Griewank and Walther 2008; Giles and Pierce 2000).

Question	Natural mode
One control, many outputs	Forward sensitivity

Question	Natural mode
Many controls, one objective	Adjoint/reverse
Few controls and few outputs	Either; choose the clearer implementation

13.3 Verification is part of the method

An adjoint can be fast and wrong. Before trusting it:

1. compare selected components with centered finite differences;
2. test several perturbation sizes;
3. inspect boundaries and clipping separately;
4. verify the objective and transition use the same time indexing.

Checkpointing trades recomputation for memory on long horizons (Griewank and Walther 2000). It is an engineering choice after correctness, not a substitute for the derivation.

13.4 Optional software lab

Use the exported ScientificIntelligence adjoint interface on the same process and objective used for forward sensitivity. Compare selected gradient components with finite differences before increasing the control dimension.

13.5 Exercises

C13-E1. Derive $\partial J/\partial u_0$ and $\partial J/\partial u_1$ for the two-step example.

C13-E2. Add a terminal cost $\Phi(x_2) = 2x_2$. Recompute λ_2, λ_1, and both control gradients.

C13-E3. For 500 controls and two objectives, explain why reverse mode is attractive. State one reason forward mode may still be useful.

C13-E4. Design a finite-difference check for three randomly selected control components.

C13-E5. Explain why a kink created by a capacity constraint can make an adjoint/finite-difference comparison direction-dependent.

CHAPTER 14

Where and When to Act

An operations manager has eight overtime blocks. Spending all eight today clears the largest visible queue immediately. Splitting them across the next two periods clears less now but prevents the queue from rebuilding.

That is the difference between choosing an action and choosing a schedule.

14.1 From a Schedule Guess to a Constrained Dynamic Policy

Continue the hospital process from Chapter 13. Regular staffing is fixed at one unit per period. Four overtime blocks must be placed over four periods. The objective penalizes waiting, fatigue, and concentrated overtime.

14.1.1 1. Evaluate every small feasible schedule

Because the teaching problem has only 35 integer schedules, enumerate them. This is ground truth, not the production algorithm. The obvious greedy policy places all overtime immediately: `(4,0,0,0)`. Its cost is 164.663488.

The unconstrained minimum is `(2,1,1,0)` with cost 159.756713. It spreads work because concentrated overtime is costly and fatigue persists. Its terminal fatigue is 1.113600.

14.1.2 2. Add the operational constraint

Require terminal fatigue at most 1.0. The unconstrained schedule is now illegal. The best legal schedule is `(2,2,0,0)`:

Policy	Schedule	Cost	Final wait	Final fatigue	Legal?
Greedy	4-0-0-0	164.663488	37.148438	0.691200	yes
Free optimum	2-1-1-0	**159.756713**	35.976562	**1.113600**	no

Policy	Schedule	Cost	Final wait	Final fatigue	Legal?
Fatigue-safe	2-2-0-0	160.142462	36.445312	0.921600	yes

The constrained policy gives up 0.385749 objective units to restore a legal terminal state. It still beats the obvious greedy policy by 4.521026.

14.1.3 3. Watch candidate updates

Update	Schedule	Cost	Terminal fatigue	Status
0	1-1-1-1	161.865450	1.740800	infeasible
1	3-1-0-0	160.802975	0.806400	feasible
2	2-2-0-0	**160.142462**	0.921600	feasible best

The Chapter 13 gradient says early overtime has the largest effect on future waiting. The candidate update therefore moves blocks earlier, but the fatigue state prevents all four from collapsing into one period. The final placement is an interaction between propagated benefit and an active state constraint.

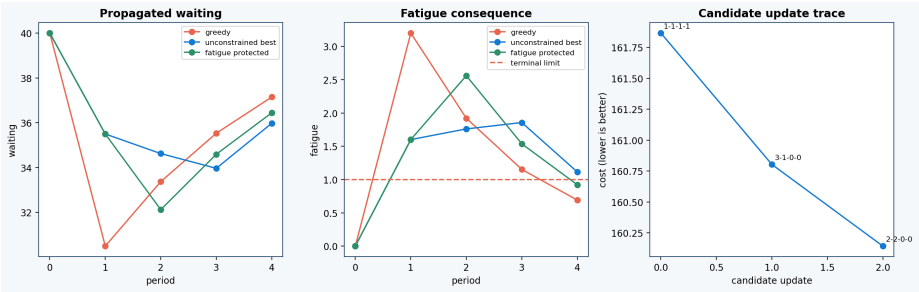


Figure 14.1: Waiting and fatigue propagate under three schedules while the candidate trace moves to the best legal policy.

This is why total budget is not a policy. Where and when the blocks enter changes both the scored waiting path and the state inherited after the objective window.

14.2 The optimization problem

Let u_t be the action at time t, x_t the persistent state, and J the chosen decision objective:

$$\max_{u_0,\dots,u_{T-1}} J(x_{0:T}, u_{0:T-1})$$

subject to

$$x_{t+1} = f_t(x_t, u_t), \qquad u_t \in \mathcal{U}_t, \qquad \sum_t c_t(u_t) \le B.$$

The transition is not bookkeeping. It is what makes timing consequential.

14.3 Decision, objective, and consequence windows

These windows need not coincide.

- The **decision window** contains times at which actions may change.
- The **objective window** contains outcomes the optimizer rewards.
- The **consequence window** extends far enough to expose deferred damage or benefit.

Optimizing a 30-day objective while simulating only 30 days can hide a depleted pipeline on Day 31. A defensible result preserves the whole baseline-versus-policy trajectory, including terminal state.

14.4 Constraints make the recommendation operational

Budgets are only one constraint. Real schedules may include period caps, minimum service levels, ramp limits, inventory floors, and fixed commitments. A mathematically superior infeasible schedule is not a candidate decision.

Under uncertainty, the same schedule should be evaluated across coherent worlds. Expected value, downside risk, robustness, and chance constraints ask different questions; none should be smuggled into an unexplained weighted score.

14.5 Optional software lab

The exported ScientificIntelligence dynamic optimizer accepts a time-indexed control schedule and returns full state and objective trajectories. Inspect the schedule and the consequence path together. A terminal objective alone cannot explain the decision.

14.6 Exercises

C14-E1. Verify that both schedules use eight overtime blocks. Why is equal total effort insufficient for equivalence?

C14-E2. If the objective is `near + long`, which schedule wins? If it is `2 * near + long`, which wins?

C14-E3. Add a constraint that at most five blocks may be used in any period. Which displayed schedule becomes infeasible?

C14-E4. Define decision, objective, and consequence windows for a hospital staffing decision made weekly but evaluated quarterly.

C14-E5. State the minimum trajectory evidence needed to claim that an early gain does not create post-window harm.

CHAPTER 15

Choosing From a Frontier

Optimization often returns several defensible policies, not one unquestionable winner. The analyst's job is to remove clearly inferior choices, identify differences too small to matter, and expose the preferences that remain.

15.1 Five hospital policies

Higher service value and reserve are better; lower risk is better.

Policy	Description	Service value	Reserve	Risk
A	Harvest	108	30	0.31
B	Balanced	101	42	0.14
C	Build capacity	92	58	0.08
D	Weak compromise	96	29	0.28
E	Stable balance	100	43	0.10

Policy D is dominated by E: E improves value, reserve, and risk. D should not consume decision attention.

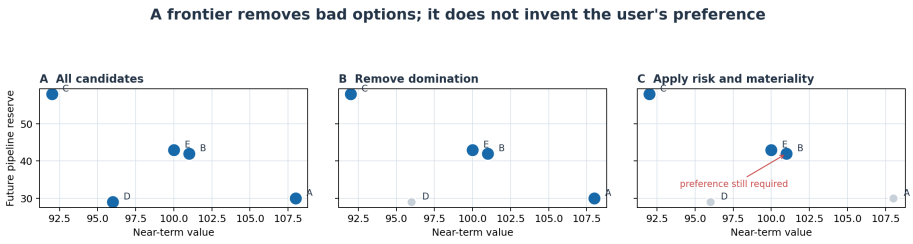


Figure 15.1: Candidate policies move from the raw pool to the Pareto frontier, then through material equivalence and an explicit risk preference.

15.2 Pareto geometry

A policy is Pareto dominated when another policy is at least as good on every objective and strictly better on at least one. Removing D leaves A, B, C, and E on the mathematical frontier.

That frontier is not yet a recommendation:

- A buys the most service value by accepting low reserve and high risk.
- C protects the most reserve but gives up value.
- B and E occupy the middle.

15.3 Material equivalence

Tiny numerical differences need not create false choices. Suppose differences smaller than two service-value points and two reserve points are operationally immaterial. B and E are then materially equivalent on those two measures: their differences are 1 and 1.

Risk still separates them: E has risk 0.10 rather than 0.14. Material equivalence is therefore objective-specific; it does not erase a consequential difference on an omitted dimension.

15.4 Preference and authority

If policy requires risk ≤ 0.15, A is inadmissible. B, C, and E remain. Choosing E over C still requires a preference between current service value and future reserve. Mathematics can present that tradeoff accurately. It cannot manufacture the organization's preference.

A useful decision record therefore preserves:

1. the candidate pool;
2. dominance exclusions;
3. material-equivalence tolerances;
4. risk or feasibility constraints;
5. the preference that selected the final policy.

15.5 When frontier claims fail

A frontier is only as credible as its evaluations. Common-reference simulation matters: every policy must face the same uncertainty worlds. Otherwise

sampling noise can masquerade as policy superiority. Near-ties also require uncertainty or stability analysis before they are ranked.

15.6 Exercises

C15-E1. Identify the objective-by-objective evidence that D is dominated by E.

C15-E2. Recompute material-equivalence groups twice: first with tolerances `(service value = 0, reserve = 0)`, then with `(service value = 2, reserve = 2)`. Ignore risk for this grouping step and state why risk must be reconsidered afterward.

C15-E3. Apply the risk constraint risk ≤ 0.12. Which policies remain admissible?

C15-E4. Write two defensible preference statements, one selecting C and one selecting E.

C15-E5. Explain why independently simulated uncertainty worlds can corrupt a close frontier comparison.

C15-E6. After dominance, materiality, and the risk constraint are applied, state precisely what preference remains unresolved between C and E.

CHAPTER 16

From Forecast to Consequence: Expected Occupancy

Depends on. Chapter 2; Chapters 9–12.

A completion forecast can be excellent and a staffing decision can still be bad. That is not a complaint about models versus judgment. It is a complaint about objects. Completions this month are a flow through a window. Staffing next month is a claim about what remains. Those two numbers can move in opposite directions on the same path of occupancy, and a dashboard that reports only the first will congratulate a policy that emptied the second.

Chapters 5–8 allocated resource as if tomorrow forgot today: trusted maps from control to outcome, no occupancy to carry. Chapters 9–16 then built the thing tomorrow remembers,

$$x_{t+1} = F_t(x_t, u_t, \theta),$$

and showed how a local control can be searched through that law. This chapter asks a narrower, prior question. If occupancy is real, what, exactly, is the *forecast of its consequences*?

Not a detached KPI. Not an ARIMA of harvest. The forecast is expected occupancy itself, issued from the information set at the origin, and two different functionals of that occupancy can disagree about whether the act improved. Age-structured expected occupancy is a first-class forecast. If you only need $E[x_t]$, you should not pay for entity noise. The public demonstration of that mean-field law is ScientificIntelligence. Matching internal mean-field arithmetic on a fixture is not a license to print another engine's internals.

Imagine the argument as three frozen frames, not as a sparkline.

In the first frame occupancy x_0 is an *area* — mass of OPEN work, drawn by age. A claims file, a patient in a queue, a unit of WIP: the same box. The area is the state. A point on a completions chart is not. In the second frame that area is pushed through F_t. Arrivals enter age 0. Incumbents age. A pulse of arrivals this week can sit in OPEN for a long time before it becomes

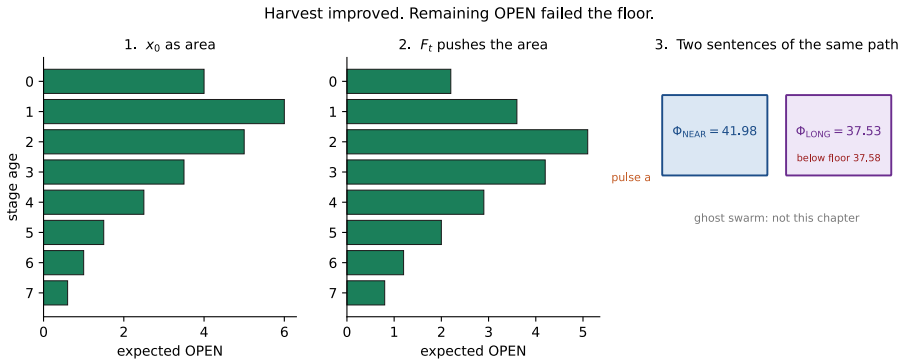


Figure 16.1: Three stills: expected OPEN by age as area; F_t aging a pulse; the same path split into $\Phi_{\text{NEAR}} = 41.98$ and $\Phi_{\text{LONG}} = 37.53$ below the floor 37.58.

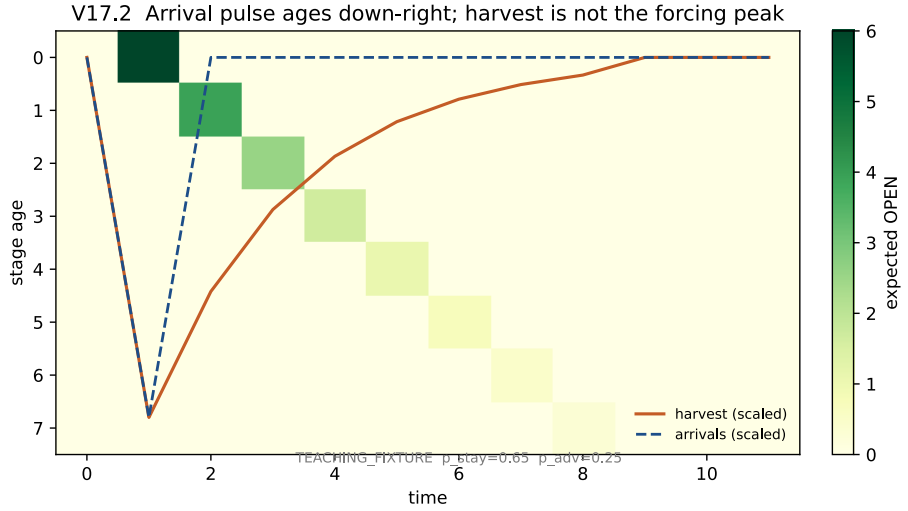


Figure 16.2: An arrival pulse enters age 0 and walks down-right; harvest (solid) peaks after forcing (dashed). Completions are not the state.

a completion. In the third frame the same expected path splits. Φ_{NEAR} reads harvest over a near window. Φ_{LONG} reads remaining OPEN at the horizon. On the unconstrained fixture those sentences disagree: harvest 41.98, OPEN 37.53, floor 37.58. A ghost swarm of entity dots hangs in the background, labeled not this chapter. Floor reliability is a property of realizations. This chapter has none. $P(\text{OPEN} \geq \text{floor})$ is a different epistemic object from $E[\text{OPEN}]$. Asking this forecast for the former is not conservative. It is a category error.

A prediction scores a row. An intervention changes a lever. Control chooses a path of levers through F_t. Allocation, in Chapters 5–8, ignores x. Optimization searches a declared mathematical problem. A recommendation is a search result plus preference, authority, and materiality. A decision is a human, or governed, act. None of those nouns is this chapter's issued object. The issued object is $E[x_{1:T}]$ and the window functionals of it. A mathematically local-optimal control is not a recommendation, and this chapter is not yet in the business of recommending.

16.1 Occupancy as a law, windows as sentences

Let $x_t(a)$ be expected OPEN mass of stage-age a at time t. A discrete step that ages, advances, and receives arrivals is the ordinary compartment update — discrete Kolmogorov, competing risks, structured population — not a product invention (McKendrick 1926; Andersen et al. 1993):

$$\begin{aligned} x_{t+1}(0) &= A_{t+1}, \\ x_{t+1}(a+1) &= x_t(a)\,(1 - p_{\text{adv}}(a) - p_{\text{exit}}(a)), \\ H_{t+1} &= \sum_a x_t(a)\,p_{\text{adv}}(a), \end{aligned}$$

with $p_{\text{adv}} + p_{\text{exit}} \leq 1$. Age is a coordinate of the state, not a reporting label glued on at the horizon. Walk a unit of mass through one step to see why the clocks disagree. An arrival A_{t+1} lands entirely in $x_{t+1}(0)$. Mass that was already age a either advances (counted in H_{t+1}), exits, or stays and becomes age $a+1$. Harvest this period is therefore a functional of *incumbent* occupancy and ADVANCE rates, not of this period's arrivals. Recurrence — a contact, a review, a reminder — emits a flow from $x_t(a)$ *without* the $(1-p)$ factor removing mass. That is why contacts are not a second occupancy, and why a later mean-field $n \cdot p$ will disagree with entity simulation on survivors. The disagreement is mechanism. It is not a bias to be LAST_ERROR'd away.

The same balance Chapter 2 wrote for a single cut holds in expectation: $\sum_s E[x_t(s)]$ must account for arrivals, ADVANCE, EXIT, and residual stay.

If twenty units are OPEN, two arrive, five advance, and three exit, expected OPEN next period is fourteen plus whatever recurrence did *not* remove. Recurrence is absent from that subtraction on purpose. A conservation failure in the engine is a defect, not a "numerical tolerance of the KPI."

This is latent-state *and* structural forecasting in Chapter 2's taxonomy. Harvest and remaining OPEN are functionals of occupancy, and occupancy has a named mechanism. It is not extrapolative. It is not a static map $Y = f(u)$. Affine reconstruction of a rate from a control is an implementation form, not a physical law; on the optimizer fixture that reconstruction matched full-evidence forcing to machine precision, which is a mapping contract, not causality.

Arrivals create age-0 mass. Incumbents age. A shock to arrivals is therefore not a shock to ADVANCE. Feeding a predictive GLM of acquisition in as causal A_t is the engine's Round-2 failure, restated as a forecasting crime: a predictive arrival model is not an identified forcing. The same crime, away from marketing, is treating next month's observed intake as if it were known at this month's origin. That object is leakage. It is not a forecast.

Windows are declared functionals, not renamed cumulatives:

$$J_{\mathrm{NEAR}} = \Phi_{\mathrm{NEAR}}(x_{1:T}, \mathrm{flows}), \qquad J_{\mathrm{LONG}} = \Phi_{\mathrm{LONG}}(x_T, ...).$$

Forecasting one does not imply the other. A "more accurate" completion forecast can worsen a staffing decision that depends on OPEN. Near harvest and long occupancy are different sentences about the same expected path. The optimizer fixture is the numerical proof of that split, not a winner's podium.

16.2 The upper triangle is the process

A visitor shop opens four Monday mornings in a row. In week 0, ten people with observed identities arrive. In later weeks the shop forecasts cohorts of people who do not yet have identities. Each cohort enters OPEN and either ADVANCEs to COMPLETE, EXITs without completing, or stays and gets one week older. Nothing in that sentence is a dashboard. It is a table with two clocks and a hard information boundary.

One clock is **calendar time** t: which Monday we are standing in. The other is **cohort** c: which Monday the file arrived. Age is not a third clock. Age is the difference,

$$a = t - c, \qquad t \geq c.$$

A visitor cannot be older than the shop, and a future cohort cannot already have identities. Those two refusals are the same geometric fact: **the process lives on an upper triangle.** Below the diagonal the cohort has not arrived. On the diagonal it enters. To the right it may still be OPEN.

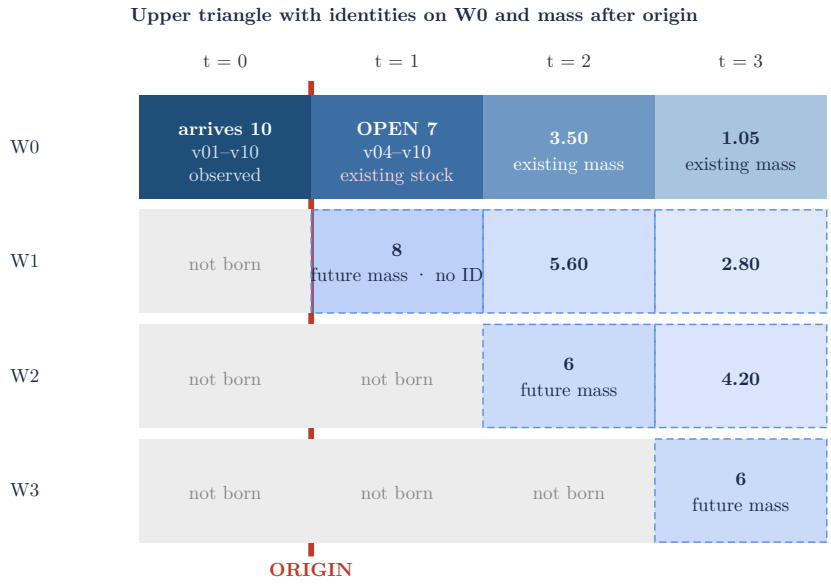


Figure 16.3: The upper triangle places observed W0 identities before the origin and unlabeled expected mass in future cohort cells.

16.2.1 The laboratory

Four arrival weeks, small enough that every cell can be inspected.

cohort	arrives at	expected size	identity at the origin
W0	$t = 0$	10	observed: `v01-v10`
W1	$t = 1$	8	none: future arrival mass
W2	$t = 2$	6	none: future arrival mass
W3	$t = 3$	6	none: future arrival mass

The W0 ledger is observed. `v01` and `v02` COMPLETE in week 0, `v03` EXITs, and `v04` through `v10` remain OPEN at the start-of-week-1 origin. W1-W3 are not lists of hidden names. They are expected future counts, $8 + 6 + 6 = 20$, to be propagated as mass.

Competing risks depend on age, not on which Monday it is. This is Chapter 16's law, with numbers a reader can hold:

age a	p_{adv}	p_{exit}	stay
0	0.20	0.10	0.70
1	0.40	0.10	0.50
2 (and older)	0.50	0.20	0.30

Arrivals land as age 0 and do **not** harvest in the week they arrive. Harvest at $t+1$ is a functional of who was already OPEN at t:

$$\begin{aligned} x_{t+1}(0) &= A_{t+1}, \\ x_{t+1}(a+1) &= x_t(a)\,(1 - p_{\text{adv}}(a) - p_{\text{exit}}(a)), \\ H_{t+1} &= \sum_a x_t(a)\,p_{\text{adv}}(a). \end{aligned}$$

The triangle is that law, written so both clocks stay visible.

16.2.2 Fill the cells

Week $t = 0$. Only W0 exists. Ten observed visitors sit on the diagonal. Calendar OPEN is 10. Harvest is 0 at entry: nobody was OPEN in a prior week. The observed week-0 realization later tells us that `v01` and `v02` COMPLETE, `v03` EXITs, and seven remain OPEN at the origin.

Week $t = 1$. W0 is now age 1. Of the ten observed visitors, $10 \times 0.20 = 2$ ADVANCE, $10 \times 0.10 = 1$ EXIT, and $10 \times 0.70 = 7$ stay. Those are exactly the observed identities above. W1 contributes expected arrival mass of eight, without visitor IDs at the origin. Calendar OPEN is $7 + 8 = 15$. Calendar harvest is 2, and every completion is from W0.

Week $t = 2$. W0 is age 2: $7 \times 0.40 = 2.8$ ADVANCE, $7 \times 0.10 = 0.7$ EXIT, $7 \times 0.50 = 3.5$ stay. W1 is age 1: $8 \times 0.20 = 1.6$ ADVANCE, stay 5.6. W2 arrives as 6. Calendar OPEN is $3.5 + 5.6 + 6 = 15.1$. Calendar harvest is $2.8 + 1.6 = 4.4$.

Week $t = 3$. W0, now past age 2, uses the age-2 rates on 3.5 remaining visitors: harvest 1.75, stay 1.05. W1 at age 2: harvest 2.24, stay 2.80. W2 at age 0: harvest 1.20, stay 4.20. W3 arrives as 6. Calendar OPEN is $1.05 + 2.80 + 4.20 + 6 = 14.05$. Calendar harvest is $1.75 + 2.24 + 1.20 = 5.19$.

The OPEN triangle, every inspectable cell:

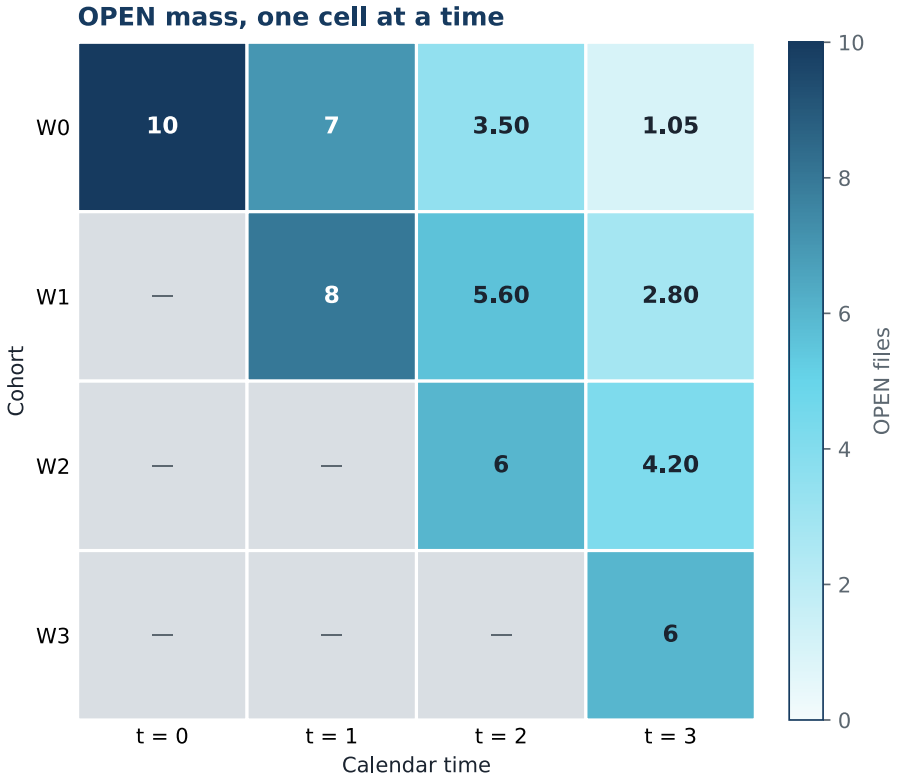


Figure 16.4: OPEN mass, one cell at a time; only the observed W0 cohort carries identities at the origin.

cohort	$t = 0$	$t = 1$	$t = 2$	$t = 3$
W0	10	7	3.50	1.05
W1	—	8	5.60	2.80
W2	—	—	6	4.20
W3	—	—	—	6
calendar OPEN	**10**	**15**	**15.10**	**14.05**

Conservation on observed W0: $2 + 2.8 + 1.75$ completed, plus $1 + 0.7 + 0.70$ exited, plus 1.05 still OPEN, equals 10. The row did not leak and no future visitor was smuggled into it.

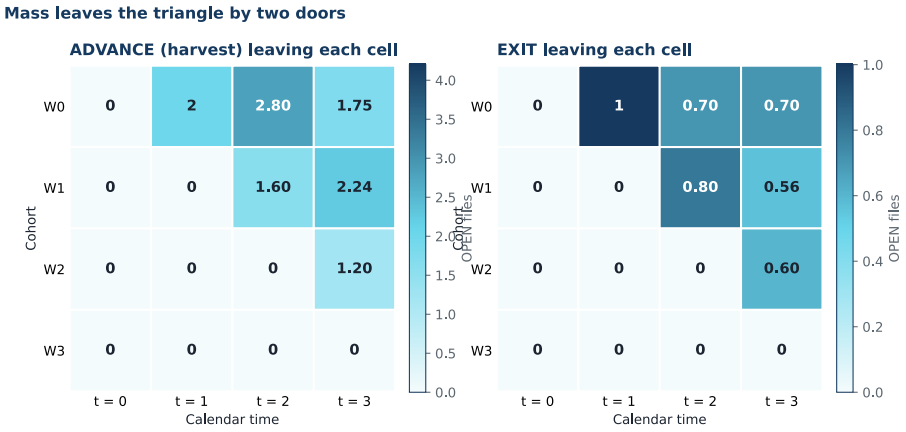


Figure 16.5: ADVANCE and EXIT leaving each cell

Harvest leaving the triangle:

cohort	$t = 0$	$t = 1$	$t = 2$	$t = 3$	**cohort total**
W0	0	2.00	2.80	1.75	**6.55**
W1	—	0	1.60	2.24	**3.84**
W2	—	—	0	1.20	**1.20**
W3	—	—	—	0	**0**
calendar harvest	**0**	**2.00**	**4.40**	**5.19**	

16.2.3 Rows are not columns

Sum a **row** and you have answered a cohort question: of the ten observed people who arrived as W0, expected cumulative completion by $t = 3$ is 6.55. That is a

statement about an entry week.

Sum a **column** and you have answered a calendar-time question: on Monday $t = 2$, expected OPEN population is 15.1 across three different ages. That is a statement about a staffing morning.

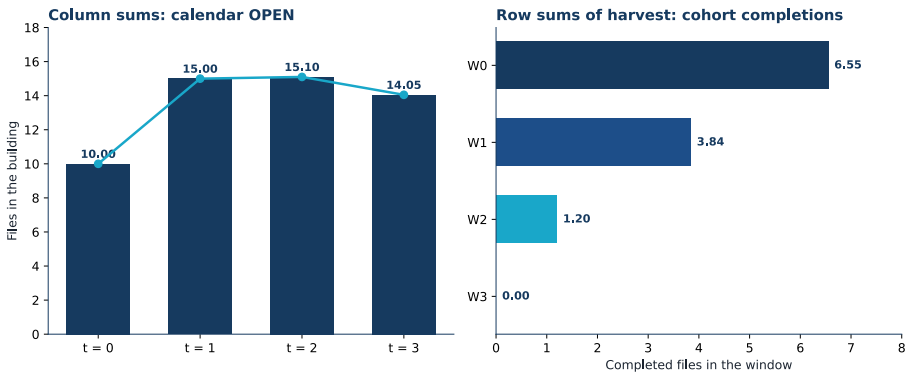


Figure 16.6: Column sums are calendar OPEN; row sums of harvest are cohort completions

A forecast that reports "next month's completions" has summed a slice of harvest columns. A forecast that reports "OPEN next Monday" has summed a column of the OPEN triangle. They are not the same functional, they are not the same decision, and the upper triangle is why they cannot be swapped by renaming a KPI. Confusing them is the crime of reading a diagonal as if it were a column, or a column as if it were a lifetime.

V17.2 already showed an arrival pulse walking down-right in *age* × calendar time. This picture is the same mass in *cohort* × calendar time. Age is the distance from the diagonal. The two pictures are projections of one law. Neither replaces the other.

16.2.4 The origin is a vertical cut

Stand at the start of $t = 1$. Week 0 has been observed. Weeks 1, 2, 3 have not.

Everything in column $t = 0$ is in the information set. Everything to the right of the coral bar is a forecast. Inside that forecast two colours remain:

- **Existing** - the continuation of W0. `v04` through `v10` are already in the building. Their future cells are not observations; they are expected mass produced by applying the process to observed stock.
- **Future** - W1, W2, W3. Those people have not arrived and have no visitor IDs. Treating next week's observed intake as if it sat in $\mathcal{I}_t$ is leakage (Chapter 2). Declaring a path for their counts is a scenario (Chapter 11).

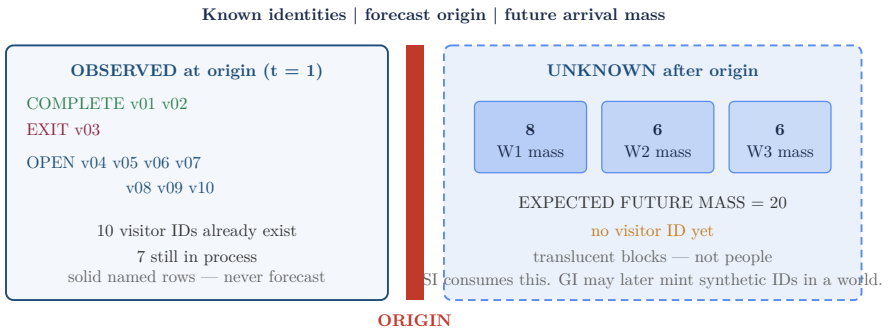


Figure 16.7: At the start-of-week-1 origin, real observed visitors remain named while future arrivals remain unlabeled expected mass.

> Identifying a lever that creates arrivals is an intervention (Chapters 3-4, 10).

The distinction is about ontology as well as time. W0 begins from observed entities. W1-W3 begin from expected counts. Both can be propagated as mass, but only a later simulated world may turn future arrival mass into synthetic future entities. Those synthetic IDs describe one possible world; they never become observed at this origin.

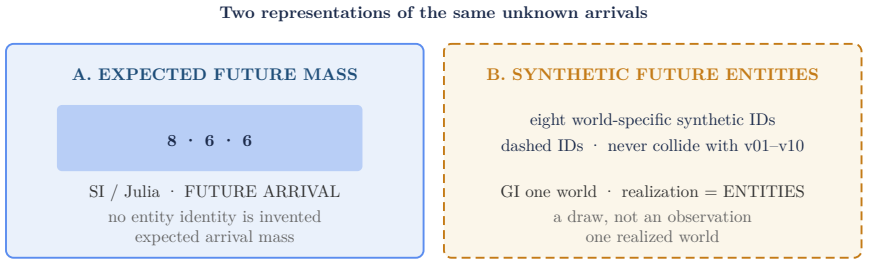


Figure 16.8: Expected future mass has no IDs. Synthetic future IDs are minted only inside a realized world and remain distinct from observed visitors.

Near harvest, $t = 1$ plus $t = 2$:

$$H_{\text{NEAR}} = 2.00 + 4.40 = 6.40 = \underbrace{4.80}_{\text{existing W0}} + \underbrace{1.60}_{\text{future W1}}.$$

Existing share $4.80/6.40 = 0.75$. Three quarters of expected completion in the next two weeks comes from people already known at the origin.

Remaining OPEN at $t = 3$:

$$x_3 = 14.05 = \underbrace{1.05}_{\text{existing W0}} + \underbrace{13.00}_{\text{future}}.$$

Existing share $1.05/14.05 \approx 0.075$. Almost all of the expected OPEN population at the horizon comes from people who had not arrived at the origin.

The same triangle, two windows, opposite stories. An arrival intervention is a LONG instrument on this laboratory. An ADVANCE intervention on visitors already OPEN is a NEAR instrument. The 0.75 and 0.075 are fixture results, not field constants. They are what existing-stock and future-arrival shares mean when every cell is visible.

16.2.5 An intervention changes the triangle, not a KPI

During the step from $t = 1$ to $t = 2$, apply an overtime pulse to visitors already OPEN. Age-0 rates do not move. Age ≥ 1 ADVANCE rises from 0.40 to 0.60 (age 1) and from 0.50 to 0.70 (age 2). EXIT is unchanged, so stay falls. Only observed W0 is age ≥ 1 at $t = 1$. W1 is still age 0 and does not feel the pulse.

W0's seven remaining visitors now yield harvest $7 \times 0.60 = 4.20$ instead of 2.80, and stay $7 \times 0.30 = 2.10$ instead of 3.50. Future rows do not move. An arrival intervention would do the opposite kind of work: it would change the 8, 6, 6 expected arrivals and leave `v04` through `v10` untouched.

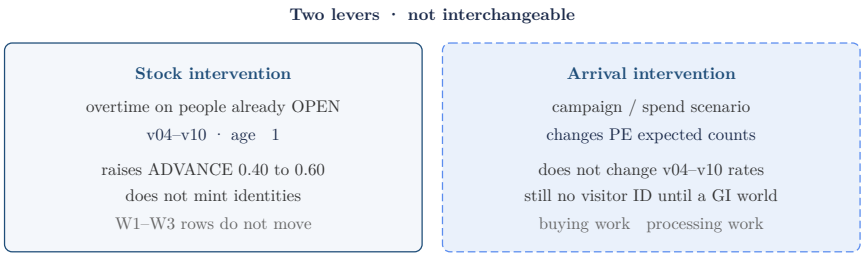


Figure 16.9: A stock pulse changes the observed W0 continuation; an arrival intervention would change future cohort mass instead.

Calendar harvest at $t = 2$ rises from 4.40 to 5.80. Calendar OPEN at $t = 2$ falls from 15.10 to 13.70. Completions and remaining stock move in opposite directions on the *same* population, now visible as a change to one cohort row.

Near harvest $t = 1 + t = 2$ rises from 6.40 to $2.00 + 5.80 = 7.80$. Remaining OPEN at $t = 3$ falls from 14.05 to 13.63, and harvest at $t = 3$ falls from 5.19 to 4.49: W0 has less mass left to complete later. A dashboard that scored only

this week's closures would call the pulse an unambiguous win. The shop that must staff the horizon would not.

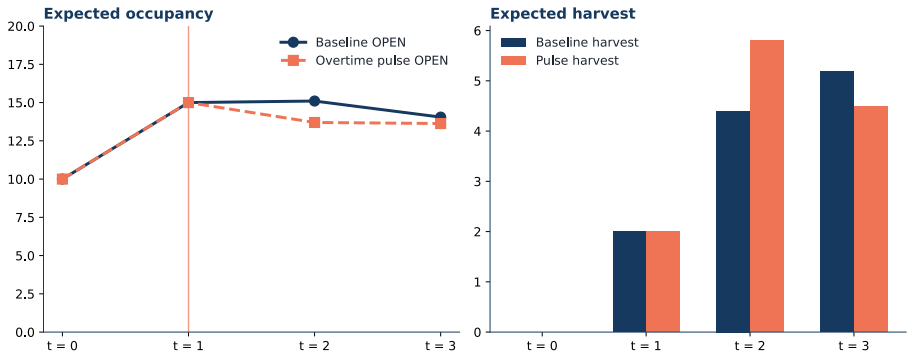


Figure 16.10: Expected occupancy and harvest, baseline versus pulse

16.2.6 Expected occupancy is the column sum. Micro is the cell, redrawn.

Every number above is an expectation. Ten observed W0 visitors, each independently ADVANCE / EXIT / STAY with probabilities 0.20/0.10/0.70, have expected stay 7 and expected calendar OPEN at $t = 1$ of $7 + 8 = 15$. The observed origin realizes the expectation as exactly two COMPLETE, one EXIT, and seven OPEN. Eight additional integer worlds, seed 7, show what could have happened:

world	ADVANCE	EXIT	W0 still OPEN	calendar OPEN if $A_1 = 8$
1	2	2	6	14
2	3	0	7	15
3	1	2	7	15
4	0	2	8	16
5	3	1	6	14
6	1	0	9	17
7	1	1	8	16
8	2	1	7	15
expectation	**2**	**1**	**7**	**15**

The swarm sits on the triangle; it does not replace it. These rows vary the outcomes of the ten existing observed people. A later world may also realize W1 arrival mass as synthetic IDs such as `F01_0000001`, but those IDs exist only

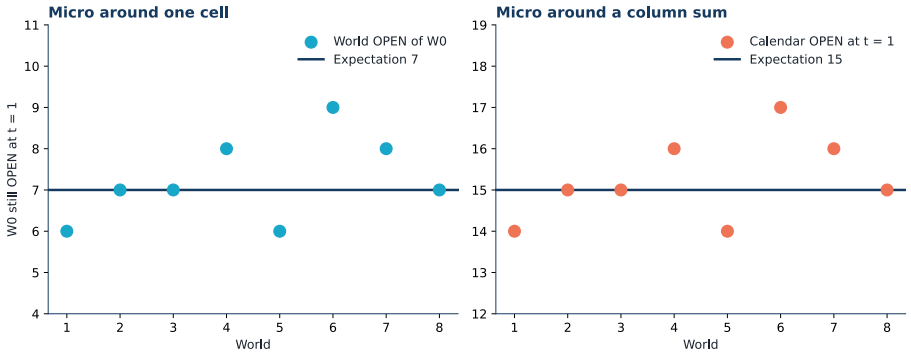


Figure 16.11: Micro around one cell, and around a column sum

in that world. Chapter 17 will perform that transformation explicitly. Floor reliability is a property of realized worlds, not of the 15.10 in column $t = 2$. Asking the mean-field triangle for $P(\text{OPEN} \geq \text{floor})$ is `UNAVAILABLE`.

16.2.7 What the reader can now reconstruct

Given arrivals, age-specific ADVANCE and EXIT, and an origin, the reader can:

1. place observed arrivals or expected future arrival mass on the diagonal;
2. propagate each cohort to the right by stay, sending harvest and EXIT out of the cell;
3. read a cohort lifetime from a row and a staffing morning from a column;
4. refuse to fill cells below the diagonal;
5. cut the triangle at the origin and distinguish known people from unlabeled future mass;
6. change one typed rate and watch which rows move;
7. create synthetic future identities only inside an integer world, without mixing that world into the expectation.

That is the process. Chapters 9–15 search and replay through F_t. Chapter 16 issues $E[x]$. Chapter 17 issues the swarm. This picture is the object those chapters have been talking about.

16.3 Three schedules, two sentences

Consider a synthetic inventory of open work — claims, patients, WIP — with a declared OPEN floor of 37.58. The search that produced the rows is assumption-conditioned, not a field spend plan. Three schedules:

schedule	NEAR harvest	LONG OPEN	on the floor?
baseline	39.16000	38.04714	yes, with slack
unconstrained SLSQP	41.98435	37.52823	**no** — below 37.58012
pipeline-protected	41.96128	37.59588	yes, barely

Read the table as a trade, not as a ranking. Baseline harvest sits at 39.16 with OPEN at 38.05. The unconstrained local solution lifts harvest to 41.98 and drops OPEN to 37.53, *below* the floor. The pipeline-protected local solution keeps almost all of that harvest lift — 41.96 — and holds OPEN at 37.60, just above the floor.

Φ_{NEAR} improved. Φ_{LONG} failed, then was protected. Accuracy of completions is not staffing quality. If the act is "do not let remaining OPEN fall through the floor," the unconstrained run is a worse act than baseline even though its near harvest is higher. If the act is unconstrained harvest, the protected run looks like a small sacrifice of $41.98435 - 41.96128 = 0.023$ harvest units for 0.068 of pipeline. Those are different problems. A single accuracy number cannot rank them.

Put the same three rows in a claims shop. Baseline leaves more files open than the floor requires and closes fewer. Unconstrained closes more this window and understaffs the remaining inventory relative to the floor the shop declared. Pipeline-protected keeps almost the extra closures and does not spend the floor. A dashboard that scored only closures would pick unconstrained and call it a win. The shop that has to assign reviewers next period would not. The fixture is synthetic and assumption-conditioned; it is not a field spend plan. It is still enough to teach that Φ_{NEAR} and Φ_{LONG} are different sentences.

The same geometry appears as soon as forcing, state, and response run on different clocks. Outdoor temperature, wall heat, and indoor comfort need not peak together; media spend, latent awareness, and conversions need not; reviewer hours, open-file age, and closures need not. The thermal-lag picture is pedagogy, not a new law, and it does not identify causality. It only forbids reading a completion peak as a forcing peak.

What this object still cannot say is as important as what it can. Expected occupancy has **no entity variance**. The mean-field OPEN of 37.60 on the protected schedule is a statement about $E[x]$. It is not $P(\text{OPEN} \geq \text{floor})$. Chapter 17 will put a swarm of worlds around that point. This chapter must not pretend the swarm is already there.

16.4 Error correction must not duplicate the state

This F_t *is* structural latent state. Ordinary statistical forecast error is a different object. Persistent residual mean against a misspecified KPI model can be calibration, or it can be a changed mechanism. Those are competing explanations, not one smoother. Chapter 19 is where that architecture is written. This chapter only forbids a reflex that looks harmless and is not.

If occupancy already moves, stacking an operational LAST_ERROR on harvest can double-count adaptation that F_t has already performed. The question to ask, before any smoother is attached, is whether the error is already a state of the process. Suppose last month's harvest residual was persistent and positive. One story is that the KPI model is miscalibrated and an intercept shift will fix it. Another is that OPEN aged into a richer ADVANCE band, so F_t will harvest more this month *because the stock moved*. Adding last month's residual to this month's process forecast then counts the aging twice: once as occupancy, once as a patch. The thermal building is the same mistake with different nouns. A late indoor-temperature peak is not a regime change of the thermostat. It is wall heat.

A regression or ARIMA of a KPI has a residual and, unless a local level is present, no latent occupancy. Intercept correction of the Clements–Hendry kind (Clements and Hendry 1998) can be a legitimate operational patch on *that* object. A Kalman filter already updates x_t from observation noise (Kalman 1960; Durbin and Koopman 2012); an external LAST_ERROR on top of the filter duplicates the update. LAST_ERROR is this book's name for that operational residual, not a named method in those sources. Structural F_t, this chapter's object, *is* occupancy. LAST_ERROR on harvest then risks counting the same motion twice: once as state, once as a residual add-on. An adaptive challenger is a different F, not a patch on this one, and must not inherit the incumbent's correction as if it were physics. The full architecture — correction versus regime versus trigger versus promotion — is Chapter 19. This chapter's prohibition is only the reflex.

Error correction and regime change remain competing explanations, not one hidden numerical blend. Do not attach operational LAST_ERROR to a process forecast because the harvest residual looks persistent. Ask first whether OPEN already moved.

16.5 Issuing the forecast

The public aggregate path is SI. Fit the age-structured process at the origin; then forecast it under declared future arrivals, context, and forcing.

```
library(ScientificIntelligence)

fit <- sci_fit_age_structured_process(
  engine, transition_rates = transition_rates,
  initial_state = origin_state, origin = 20,
  recurrence_rates = recurrence_rates
)
fc <- sci_forecast_age_structured_process(
  fit, arrivals = future_arrivals, future_context =
 ↪  future_context,
  horizon = 28, exogenous_forcing = forcing, summary_spec =
   ↪  summary
)
```

Forecast-intelligence adapters exist when the object is an FI *revision*. They do not replace SI as the process engine. Do not wrap `fi_operational_forecast` around this path in this chapter's examples: that call assimilates operational error on a forecast revision, which is the double-count refused earlier in this chapter.

`future_arrivals` and `future_context` are elements of $\mathcal{J}_t$, or they are declared scenarios. If they are next month's *observed* arrivals, the object is leakage, not a forecast. The origin bar in the first frozen frame is the test: anything to the right of that bar is not in the information set, however accurate the resulting "prediction" becomes. A claims shop that "forecasts" next month's OPEN using next month's intake tape will report an impressive score and will not have improved the staffing act that had to be chosen before the tape existed. The same crime in quieter clothes is a GLM of acquisition treated as identified A_t: a predictive arrival model can be a legitimate *conditional* input, and it is still not a causal forcing.

On the same optimizer fixture, affine reconstruction of typed forcing matched SI full-evidence forcing with maximum absolute difference zero, and a mean-field harvest comparison against SI sat at 10^{-8} on four retained candidates. That is why SI is the public demonstration. It is not a reason to print unexported mean-field internals from another package, and it is not a field claim about spend.

A claims shop forecasting expected open files by age of file is the same object as the fixture, with different nouns. So is expected WIP by hours on the floor, or expected patients by time in stage. Recast the nouns; do not invent new decimals. Non-marketing optimizer qualification remains an evidence gap: teach the *claims* of this chapter, not a second decimal table.

Walk the split on a three-age toy before the optimizer table. Mass at ages $0, 1, 2$ is $(8, 6, 4)$, total OPEN 18. ADVANCE probabilities $(0.25, 0.40, 0.50)$, EXIT $(0.10, 0.15, 0.20)$, remainder stays (age 2 stays at 2). Arrivals 3. Harvest this step is $8 \cdot 0.25 + 6 \cdot 0.40 + 4 \cdot 0.50 = 6.4$. Next OPEN is $3 + 8 \cdot 0.65 + 6 \cdot 0.45 + 4 \cdot 0.30 = 12.1$. Raise incumbent ADVANCE by staffing the old files, $(0.25, 0.60, 0.70)$: harvest 8.4, next OPEN 10.1. Completions and remaining stock move in opposite directions on the *same* occupancy. The optimizer table below is that geometry on a larger fixture; the toy is what "near versus long" means in mass.

The same split, read as a failure rather than as a table, is the chapter's close. Harvest rose from 39.16 to 41.98 (about 7%) while remaining OPEN fell from 38.05 to 37.53 (about 1.4%, and through the floor): Φ_{NEAR} moved one way and Φ_{LONG} the other. A more accurate completion forecast, scored on harvest, can worsen the staffing act that depends on remaining OPEN. The dashboard did not lie about harvest. It answered the wrong sentence. The unconstrained local solution is the numerical version of that failure. It beats baseline harvest and violates the OPEN floor *in expectation*. Accuracy of J_{NEAR} improved; decision quality for a floor-constrained act worsened. Optimization, in the solver's sense, occurred. The act, if the act is the floor, did not improve.

If someone now asks this chapter for $P(\text{OPEN} \geq \text{floor})$, the honest status is `UNAVAILABLE`. There is no entity variance on $E[x]$. That question is Chapter 17.

16.6 Exercises

C16-E1. Mathematical. Write F_t for OPEN with age, ADVANCE, EXIT, and arrivals. Where does recurrence enter, and why does it not remove mass?

C16-E2. Interpretation. A dashboard shows completions up and OPEN down. Which window functional moved? Which decision, if any, should change?

C16-E3. Interpretation. Two engines agree on expected harvest to machine precision on this laboratory. Why is that still not a license to treat one engine's internals as the public forecast?

C16-E4. Failure diagnosis. When should you *not* attach operational LAST_ERROR to this forecast? What competing explanation does Chapter 19 still owe you?

C16-E5. Conceptual. In one sentence each: a locally optimal control; a recommendation; an authorized action. Why are they not the same object?

C16-E6. Decision reasoning. Unconstrained harvest 41.98, OPEN 37.53, floor 37.58. Did "optimization" improve the act? If the floor is the act, what is the honest answer?

C16-E7. Assumption criticism. At the start-of-week-1 origin, someone assigns visitor IDs to the expected 8, 6, and 6 future arrivals. What is wrong? When may a synthetic future ID legitimately be created?

C16-E8. Epistemic restraint. You are asked for the chance remaining OPEN clears the floor. This chapter issued an expectation. What is missing, and why can neither an expected count nor a list of synthetic IDs from one world answer it?

16.7 Transition

Chapter 17 realizes the same process as integer events. The mean can sit above the floor while a third of worlds fall through. Expected occupancy remains a legal forecast. It is no longer the act.

CHAPTER 17

When Expectation Is Not the Act: Micro Realization

Depends on. Chapter 16.

Chapter 16 forecast *expected* occupancy. Many acts do not live in an expectation. They live in a world. Will OPEN be below the floor *this quarter*? Will this clinic's twenty open files, not a continuum of mass, actually advance before Friday? Integer events, small N, recurrence that fires only on those still present: the mean can sit above the floor while a third of worlds fall through.

That is not a failure of Chapter 16. It is a different functional of the same F_t. Chapter 16 issued $E[Y \mid u, \theta]$. This chapter draws from $P(Y \mid u, \theta)$. In Chapter 2's taxonomy the second object is simulation-based forecasting. The law is the process; the issued statement is a realization.

Entity draws d are aleatoric given frozen θ. They are not a posterior. They are not a credible interval for a rate. Collapsing d into "the uncertainty" with parameter draws b is Chapter 19's crime; this chapter only forbids calling a quantile of d a confidence interval for θ. Two number lines, never one ribbon.

17.1 Six Claims Become One Realized World

This uses the same W0 competing-risk law as the cohort triangle.

Take six named OPEN claims. ADVANCE probability is 0.20, EXIT probability is 0.10, and STAY_OPEN receives the remaining 0.70. Recurrence probability is 0.30, but only among entities that remain OPEN. In the compact event table, STAY means STAY_OPEN.

17.1.1 1. Resolve individual competing events

Entity	Event u	Event	OPEN after event?	Recur. u	Recurs?
C1	0.08	ADVANCE	no	0.90	no
C2	0.24	EXIT	no	0.20	no
C3	0.42	STAY	yes	0.12	yes
C4	0.77	STAY	yes	0.55	no
C5	0.18	ADVANCE	no	0.61	no
C6	0.95	STAY	yes	0.25	yes

The realized world contains two ADVANCE, one EXIT, three survivors, and two recurrence events.

17.1.2 2. Expose the wrong event order

The correct recurrence expectation is based on three survivors. The shortcut (n p=6(0.30)=1.8) treats the three departed entities as still eligible. In this realized world, it can even assign a contact to C2 because C2's recurrence uniform is below 0.30, although C2 already exited.

The wrong method did not merely approximate a count. It changed the event law by creating ghost contacts.

17.1.3 3. Repeat worlds around the expectation

The fixed W0 fixture supplies eight count worlds. After eight new arrivals are added, realized calendar OPEN is

$$(14, 15, 15, 16, 14, 17, 16, 15),$$

centered on expected OPEN 15.

Expectation is the center of the process law, not the result in every world. Reuse the same draw identity under a second policy and the difference becomes a paired experiment; change draw identity by candidate and policy differences become entangled with different worlds.

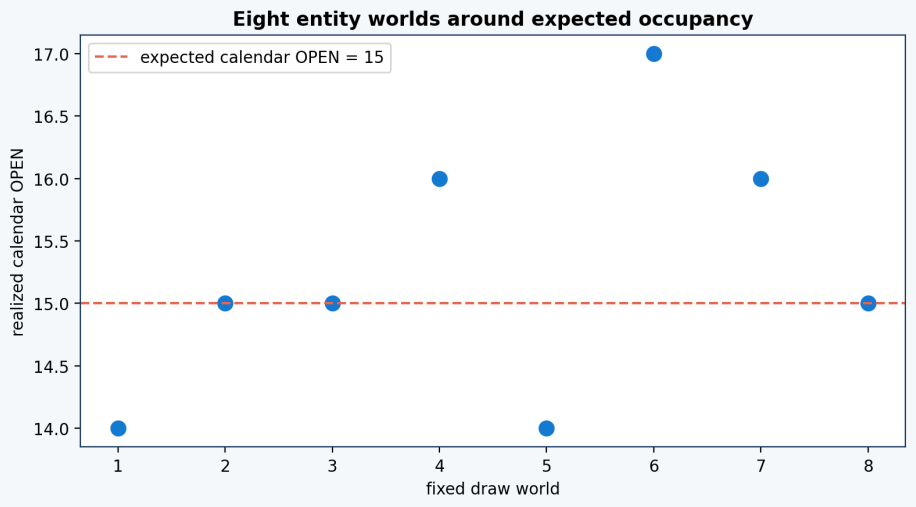


Figure 17.1: Eight countable entity worlds vary around the expected occupancy produced by the cohort triangle.

17.2 Pairing is the experiment

Comparing two policies under independent noise is legal and wasteful. If $J_A(\omega)$ and $J_B(\omega)$ are positively associated — same arrivals, same clocks, same unlucky week — the variance of the contrast is

$$\mathrm{Var}(J_A - J_B) = \mathrm{Var}J_A + \mathrm{Var}J_B - 2\mathrm{Cov}(J_A, J_B),$$

which is at most the independent-stream total, with equality only when the covariance is zero. Common random numbers keep the covariance. Independent streams throw it away. That reduction is Glasserman and Law, not a product theorem (Glasserman 2004; Law 2015). Ranking-and-selection and paired experiments are older still (Bechhofer 1954; Chick and Inoue 2001). What has to be engineered is the identity of ω.

Let $G(\text{seed}, \text{origin}, d)$ be the uniform stream (or equivalent) that drives competing exponential clocks or multinomials. Policy A and policy B must consume $G(\cdot, d)$ in the same event order wherever they share structure. Candidate identity is not an argument of G. Worker identity is not an argument of G. If A has an extra transition, CRN theory does not magically pair the unmatched events; it pairs the shared ones. Parallelism that reseeds per worker is not CRN. It is a race. The "pair" then becomes an artifact of thread scheduling, and "A beat B" is no longer an experiment.

A toy of six paired draws, frozen θ, teaching table rather than field sample, makes the covariance visible.

d	J_A CRN	J_B CRN	Δ CRN	J_A indep	J_B indep
1	40.2	40.0	0.2	41.8	38.1
2	39.1	38.8	0.3	37.4	41.6
3	41.5	41.1	0.4	42.9	39.0
4	38.6	38.4	0.2	39.2	42.4
5	40.8	40.5	0.3	36.7	38.8
6	39.9	39.7	0.2	43.1	40.2

The CRN differences are $\{0.2, 0.3, 0.4, 0.2, 0.3, 0.2\}$. Their sample variance is 0.0067. The independent columns, unpaired worlds with similar *levels*, have sample variances adding to 10.75. The covariance *is* the experiment. Independent streams paid two orders of magnitude more contrast variance to learn a difference that CRN learned at 0.0067. The toy is six draws, frozen θ, teaching arithmetic, not a universal reduction factor. Both policies read the same ω_d; candidate id does not enter G; a worker-seeded path is unpaired. A clinic comparing two reviewer calendars on different random weeks is not comparing the calendars. It is comparing the weeks.

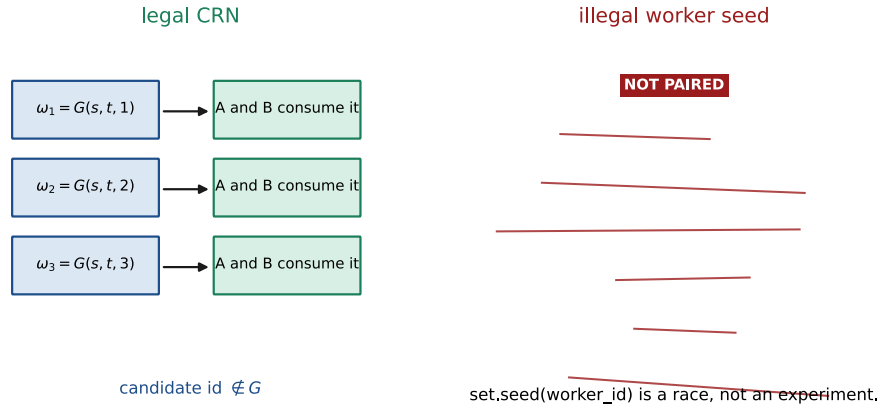


Figure 17.2: Two policies consume the same draw stream $\omega_d = G(\text{seed}, \text{origin}, d)$. Candidate identity is not an argument of G. Seeding by worker id is a race, not a paired experiment.

The public construction of that identity is small, and it is the whole point of the evaluator.

```
library(GenerativeIntelligence)

ids <- gi_policy_draw_ids(n = 24L)
gi_policy_draw_stream(seed = 7, origin = 0, draw_id = ids[1])
```

Same seed, same origin, different `draw_id`: the streams must differ. Same `draw_id`, two candidate ids: the stream must match. Probability-engine CRN (`pe_crn_sample`, `pe_crn_plan`) is the same idea on a probability space, not a policy-shortlist tool. Do not ask it to rank schedules.

17.3 Most of the near harvest is already in the building

Origin splits mass that is already present from mass that has not yet arrived. That split is a forecasting fact, not a footnote about accounting.

Walk it on Chapter 16's three-age toy. Origin occupancy is $(8, 6, 4)$, ADVANCE is $(0.25, 0.40, 0.50)$, EXIT is $(0.10, 0.15, 0.20)$, and arrivals are 3 each step. Near is two steps.

Step 1 harvest is all incumbent:

$$8 \cdot 0.25 + 6 \cdot 0.40 + 4 \cdot 0.50 = 6.4.$$

What remains of that stock is $8 \cdot 0.65 = 5.2$ now age 1 and $6 \cdot 0.45 + 4 \cdot 0.30 = 3.9$ at age 2, plus three arrivals at age 0. Step 2 harvest splits:

$$\underbrace{5.2 \cdot 0.40 + 3.9 \cdot 0.50}_{\text{4.03 existing}} + \underbrace{3 \cdot 0.25}_{\text{0.75 future}} = 4.78.$$

Near harvest is 11.18, of which 10.43 came from the origin stock. Its share is

$$\frac{10.43}{11.18} \approx 0.933.$$

The optimizer laboratory showed the same geometry at a different scale: $38.375/40.25 \approx 0.953$ for one candidate. That is illustrative fixture evidence, not a field constant or an authority threshold. The walk above is what the ratio means. The figure draws that split as area, not as a stamped KPI.

If ninety-five percent of NEAR harvest is already in the building, an arrival intervention is mostly a LONG instrument. An ADVANCE intervention can still move NEAR, because it acts on stock that is already OPEN. Recurrent contact may move a flow without moving occupancy. A completion-only KPI forecast will look "driven by spend this week" even when it is driven by last

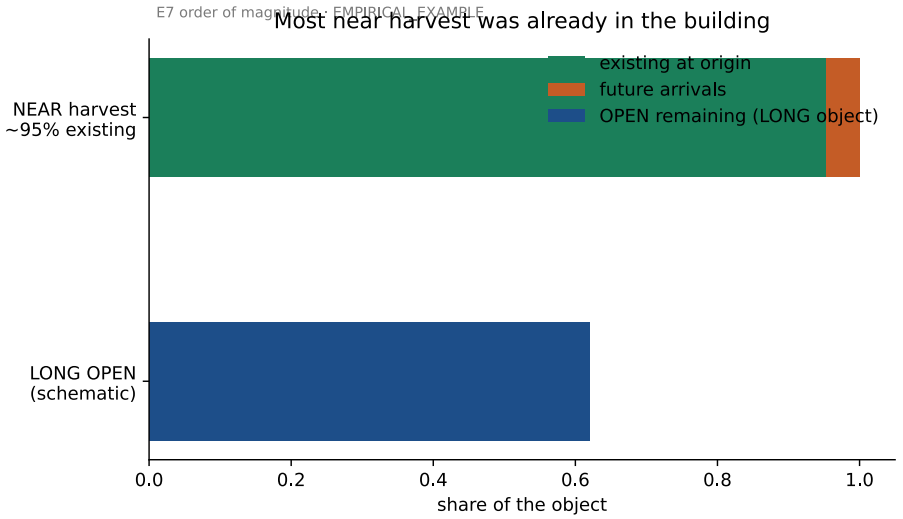


Figure 17.3: Existing versus future origin. Near harvest is almost entirely stock already present; remaining OPEN at LONG is a different object.

month's stock. That is Chapter 2's stock/flow contract, now with a number, and it is why Chapter 16's OPEN at LONG is a decision object rather than a leftover.

The same number tells you which forcing class is even in play. ARRIVAL barely moves NEAR on this fixture. ADVANCE can. If a claims shop responds to a bad near-harvest week by buying more intake, it is mostly buying next window's work. If it changes how existing files advance — a specialist hour, a faster review, a competing-risk shift from EXIT into ADVANCE — it is acting on the object the near functional can see. Recurrence still does not fill the gap: contacts fire on survivors, so a mean-field $n \cdot p$ of "outreach events" will overstate the flow whenever OPEN has already thinned. Preserve that disagreement. It will reappear as a GI mean that does not match an SI mean in Chapter 18, and it will not be a bug in either.

A non-marketing twin is immediate. A ward with twenty patients already in stage, and a trickle of admissions, does not staff the near shift from an admissions forecast. It staffs from the patients in the beds. Future admissions are a LONG instrument unless ADVANCE is so fast that age-0 mass harvests inside the near window — which is a statement about F_t, not a hope about the dashboard. The origin bar in Chapter 16's first frame is doing the same work here: mass to the left of the origin is existing; mass not yet arrived is future. Confusing them is how a near-window KPI gets attributed to this week's lever.

17.4 Frozen θ, and what the evaluator is allowed to say

Uncertainty does not come in one size. If θ is known and N is huge, micro noise vanishes and Chapter 16 was enough. If θ is weakly identified and N is huge, parameter uncertainty dominates: more entity draws will not shrink Θ_I. If θ is pinned and $N = 20$ files, micro dominates: the floor lives in d.

Crossing every parameter world b with every entity world d is Chapter 19's stress test. This chapter keeps θ frozen. Current GI validation is **conditional micro**: one declared posterior row per policy. That is an honest statement of $P(Y \mid u, \theta)$. It is not $P(Y \mid u, \text{data})$. Asking this evaluator for the mixture over the posterior is `UNAVAILABLE`. More draws of d do not mix b. A tenth quantile of those draws, labeled "90% CI for the demand mean," is the same error in a ribbon: d is aleatoric given θ, and a confidence interval for a mean is an epistemic statement about θ. The two number lines do not become one interval by being plotted together.

The public evaluator is built from a process spec, caller-supplied bindings and forcing, a seed, and an origin. Absence of a helper for the bindings table is a pedagogical gap, not a license to reach into internals.

```
library(GenerativeIntelligence)

spec <- gi_age_structured_micro_spec()
ev <- gi_make_process_policy_evaluator(
  spec, baseline_forcing, bindings, seed = 7, origin = 0
)
```

Bindings and forcing tables are caller-supplied. The simulator you do not see is wrapped by this evaluator on purpose. An unexported entity stepper would not teach the identity of d. It would teach a private name. Probability-engine CRN remains available for a different job: sampling a probability space under shared uniforms. It is not how a policy shortlist is paired.

The evaluator will be misread in the ways Figure 18.1 already stamped. Seeding by worker id makes the paired difference illegal: "A beat B" is then an artifact of thread scheduling, because independent worker streams set $\text{Cov}(J_A, J_B)$ to an accident. High expected harvest with a missed floor is not a paradox. Mean harvest 41.98 on the optimizer-scale fixture coexists with $P(\text{OPEN} \geq \text{floor}) \approx 2/3$ under micro. Two thirds is not a theorem and not a service level; Chapter 18 stamps it onto the swarm. Maximizing expected harvest without the chance

functional is a hidden risk-neutral preference. The mean can sit above the floor while a third of worlds do not, and those worlds are the act.

A preview, not this chapter's object: policy C is noisy at small N and would change the act if its tail were known. Acquiring more micro draws — or more data on θ — can have value even if the current mean ranking looks settled. That is value of information, and it is Chapter 19's. The present evaluator, with θ frozen, cannot answer it.

17.5 Exercises

C17-E1. Mathematical. Show that $\mathrm{Var}(J_A - J_B)$ under CRN is at most the independent-stream variance when the outcomes are positively associated. Where does the covariance go if each worker reseeds?

C17-E2. Failure diagnosis. `set.seed(worker_id)`. Why is the paired difference illegal? What experiment do you think you ran?

C17-E3. Interpretation. From the three-age walk, existing near harvest is 10.43 of 11.18. Which forcing class still moves NEAR a lot? Which barely moves it? Why does a completion-only dashboard mislead?

C17-E4. Method selection. Frozen θ, $N = 20$ versus $N = 10^6$: which uncertainty dominates in each case? What changes if θ is weakly identified and N is huge?

C17-E5. Assumption criticism. A tenth quantile of d is presented as a "90% CI for the demand mean." Name the axis error. Why is one ribbon the wrong picture?

C17-E6. Public API. Construct two draw streams with the same seed and origin and different `draw_id`. They must differ. Same `draw_id`, two candidate ids: the stream must match. What is *not* an argument of G?

C17-E7. Epistemic restraint. You want the chance of the act given the data, mixing over unknown rates. This chapter froze the rates. What would more entity draws fail to provide?

17.6 Transition

Chapter 18 takes a *supplied* shortlist and validates it under these streams. GI does not search. A paired tie is not a winner. The label you are allowed to print is narrower than "optimal," and it is the whole point of the split.

CHAPTER 18

Validation Is Not Search, and Feasible Is Not Reliable

Depends on. Chapters 15–17.

Chapter 16 forecast expected occupancy. Chapter 17 realized it. Someone now holds a handful of schedules — a baseline, a local NLP solution, perhaps a human alternative — and wants to know which one to trust under the worlds of Chapter 17. The temptation is to hand that question back to the solver: put Monte Carlo inside SQP, call the result optimal, and go home.

That temptation confuses two computational objects.

Search proposes. SI, or a frontier optimizer, or a human, searches a declared mathematical problem and emits candidates. The cheap object is $E[x]$, a map, an adjoint. Validation proposes nothing. It takes a *supplied* shortlist and spends compute on paired laws: distributions of J, paired differences, regret, constraint frequencies, rank. Decision intelligence then compares those laws with a materiality ε and a declared risk preference. A human authorizes. A mathematically local-optimal control is not a recommendation. A recommendation is not an authorized action. GI does not find the optimal policy. The validator may say which supplied candidate looked best under the paired worlds. It may not say the policy is globally optimal.

Freeze that split as three frames.

In the first frame a search box — sequential quadratic programming, an evaluation-limited run, a greedy frontier step, a person with a pencil — emits points in control space. Occupancy is still an area if the search used F_t; it may be nothing of the kind if the search used static maps. The frame does not care. It only produces candidates. In the second frame those candidates are frozen as a shortlist. No new point is minted. In the third frame GI consumes the shortlist under common random numbers. Shared streams, paired harvests, a rank that can refuse to separate. A stamp on the output reads that this was the best of the *supplied* shortlist. Another stamp sits on "globally optimal." There is no arrow back into search. If there were, the random worlds would chase a moving candidate set, and "optimal" would mean a noisy inner loop.

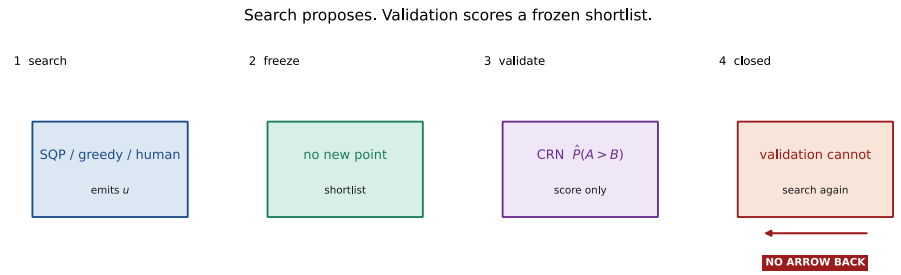


Figure 18.1: Three stills: search proposes; the shortlist freezes; GI validates under CRN. Stamp: no arrow back into the solver.

This is the chapter where a forecast *becomes* decision evidence: not because the mean moved, but because paired realizations, ranks, and constraint frequencies exist for a *named shortlist*.

18.1 What paired Monte Carlo actually adds

Ranking-and-selection, paired experiments, and indifference-zone procedures are old (Bechhofer 1954; Chick and Inoue 2001). Simulation does not become a new theory by being applied to OPEN files. What the method adds, relative to Chapter 16, are objects the mean-field forecast cannot issue: a distribution of J, a paired Δ, a regret, a constraint frequency, a rank. What it does not add is identification. A million draws of an unidentified causal object remain unidentified. Simulation does not promote `PRIOR_ONLY`.

For two candidates A and B on shared draws ω_d,

$$\hat{P}(A > B) = \frac{1}{D} \sum_{d} \mathbf{1}\{J_A(\omega_d) > J_B(\omega_d)\}.$$

Ties are first-class. $\hat{P}(\text{tie})$ is not a rounding error, and it is not a license to declare the two policies decisionally equivalent. Equivalence requires a declared ε. Without it you have two refusals, not a winner: the paired experiment has not separated them, and you have not been told how close is close enough.

Adaptive depth is a screen. Staged $6 \to 12 \to 24$ on Wave 1, and common-depth 512 on a later hardening pass, are empirical schedules on particular fixtures. They are not GI theorems, not recommended defaults, and not a contribution. Do not cite 6/12/24 or 512 as if a sequential-sampling paper had been proved

in this book. Final evidence uses common depth and shared streams. Adaptive GI, used honestly, only decides which candidates deserve that common depth.

Name every reliability claim with the axis, the draw count D, the target, a bandwidth if a surrogate was used, and the Monte Carlo precision. A bare "about two thirds" is a classroom number. A production sentence needs the rest.

New failures come with the new objects. Putting the validator inside the searcher destroys the split and typically wrecks pairing as the candidate set moves (Chick and Inoue 2001; Bechhofer 1954). Simulation does not identify a lever Chapter 4 left unidentified. Ties without a declared tolerance are not equivalence (Chapter 21). Search is allowed to be cheap. Validation is allowed to be expensive. Neither is allowed to impersonate the other.

18.2 Two depths, two objects, no winner

On the Wave 1 optimizer shortlist, at common depth $D = 24$, GI mean NEAR harvest and the SI source that proposed the candidates are different objects:

candidate	GI mean	SI source
eval-limited SLSQP	40.50	42.32893
pipeline-protected	40.29	41.96128
unconstrained SLSQP	40.25	41.98435
baseline	38.00	39.16

The numbers do not match because they should not. SI issued $E[x]$. GI issued a micro mean of integer events on survivors. Spearman rank correlation is about 0.8; there is one pairwise reversal, unconstrained versus protected. Top-1 is shared: eval-limited. That last fact is a fixture observation, not a law that evaluation limits beat converged SQP. The eval-limited protected run that ranked first is one fixture, two starts, `maxeval=60`. Do not promote it.

Unconstrained versus pipeline-protected, paired at $D = 24$: $\hat{P}(\text{tie}) = 0.958$. Picture the histogram of harvest differences piled on zero: almost every shared world, the two schedules produce the same integer harvest; on a handful, protected is ahead; unconstrained is never ahead. That is not "they are the same policy." It is "at this D, on this harvest functional, the paired experiment cannot tell them apart." With ε unspecified, you also cannot say they are decisionally equivalent. Two refusals. No winner crown. $P(\text{OPEN} \geq 37.58) \approx 0.667$ under the optimized candidates, on axis `MICRO_ENTITY`. Risk preference is unresolved.

Business materiality is unspecified. Accuracy of one object is not quality of the other: a GI rank that disagrees with an SI rank is not a bug in either engine. SI ranked unconstrained slightly above protected on expected harvest; GI reverses that sliver and then, in the paired experiment, mostly cannot see a sliver at all. Different objects.

A larger laboratory rerun with common streams moved the decimal estimates but did not reverse the qualitative ordering or the downstream decision. That is qualification provenance, not a recommended simulation depth. The teaching result is narrower: more draws can tighten evidence without minting a new act, and statistical resolution is not business materiality. Without a declared ε, preference remains `INSUFFICIENT_INFORMATION`.

The long-horizon tape remains an evidence gap. This chapter's six-period and twenty-eight-step APIs are worked illustrations. Do not generalize them to a year of daily occupancy without that tape.

18.3 Ingest adds an axis; it does not replace one

The public path is ingest a shortlist, evaluate it under paired worlds, then validate. Entity noise is an added axis. It does not replace parameter uncertainty. Two axes remain two axes. A later overlay that mixed them into one ribbon would not be conservative. It would be false.

The validator may print that the ranking is not a global optimum, and that preference is unresolved if nobody supplied one. Those fields exist so a later reader cannot launder the shortlist into a winner. If the paired evidence is missing, do not invent harvests. If someone asks which of two statistically adjacent means is "best" with no ε and no risk mode, say that materiality is unspecified and preference is unresolved. Those are answers. Inventing a winner is not. Means 40.29 versus 40.25, or expected-value versus deterministic at a paired difference of 0.00586, are the same kind of question: a number moved, and nobody has said whether the move is an act.

Two replenishment calendars validated on entity demand, two reviewer-staffing calendars validated on open claims, two clinic templates validated on patients still in stage: the same path, different nouns. Recast the nouns. Do not mint a second decimal table.

The three-frame picture already refused the readings that usually follow a table of means. A 95.8% tie is not a photo finish that needs a crown; it is unresolved

at this D without ε. GI rank differs from SI rank because micro and mean-field are different objects, and neither is wrong; accuracy of harvest is not quality of a floor. Eval-limited can win SI J and GI mean on this fixture and still be the wrong act if the act is the floor — look at $P(\text{OPEN} \geq \text{floor})$, not at the mean, when the floor is the decision. Calling GI from SQP puts validation in the inner loop: the search/validate split dies, and CRN identity typically dies with it. There is no arrow from the third frame back into the first.

If you have one candidate and you need expected occupancy, you do not need this chapter. You need Chapter 16. Validation is for a shortlist, under realizations, when the act cares about a law the mean cannot issue.

18.4 Feasible on the mean is not reliable in the worlds

The protected schedule sits just above the OPEN floor in expectation. About a third of the paired worlds miss it. Those sentences can both be true. They are not the same sentence (Charnes and Cooper 1959).

A two-point cartoon is enough. Remaining OPEN is 37.70 or 37.46 with equal probability, floor 37.58. The mean is 37.58, which can be dressed as feasible with a rounding, and half the worlds miss. Jensen bounds an *expectation of a concave payoff*. It does not turn a floor into a probability. The indicator of a mean is not the mean of an indicator.

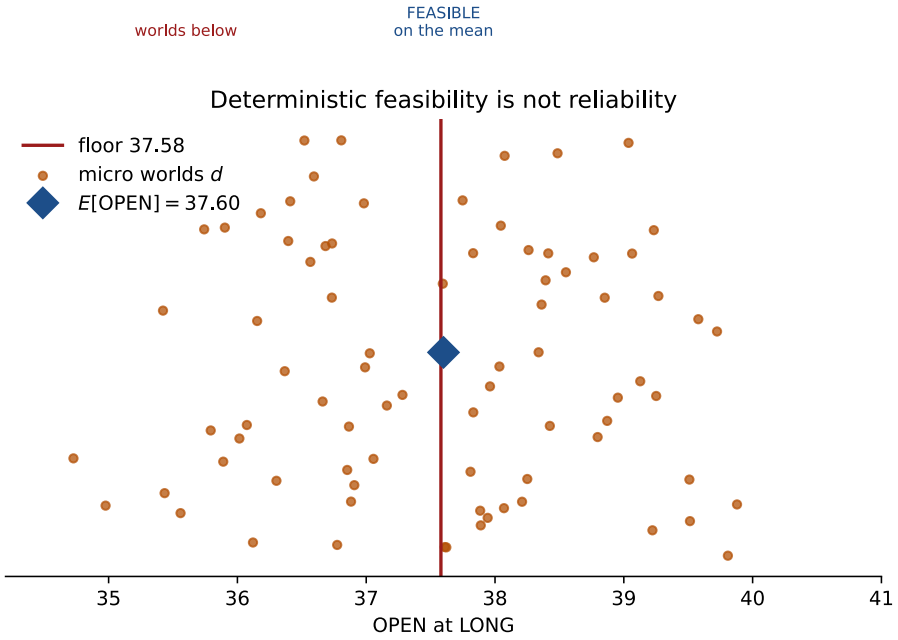
On the shortlist fixture, the protected local solution is feasible on the mean-field object and still misses the floor on about one in three micro worlds at $D = 24$. The unconstrained local solution is infeasible even on the mean. A schedule can be locally optimal, mean-field feasible, and unreliable on the chance object. If the act is the floor, look at the frequency, not at the mean. Do not invent a 33% spending buffer: that would be a preference smuggled in as arithmetic (Chapter 21).

18.5 Exercises

C18-E1. Conceptual. Why can SI rank and GI rank differ without either being wrong? What objects are being ranked?

C18-E2. Failure diagnosis. $P(\text{tie}) = 0.958$. What is the only honest label, and why is "they are the same policy" not it?

C18-E3. Interpretation. After GI ingest, why do `PARAMETER` and `MICRO_ENTITY` stay two axes? What would mixing them into one P claim?



TEACHING swarm; 0.667 frequency is the chapter's EMPIRICAL_EXAMPLE

Figure 18.2: Mean-field OPEN sits just above the floor. Micro worlds miss the floor about a third of the time.

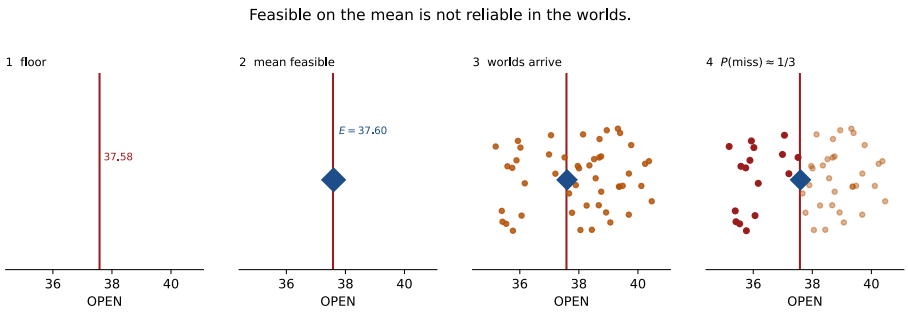


Figure 18.3: Four stills: the floor; the mean just above it; worlds arrive; about one third miss.

C18-E4. Failure diagnosis. Remaining OPEN is 37.70 or 37.46 with equal probability, floor 37.58. Is the mean feasible? What fraction of worlds miss?

C18-E5. Decision reasoning. Means 40.29 versus 40.25, no business tolerance declared. Can you pick a winner? Why is a t-test not a substitute?

C18-E6. Method selection. You have one candidate and need expected occupancy. Do you need this chapter?

C18-E7. Hostile. A schedule is feasible on the mean and misses the floor on one world in three. What buffer is *not* implied?

CHAPTER 19

Not All Uncertainty Is the Same

Depends on. Chapters 2, 4, 8, and 16–18.

19.1 Two intervals of width four

A clinic must staff next week before the waiting room is observed. Two vendors each return a 90% interval of width 4 around the same point: expected occupancy is 12 FTE, and the issued band is $[10, 14]$. Purchasing looks at the pictures and says they agree. They do not. The pictures have the same width and opposite implications for the act.

Call this pair a TEACHING_FIXTURE, not a field study. Forecast A is almost entirely aleatoric given rates that are already known. Arrival intensity, service time, and no-show probability have been identified from a long internal series; what remains is Poisson occupancy of a known-rate process. The interval says: in nine weeks out of ten, realized FTE demand will land inside four heads of the mean *even if nothing about the mechanism is still in doubt.* The justified act is to buy a buffer. Staff to 14, or keep a float pool of two. The lever is not the question. Noise is.

Forecast B issues the same $[10, 14]$, but almost all of the width is unidentified elasticity. Outreach spend is supposed to move arrivals; the data have never seen an experiment that identifies that map (Chapter 4). The interval is the image of an identified *set* Θ_I under a staffing function, not the Poisson scatter of a known θ. Staffing to 14 as if the lever were known treats a set as a point. The justified act is `PRIOR_ONLY`: do not lock a roster as though elasticity had been measured. If the act is irreversible — a union roster, a plant closure — refuse it. If it is reversible and the lever's materiality M is large relative to ε, the next justified move may be an experiment, not a tighter ribbon.

Decompose the width, or the pictures will keep lying. Write the 90% interval as a sum of contributions that are not commensurate and therefore must not be added. For Forecast A, almost all of the four FTE is Poisson occupancy given θ: if you could freeze the rates, next week's draw would still wander by about

two heads either side of 12. For Forecast B, almost all of the four FTE is the image of Θ_I. Freeze a selection $\widehat{\theta} \in \Theta_I$ and the remaining Poisson width might be a few tenths of an FTE; unfreeze the set and staffing demand ranges from "the campaign barely moves arrivals" to "the campaign doubles them." Adding those widths into one band is how purchasing concludes that the vendors agree.

Same width. Opposite acts. A single shaded band that does not name its axis cannot tell purchasing which sentence it is looking at. That is the subject of this chapter.

What it means to know something about the future is therefore not "to have a number," and not "to have a shaded interval." It is to issue, from an information set $\mathcal{I}_t$, a **named object** — a mean, a tail probability, a law, a set — under **named assumptions**, that would **change or refuse** an act. If the act is a threshold on Y, a change in $P(Y > \tau \mid \mathcal{I}_t)$ that exceeds the organization's materiality ε should move the act. It should not move the act when the difference lives inside ε; when authority is `PRIOR_ONLY` and the act needs causal identification; when the interval is on the wrong axis; or when every plausible statement, once resolved, issues the same α^*.

Chapter 2 already forbade mixing kinds of not-knowing. Chapter 18 showed that a mean-field feasible schedule can fail a floor on a third of micro worlds. This chapter is not a second four-axis lecture. It asks what to *do* after a miss, and when a sharper forecast is worth buying.

19.2 Four number lines, never one ribbon

The temptation is visual and almost irresistible: draw a point, shade a 90% band, call the band "uncertainty." The shade then quietly concatenates posterior θ, entity noise d, a declared scenario z, and last week's residual. That is not conservative. It is false. Each of those objects lives on its own line, with its own owner, and a quantile on one line is not a quantile on another.

Four axes, not one band (reprise of V2.2). Aleatoric d, epistemic b, scenario z, and structural residual are separate number lines. A vertical "90% interval" that crosses all four is a category error, not a conservative summary.

The four lines, and two cousins that are not lines at all:

The **aleatoric** line is $P(Y \mid u, \theta)$. Given a world θ and a control u, remaining randomness in the observable — entity Poisson, coin-flips of ADVANCE, who shows up on Tuesday. GenerativeIntelligence owns this line as micro draws d. A quantile here is a quantile of occupancy *given* θ.

The **epistemic** line is $P(\theta \mid \text{data}, \text{prior})$, or, when the likelihood does not inform a coordinate, a `PRIOR_ONLY` **set**. ResponseIntelligence and belief draws b live

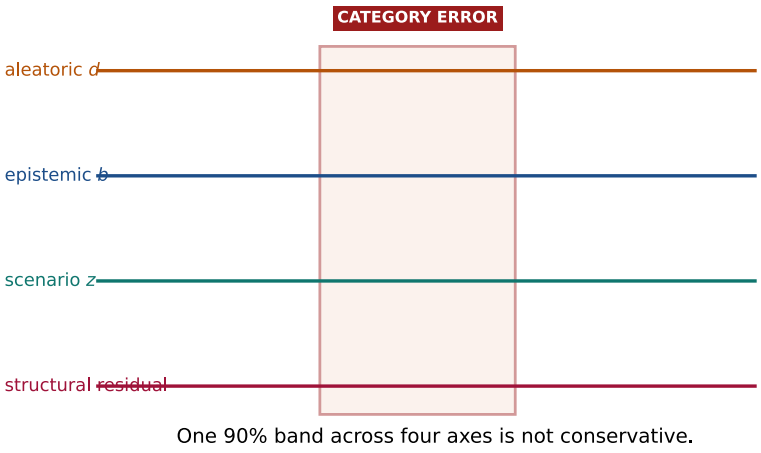


Figure 19.1: Four parallel number lines for aleatoric, epistemic, scenario, and residual uncertainty; a vertical band crossing all four is marked a category error.

here. A quantile here is a quantile of the *parameter*, not of next Tuesday. Collapsing the set to a posterior mean and then quoting a micro interval around that mean is how Forecast B impersonates Forecast A.

Identification is not a number line. It is the set Θ_I of θ compatible with the assumptions and the data (Manski 2003). A singleton is earned, not computed. Draws do not shrink Θ_I. Chapter 4 already said this; the operational insult is that a tight interval on b can be a costume of a wide set.

Structure F is the mechanism itself: which process, which lag, which competing risk. Model uncertainty is not sampling uncertainty. ScientificIntelligence tournaments and ForecastIntelligence challengers live here. A "quantile of models" is not a predictive interval.

The **scenario** line is a declared future z: "if supply is cut 20%." It is not a residual spike that someone later named COVID. Scenario replay (Chapter 11) holds F fixed and changes z. Treating last week's error as a scenario is a category error in the other direction.

The **operational residual** is $y - \hat{y}^{\text{base}}$ versus $y - \hat{y}^{\text{final}}$ after a correction has been applied. It is calibration of a published forecast, not a new z and not a new F. ForecastIntelligence assimilation owns this object. It is never an exogenous regressor.

`PRIOR_ONLY` is a set. Until a campaign has been run, FO's honest report on decision uncertainty is `UNAVAILABLE`. Absence of a campaign is not $\sigma = 0$.

Aleatoric versus epistemic, partial identification, model uncertainty versus sampling uncertainty: none of this is a system invention (Der Kiureghian and

Ditlevsen 2009; Manski 2003). The four-axis layout (aleatoric, epistemic, scenario, residual) and the refusal to add them into one ribbon are this book's composition. What the rest of the book insists on is that the axes remain *separately governed objects* through forecast, process, micro, and preference, and that mixing them is a refusal, not a darker shade of grey.

Uncertainty must survive the pipeline unmixed. A local SI solution is not allowed to arrive at the decision layer as a point with forgotten b. A GI ingest is not allowed to overwrite a PARAMETER sentence with a MICRO_ENTITY frequency. A LAST_ERROR patch is not allowed to pretend it is a scenario. Chapter 18 treats a different interchange — $\nabla E_\theta[J]$ versus $E_\theta \nabla J$ — and will not be asked to rescue a ribbon that already mixed the wrong objects.

19.3 A Champion Does Not Promote Itself

Public API: `fi_fit_forecast()`, `fi_publish_forecast_revision()`, `fi_adapt_revision()`, `fi_promote_forecast_challenger()`, and `predict()`.

The fixture is the qualified 80-day state-space example. A linear champion is fitted through Day 40. The remaining 40 observations are supplied in three different continuations.

19.3.1 1. Publish once

The fitted champion is frozen as a revision. Later observations do not train or silently mutate it. Initial and future forecasts use the same public prediction semantics.

19.3.2 2. Adapt three continuations

Later evidence	Mean residual	P(SHIFT)	Operational action
No shift	0.0255	0.0333	Keep model
Level shift +3	0.8102	0.9959	Recalibrate level
Slope change	0.0030	0.1460	Keep model

Later evidence	Challenger built?	Recommendation
No shift	No	Keep champion
Level shift +3	Yes	Promotion recommended
Slope change	Yes	Promotion recommended

The operational evidence is not one score. The no-shift path refuses unnecessary change. The level-shift path supports a 3.518260 level recalibration. The slope-change path demonstrates why regime action and forward challenger evidence remain separate fields: `KEEP_MODEL` is not identical to "no challenger can ever be competitive."

The public call stays small because the champion already contains its fitted state and lineage:

```
adapted <- fi_adapt_revision(
  champion = champion,
  later_data = later_data,
  error_assimilation = assimilation,
  request = request,
  classes = "LEVEL_RECALIBRATION"
)

adapted$diagnostics
adapted$regime_belief
adapted$comparison
adapted$recommendation
```

The method outputs remain inspectable. The call hides attachment and evidence routing, not the evidence itself.

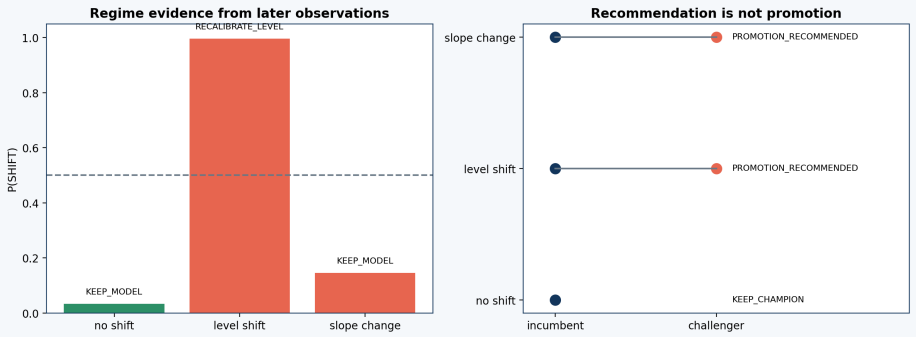


Figure 19.2: Actual ForecastIntelligence outputs separate regime evidence, challenger availability, and recommendation.

19.3.3 3. Recommendation is still not promotion

On the qualified level-shift lane, `fi_adapt_revision()` returns:

- diagnostics;
- regime belief;
- a challenger;
- four comparison rows;
- `PROMOTION_RECOMMENDED`;
- `automatic_promotion = FALSE`.

The champion is still incumbent. Promotion requires the champion, challenger, comparison, rationale, and expected fingerprints. Stale evidence fails before state changes.

19.3.4 4. Promote explicitly and replay

The level-recalibrated challenger is promoted through `fi_promote_forecast_challenger()`. The resulting revision stores the admitted adjustment. `predict()` applies it once. After serialization and reload, the forecast is identical without refitting.

Stage	Level delta	Promoted?	Replay equal?
Published champion	0.000000	no	
Adapted challenger	3.518260	no	
Explicitly promoted	3.518260	yes	
Serialized/reloaded revision	3.518260	yes	**yes**

The promoted two-step forecast is:

Horizon	Prediction	5%	Median	95%
1	3.4939	2.5070	3.4628	4.6235
2	3.6079	2.6209	3.4993	4.7374

Point forecast and quantiles move with the admitted revision. The stored level shift is not added again during replay.

That sequence is the governance contract:

published champion $\rightarrow$ new observations $\rightarrow$ evidence $\rightarrow$ challenger $\rightarrow$ recommendation $\not\Rightarrow$ promotion.

Only the explicit governed call crosses the final arrow.

Executable source: `tools/capture_adaptive_lifecycle.R`.

Captured evidence: `arithmetic/adaptive_lifecycle_lanes.csv`, `adaptive_promotion_trace.csv`, `adaptive_winning_forecast.csv`, and serialized replay artifact.

19.4 Crossed parameter and micro

The strongest remaining analytical limitation in this stack is a fully crossed $Y_{b,d}$. The two axes mean different things, and the complete predictive object for an act is their mixture.

Let b index epistemic or parameter worlds and d index aleatoric realizations given a world. Then

$$P(Y \in \cdot \mid u, \text{data}) = \int P(Y \in \cdot \mid u, \theta)\, P(\theta \mid \text{data})\, d\theta,$$

which simulation approximates by a grid $\{Y_{b,d}\}$. The inner law is GI's object given frozen θ. The outer mixing measure is a posterior, a belief, or — when identification fails — a set. They are not substitutes.

Collapsing the grid is dangerous in both directions. Evaluating micro only at $\widehat{\theta}$, or at one declared posterior row, understates tails whenever a bad θ and a bad d co-occur: ruin lives in a corner of the rectangle that a single slice never visits. Evaluating only $E_d[Y \mid b]$ and then mixing b can hide chance-constraint violations that live entirely in d: the mean of each row clears the floor and a third of the cells in that row do not (Chapter 18). The two collapses fail oppositely. Neither is a conservative version of the other.

Crossed $Y_{b,d}$. Rows are parameter worlds b, columns are micro realizations d, color is the outcome. A horizontal slice is the qualified GI path: honest *given* one θ. The full rectangle is the mixture. Ruin cells in a corner that a slice misses are why unconditional posterior reliability is `UNAVAILABLE`.

Conditional micro-validation is sometimes sufficient. If $P(\theta \mid \text{data})$ is tight, or the act depends on $E[Y \mid \theta]$ nearly linearly, or ε is large relative to both axes, a declared row plus the MICRO axis can be the right object. A larger laboratory rerun did not reverse the decision-layer rankings. That is fixture evidence, not a theorem. Full crossing matters when the act is a chance constraint, when θ is weakly identified, when N is small, or when Φ is nonlinear — floors, maxima, ruin.

Nested or adaptive sampling can approximate the rectangle: screen b, deep-sample d on the remaining b, or run inner–outer Monte Carlo. Naive cost scales

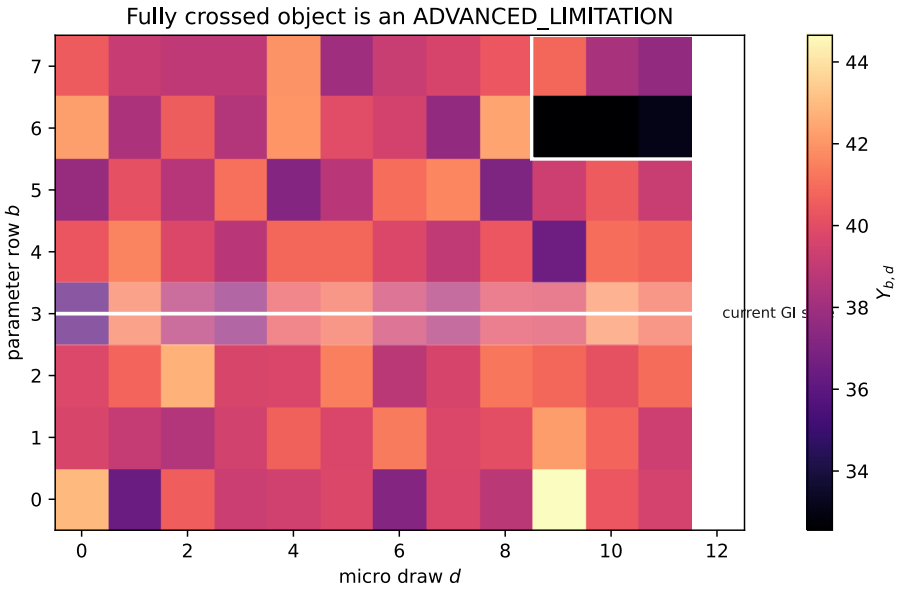


Figure 19.3: Heatmap of outcome on a grid of parameter worlds by micro realizations; a single horizontal slice is the current GI path, and ruin can sit in a corner the slice never visits.

as $|B| \times |D|$. Nested schemes trade bias in the mixture for cost. They do not change the estimand. They are not a new named primitive in this book.

Conditional on a declared θ-row, micro frequencies are honest. Unconditional reliability over the posterior,

$$P(J \geq \tau \mid \text{data}),$$

issued from GI at one posterior row, is `UNAVAILABLE`. That sentence is load-bearing. It is also why $Y_{b,d}$ is an **advanced limitation**, not a book-blocking gap and not a stealth modeling wave. The book can state the mixture, refuse the unconditional claim, and teach the conditional architecture that is qualified. Composition of existing exports can already produce honest *conditional* evidence; that is why the limitation is advanced rather than an unbounded missing function.

Could a campaign over parameter worlds, nested with a shortlist validation under entity worlds, already *be* the crossed object? Only if, for each retained parameter, a full paired validation were run on that frozen θ-row, and the decision layer mixed the resulting laws. That nest is conceivable. It is not a closed path in this book, has no shared-draw contract across the nest, and would be $|B| \times |D|$ compute. Conceivable composition is not a shipped capability. It is why the limitation is advanced rather than a missing named function. It is also why we still must not write $P(J \geq \tau \mid \text{data})$ as if the entity validator already mixed b.

When does model uncertainty dominate micro, and when the reverse? A numerical sketch, not a theorem. Let outreach elasticity range over a `PRIOR_ONLY` interval $[0.1, 2.0]$. Two spend schedules reverse their ranking of expected harvest as b moves across that interval, while entity Poisson at any frozen θ moves harvest by a fraction of a file. Model uncertainty dominates; more micro draws will not settle the ranking. Conversely, let θ be tightly identified and the act be $P(\text{OPEN} \geq 37.58) \geq 0.9$. Then the chance lives in d: Chapter 18's micro frequencies at frozen θ are the right object, and a campaign over b is almost idle. One ribbon cannot tell those problems apart.

19.5 When information has value — and when it does not

Uncertainty is not only a tax. If resolving an unknown would change the preferred act by more than ε, the information has value. That sentence is older than this book (Raiffa 1968; Howard 1966). It is the right next question after

the axes have been named, and it is the wrong question to smuggle into a control solver.

Let α^* be the act under current information, and $\alpha^*(s)$ the act that would be preferred if a signal s arrived. The expected value of perfect information is the expected improvement from observing the unknown itself. The expected value of sample information is the same object for a noisy experiment. If $\alpha^*(s) = \alpha^*$ for every signal in the support of s, then EVSI $= 0$ even if the forecast gets sharper. Intervals can shrink by half and the act does not move. The extra accuracy is scientifically real and decisionally worthless.

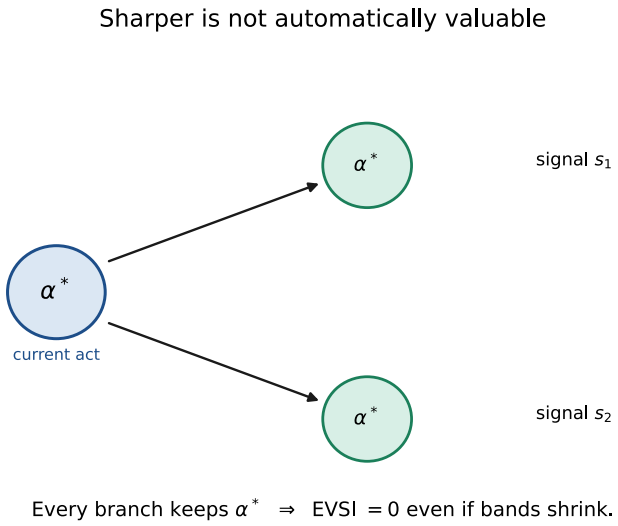


Figure 19.4: A decision tree in which every signal leaves the current act unchanged, so the expected value of sample information is zero even as the forecast band shrinks.

Value of information can be zero. The current act is α^*. Signals s_1 and s_2 branch. If every branch keeps α^*, EVSI $= 0$ even if the predictive interval shrinks. Sharper is not automatically valuable.

A two-act, two-state table where each act is best in one state. States Low and High, equal prior 1/2. Payoffs in the same units as J:

	Low	High	prior expected
staff 12	40	10	25
staff 14	38	36	37

Staff 14 is the current act: $37 > 25$. If you learn Low, you switch to 12 ($40 > 38$). If you learn High, you stay at 14 ($36 > 10$). The expected payoff with the signal

is $0.5 \cdot 40 + 0.5 \cdot 36 = 38$. EVPI is $38 - 37 = 1$. Variance of occupancy can still be large either way. The value is the *switch*. A table in which one act dominates both states has EVPI of zero, and that is a different sentence: you already know the act. Chapter 20 treats the same interchange when a control must be committed before θ is seen. This chapter only insists that a sharper interval is not, by itself, a reason to buy the model.

Connect this to the rest of the book without turning every problem into reinforcement learning. Experimentation earns its keep when identification is `PRIOR_ONLY` and M of the lever is large: Forecast B, not Forecast A. Sequential learning and bandits are for when the act *is* the experiment and regret is the loss; that is a different problem (Sutton–Barto), and it must not be smuggled into SI control. Challenger evaluation is value of information against *model* uncertainty: is a new F worth promoting, given ε? Weakly identified response is the hostile twin: more data in the unidentified region may have no value of information at all, because it does not shrink Θ_I. Decision sensitivity is the other twin: if ε is large, even a perfect experiment can have EVSI $= 0$.

Do not build a second value-of-information engine inside the decision layer. Units must match the objective; there is no universal money. A hostile case in the other direction: policy C is statistically noisier than A and B. A risk-neutral mean ranking picks A. If C's tail would flip a floor decision, the value of more draws — or of more data — can exceed the value of "picking a winner now." Chapter 21 will put ε and declared risk on that sentence. This chapter only insists that a sharper forecast is not, by itself, a reason to act, and that a wide forecast is not, by itself, a reason to wait.

19.6 Exercises

C19-E1. Mathematical. Write $P(Y \mid u, \theta)$ versus $P(\theta \mid \text{data})$ versus an identification set. Which of the three is Forecast A's interval? Which is Forecast B's?

C19-E2. Failure diagnosis. A $\pm 2\hat{\sigma}$ LAST_ERROR ribbon is drawn on an SI occupancy forecast. Name two distinct errors in that picture.

C19-E3. Interpretation. High $P(\text{SHIFT})$ after a level shift is not `REFIT_STRUCTURAL_MODEL`. Why not?

C19-E4. Method selection. When does model uncertainty dominate micro? When the reverse? Give a numerical sketch.

C19-E5. Decision reasoning. If every signal leaves the same act, EVSI is?

C19-E6. Assumption criticism. Is $Y_{b,d}$ a book-blocking gap or an advanced limitation? Argue in six sentences.

C19-E7. Method. After a miss, does attaching later data to the champion train a new model, or issue a forecast from the same revision?

C19-E8. Hostile. Apparent calibration δ with $\mathrm{corr}(e, x) \neq 0$. Correction or challenger?

C19-E9. VOI. Using the Low/High table, compute EVPI. Which act is current? When do you switch?

C19-E10. VOI. Change the table so one act is better in both states. What is EVPI then, and why is that uninteresting for buying a signal?

CHAPTER 20

Deciding Before and After Information

A hospital must reserve overflow beds before tomorrow's demand regime is known. It may reserve zero, one, or two units. Demand will be low or high with equal probability.

Reserved units	Low-demand payoff	High-demand payoff
0	0	-40
1	-5	-10
2	-20	0

Payoffs are negative costs, so larger is better.

20.1 The decision made today

Before demand is observed, one action must work in both worlds. Its expected payoff is

Reserved units	Expected payoff
0	-20.0
1	-7.5
2	-10.0

The best fixed decision is one unit, with expected payoff -7.5:

$$\max_u E_\theta[J(u,\theta)] = -7.5.$$

The maximization is outside the expectation because the hospital chooses before θ is known.

20.2 The decision made after learning

Now imagine perfect information arrives before the reservation. Under low demand, choose zero units for payoff 0. Under high demand, choose two units for payoff 0. Thus

$$E_\theta[\max_u J(u, \theta)] = 0.$$

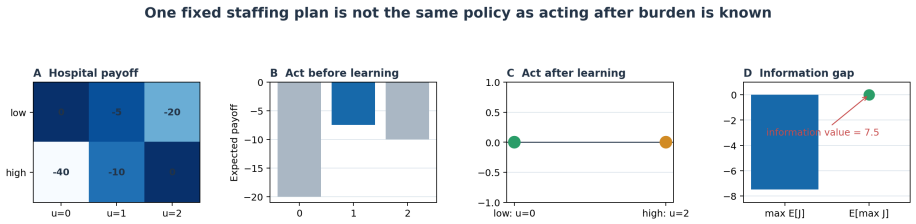


Figure 20.1: Four panels show the same policy evaluated across uncertainty worlds, the best fixed policy, the impossible world-by-world oracle, and the value of information.

The 7.5-point gap is the expected value of perfect information. It is not free performance. It is the value of changing when the decision can use information (Raiffa 1968; Howard 1966).

This is why

$$\max_u E_\theta[J(u, \theta)] \neq E_\theta[\max_u J(u, \theta)]$$

in general. The right side allows a different action in each world. Using it to describe a decision that must be made today gives the decision-maker impossible foresight.

20.3 Averaging the World Changes the Decision

Demand is 4 with probability 0.8 and 14 with probability 0.2. Mean demand is 6. The hospital can reserve capacity 4, 8, or 12. Reservation costs 1.2 per unit; shortage costs 5 per missing unit.

20.3.1 Wrong operation: optimize at mean demand

Capacity	Cost when demand is replaced by 6
4	14.8
8	**9.6**
12	14.4

The mean-parameter shortcut chooses capacity 8.

20.3.2 Correct operation: optimize expected consequence

Evaluate each action in both coherent worlds, then average costs:

Capacity	Low-world cost	High-world cost	Posterior expected cost
4	4.8	54.8	**14.8**
8	9.6	39.6	15.6
12	14.4	24.4	16.4

Expected-cost optimization chooses capacity 4. The operations differ because shortage cost is nonlinear at the capacity boundary:

$$\arg\min_u C(u, E[\theta]) = 8, \qquad \arg\min_u E[C(u, \theta)] = 4.$$

Squared error is the useful special case: its optimizer may remain at the mean even though expected loss includes variance. The special case does not license mean substitution for thresholds, asymmetry, constraints, or nonlinear state transitions.

20.4 One policy across coherent worlds

Posterior-aware optimization evaluates each candidate policy against the same sampled worlds:

World	Probability	Policy P payoff	Policy Q payoff
Mild demand	0.50	8	5
Severe demand	0.50	-12	3

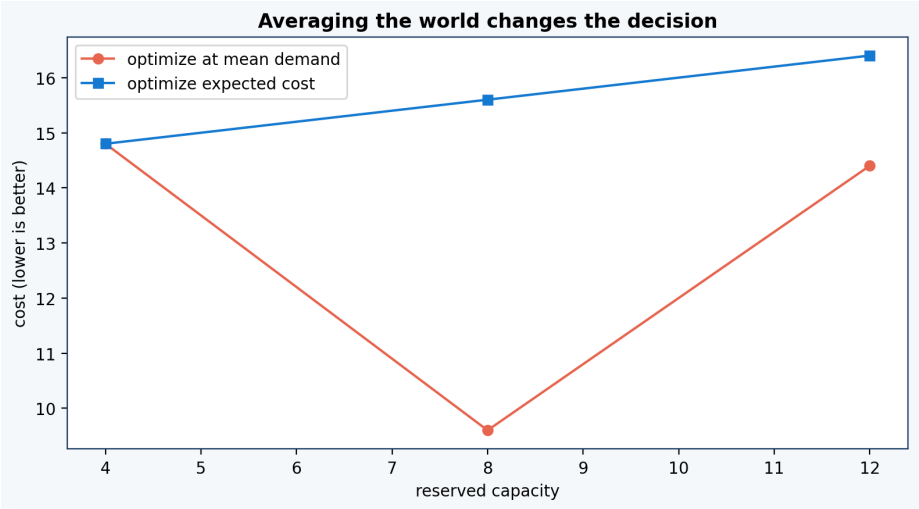


Figure 20.2: Replacing demand by its mean chooses capacity 8; averaging consequences across demand worlds chooses capacity 4.

Policy P has expected payoff −2; Q has expected payoff 4. P wins in the mild world but fails badly in the severe world. Because both policies faced the same worlds, their difference is attributable to policy rather than Monte Carlo luck.

Different decision rules can now inspect the same consequence distribution:

- expected value asks for the highest mean payoff;
- downside criteria ask how bad adverse worlds become;
- chance constraints ask how often a threshold fails;
- robust rules ask how a policy performs under a declared ambiguity set (Rockafellar and Uryasev 2000; Ben-Tal et al. 2009).

These are distinct preferences over shared evidence, not interchangeable labels.

20.5 The Action Cannot Know Which Branch Comes Later

Before a signal arrives, the hospital chooses first-stage reserve (a_0 {0,1}). After a calm or stress signal, it may add (a_1 {0,1}). Terminal demand is either 0 or 2. Reserve costs 3 initially, an added unit costs 4, and shortage costs 10 per unit.

The four worlds are:

World	Signal	Demand	Probability
calm-low	calm	0	0.40
calm-high	calm	2	0.10
stress-low	stress	0	0.10
stress-high	stress	2	0.40

20.5.1 Legal policy

Before the signal, all worlds share the same history. They must share (a_0). Enumerating the eight legal policies gives the minimum:

$$a_0 = 1, \qquad a_{1,\text{calm}} = 0, \qquad a_{1,\text{stress}} = 1,$$

with expected cost 6.0.

20.5.2 Illegal clairvoyant schedule

If the optimizer is allowed to choose (a_0) separately after seeing each terminal world, it chooses no initial reserve in low-demand worlds and initial reserve in high-demand worlds. Expected cost falls to 3.5.

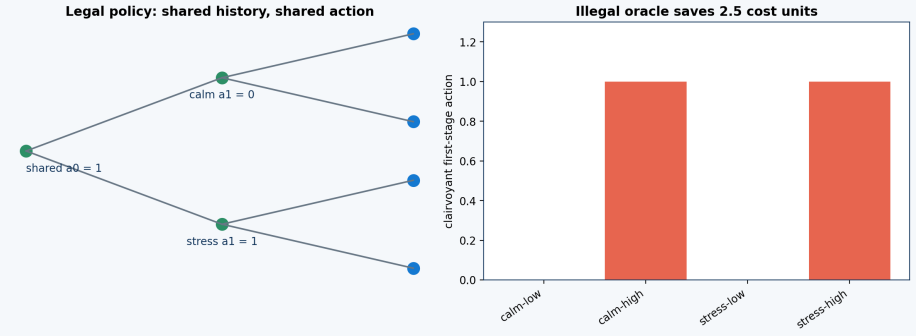


Figure 20.3: The legal tree shares its first action; the clairvoyant solution changes that action using information that does not yet exist.

The apparent 2.5-unit improvement is the value of impossible foresight, not a better implementable policy. Information arriving at Stage 2 can legitimately change (a_1). It cannot travel backward and alter (a_0).

The audit is mechanical: if two scenario-tree nodes share the same information history, their actions must be equal. Any reported advantage obtained by breaking that equality is anticipative bias.

20.6 From mathematical optimum to governed decision

A posterior mean is not a decision. Retain, for each policy:

1. its full consequence distribution under common-reference worlds;
2. feasibility and support checks;
3. downside and threshold behavior;
4. the information available when each action is chosen;
5. the explicit decision rule used to prefer it.

When policies are materially equivalent, say so. When uncertainty changes the ranking, expose the disagreement. When a recommendation depends on information that will not exist at action time, reject it.

20.7 Exercises

C20-E1. Recompute all three expected payoffs in the hospital example.

C20-E2. Compute the best action in each demand state and the expected value with perfect information.

C20-E3. State in words why the two optimization/expectation expressions differ.

C20-E4. If high demand has probability 0.10, recompute the best fixed reservation.

C20-E5. At probability 0.10, compute the expected value of perfect information.

C20-E6. Show algebraically why expected squared error includes $\mathrm{Var}(\theta)$.

C20-E7. Give a nonlinear loss for which optimizing at the mean parameter can select a different action.

C20-E8. Compute the expected payoffs of policies P and Q.

C20-E9. If the organization requires payoff at least zero in every displayed world, which of P and Q is admissible?

C20-E10. Explain why evaluating P and Q on independent simulation draws is inefficient for comparison.

C20-E11. Give one example of information in I_t and one future quantity that must not enter u_t.

C20-E12. Two scenario-tree worlds are identical through Week 2 and diverge in Week 3. What equality must their Week 2 actions satisfy?

C20-E13. Write a short decision note recommending one hospital reservation while stating the information timing and uncertainty limitation.

CHAPTER 21

Materiality, Risk, and the Decision Frontier

Depends on. Chapters 8 and 18–20.

21.1 When a statement should change an act

A forecast is a statement from an information set (Chapter 2). Static allocation is a different object from a process (Chapters 5–8 versus 10–17). Expectation is not a floor (Chapter 17). Search is not validation (Chapter 18). Axes do not mix (Chapter 19). Error correction is not regime change.

The last question those distinctions were for is not "which package wins?" It is:

> When should a statement about the future change an act? When should it not? When must a machine refuse, so that a human still owns the preference?

That is decision-conditioned forecasting in Chapter 2's taxonomy: the issued object is a *functional chosen for an act*, plus the right to issue no act. Chapter 8 already required that preference be declared or unresolved, and already forbade averaging maps before optimizing. Chapter 20, the interchange chapter, will treat $\nabla E_\theta[J]$ versus $E_\theta \nabla J$ as mathematics. This chapter owns what those objects are *for*: business materiality ε, declared risk, bandwidth as an assumption rather than a default, value of information as a reason to wait, and refusal as a scientific output.

The modeling layer estimates consequences and uncertainty. It does not contain a unique "best." A sure 40.0 versus a 40.2 with a 5% chance of 10 is not a scientific controversy. Risk-neutral preference may pick the second; downside-protected preference the first. Those are different people. The machine does not get to pick which person you are. Inventing risk-neutral because it is easy is how quadratic penalties and default bandwidths impersonate ethics.

> Lower variance is not automatically better.

If no risk mode is supplied, `RISK_PREFERENCE_UNRESOLVED` is a legal, scientific answer. It is not a bug, and it is not a missing row in a ranking table.

The vocabulary of the book, restated here because this is where the words can no longer be collapsed: a **prediction** is not an act. An **intervention** is a change to a lever, not yet justified. **Control** is a path through F_t. **Allocation** is a static portfolio. **Optimization** is search on a stated problem. A **recommendation** is search plus validation plus preference plus authority. A **decision** is an authorized act. A mathematically local-optimal control is not a recommendation. A recommendation is not an authorized action.

21.2 ε is business materiality, not a t-statistic

Two staffing plans that differ by two minutes of expected wait are not a decision if the organization already called two minutes noise. Two plans that differ by two minutes *are* a decision if the organization called one minute material. ε is that sentence, written as a number. It is not the standard error of a Monte Carlo.

For a maximize-sense objective,

$$B \approx_\varepsilon A \iff J(B) \geq J(A) - \varepsilon.$$

Nearness is not symmetry of indifference once several objectives, or several senses, are in play: each coordinate can carry its own ε_k. The near set is then a rectangle in consequence space, not a point. Statistical equivalence testing (TOST) is a cousin and **not the same object** (Schuirmann 1987). A statistical ε is a tolerance on a *parameter* under a sampling model. A business ε is a tolerance on *consequences* under an act. Using one as the other is how a t-statistic impersonates a staffing policy.

The standard error **cannot** be ε. The public constructor `aq_opt_epsilon_spec` rejects that substitution. The standard error **can** block a difference claim: on the qualification fixture, expected-value minus deterministic harvest was 0.00586, and the paired 95% interval $[-0.00076, 0.01248]$ included zero (Chapter 18). That is statistics saying "not resolved at this D." It is not a business tolerance, and it does not become one by being typed into the ε slot.

Without ε, the near set *is* the mathematical set, and there is **no practical equivalence**. Two forecasts can differ statistically and not materially. Two can differ materially with overlapping confidence intervals. Those are different sentences. Collapsing them into "significant, therefore act" or "overlapping, therefore ignore" is the crime this section exists to name.

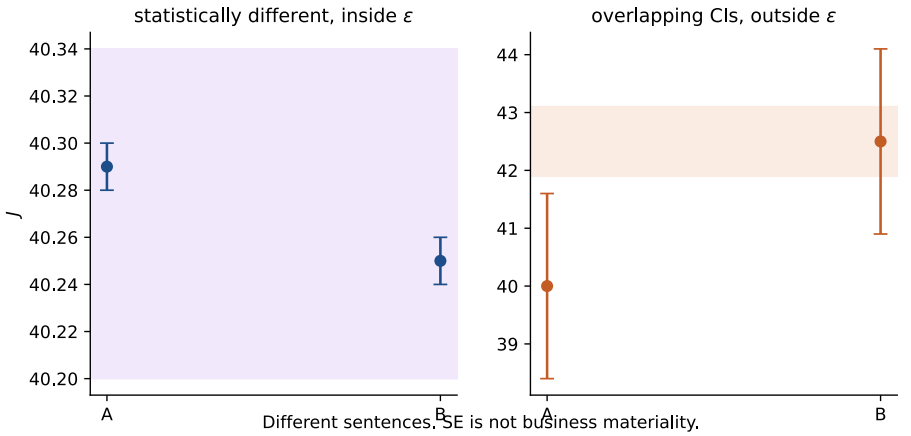


Figure 21.1: Two pairs of consequence estimates: statistically resolved but inside materiality on the left; overlapping intervals but a material gap on the right.

Statistical difference and material difference are different sentences. Left: plans A and B, expected wait 40.22 versus 40.28 minutes. A paired experiment resolves the gap — the intervals do not overlap — and the organization already called half a minute noise ($\varepsilon = 0.50$). Known, and not worth acting on. Right: harvest 40.1 versus 42.9, wide overlapping intervals, $\varepsilon = 0.60$ on the harvest scale. The *point* gap is outside ε, so if the difference is real it is material; the intervals still overlap, so it is not yet known. "Not statistically resolved" is not "not worth acting on," and it is not "act now."

The left pair is Forecast-A thinking applied to preference: the science is sharp and the act should not move. Staff 12 or staff 12.1 is not a decision if the organization already called half a minute noise, no matter how small the Monte Carlo error is. The right pair is the opposite: the act *would* move if the science were sharp, and the honest output is not a winner. It is "buy more draws, or declare that you will treat the point gap as real." Those are different human sentences. The machine may not pick one. Statistical significance is not materiality; overlapping intervals are not permission to ignore a gap the organization already called large.

21.3 Overlay order: $\mathcal{F}$, then near, then risk

Preference has to speak in a definite order, or it becomes a hidden quadratic.

The mathematical frontier $\mathcal{F}$ of a *supplied* set is the undominated subset (Chapter 8). It is not a global optimum of an unenumerated universe, and

`globally_optimal` is hard-false on that object. The ε-near set $\mathcal{N}_\varepsilon$ is the subset of supplied states that are within ε of $\mathcal{F}$ in the declared sense. That construction is this book's overlay, not the ε-constraint method of multiobjective NLP (Miettinen 1999), and not TOST. Risk preference — risk-neutral, reliability-first, downside-protected, upside-seeking, robust over a declared set, or a custom functional — is then applied **inside** $\mathcal{N}_\varepsilon$.

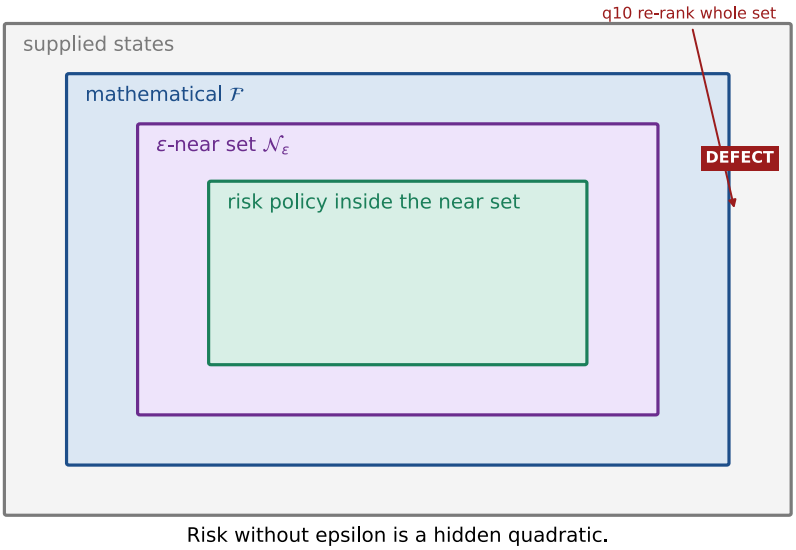


Figure 21.2: Nested sets from supplied states to the mathematical frontier to the epsilon-near set to a risk policy inside the near set; a side door that re-ranks the whole set by a tail quantile is marked a defect.

Overlay order (reprise of V8.2). Supplied states, then mathematical $\mathcal{F}$, then $\mathcal{N}_\varepsilon$, then a declared risk policy *inside* the near set. A side door that q10-re-ranks the whole set is a defect: risk without ε can pick a "safer" point that is materially worse in expectation.

Why this order. Risk without ε can select a safer-looking point that is worse than $\mathcal{F}$ by more than the organization was willing to pay for safety — a hidden quadratic penalty. ε without risk leaves a set, which is honest. Risk *inside* the near set is how preference speaks only among acts the user already called close enough. Whole-set re-ranking of a campaign by a tail quantile is a documented limitation of the campaign object, not the philosophy. Prefer the decision-layer overlay when comparing supplied states.

The named risk modes are **inputs**, not discoveries. Twenty equally likely weeks. Plan Sure pays 40 every week. Plan Gamble pays 42 in nineteen weeks and 10 in one floor-miss week.

$$E[\text{Sure}] = 40, \qquad E[\text{Gamble}] = \tfrac{19}{20} \cdot 42 + \tfrac{1}{20} \cdot 10 = 40.4.$$

Risk-neutral, maximizing expected payoff, picks Gamble. Conditional value-at-risk at 5% is the mean of the worst 5% of weeks — here, the single miss week (Rockafellar and Uryasev 2000):

$$\text{CVaR}_{0.05}(\text{Sure}) = 40, \qquad \text{CVaR}_{0.05}(\text{Gamble}) = 10.$$

Downside-protected preference picks Sure. Those are not two scientific readings of the same table. They are two organizations. The pick *moved*. A chance constraint $P(\text{miss}) \leq 0.02$ would also pick Sure (Charnes and Cooper 1959); Gamble misses in $1/20 = 0.05$ of weeks. If nobody will say which person they are, stop. Inventing expected-value because the ranking table looks empty is not a default. It is a preference.

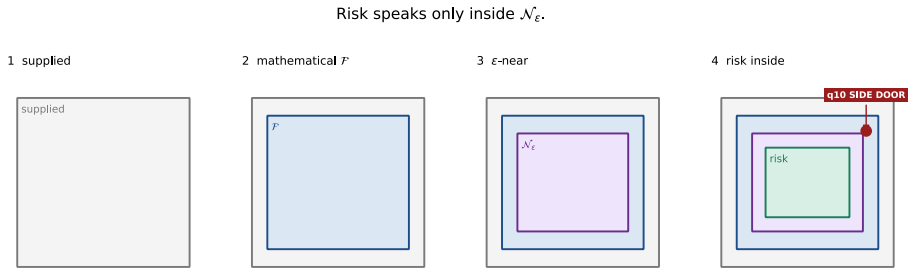


Figure 21.3: Four stills: supplied states, mathematical frontier, near set, risk only inside the near set.

21.4 A numerical h that changed the problem

A parameter that looks numerical can change **what the problem is**. SI reliability generation used a smooth empirical surrogate, not an exact chance constraint. Bandwidth h is **decision regularization**. It is not "a kernel bandwidth the user can ignore."

The hardening grid below is the same posterior, the same process, the same target, the same solver. The axis is PARAMETER-style evaluation of the retained posterior, not GI micro. Do not paste a MICRO frequency such as Chapter 18's floor probability into this table and call it the same P. Name every reliability sentence with **axis, D, target, h, Monte Carlo precision**.

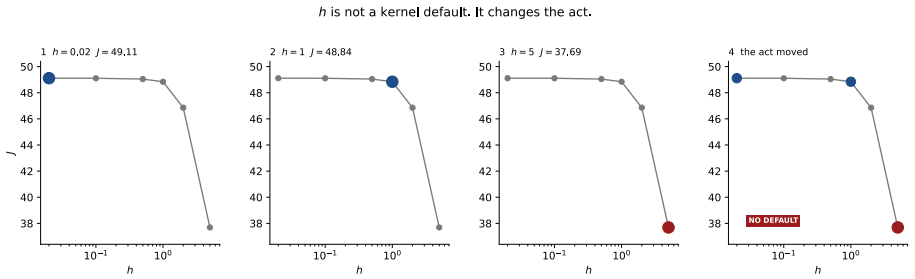


Figure 21.4: Three bandwidths on the same problem: $h = 0.02$, $h = 1$, $h = 5$. The act moves.

h	Finite-draw reliability	Full-160 expected J	L1 from $h = 1$
0.02	0.75	49.1132	0.0895
0.10	1.00	49.1071	0.0861
0.50	1.00	49.0465	0.0579
1.00	1.00	48.8412	0.0000
2.00	1.00	46.8582	0.0755
5.00	1.00	37.6898	0.2178

Read the spine of the table as three rows if the others are noise: at $h = 0.02$, expected $J = 49.1132$; at $h = 1.00$, $J = 48.8412$; at $h = 5.00$, $J = 37.6898$. Moving h from 1 to 5 moved expected objective from 48.84 to 37.69. That is not a rounding error. It is a different act. Sensitivity analysis exposed it. **No automatic default was invented** — that would have been a false authority. An unresolved explicit h is more scientifically honest than a convenient default sitting on the $h = 1$ row because the table has a row there.

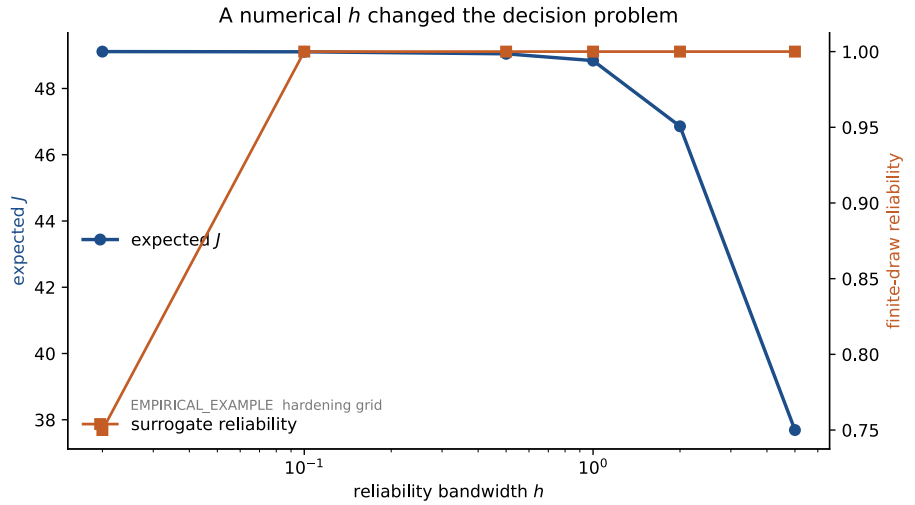


Figure 21.5: Expected objective and surrogate reliability as functions of bandwidth h; a numerical h moves the decision problem, and no default is marked recommended.

Reliability bandwidth as a hidden assumption. Horizontal axis h; expected J falls from about 49 to about 38 as h goes from a tenth to five. Surrogate reliability saturates. The slider moves the *problem*, not just a number. No tick is marked "recommended."

is only 0.75, expected J is 49.1132, and the solver is still trying to sit on a knife-edge. At $h = 1$ the chance has been smoothed; reliability saturates at 1.00 on the finite-draw object and expected J has moved only to 48.8412 — inside a synthetic $\varepsilon = 0.60$, not a reason to change the act by itself. At $h = 5$ the surrogate is so smooth that the "reliability" problem has become a different, much more conservative harvest problem: $J = 37.6898$. That is below 48.84 by more than eleven units. No organization that called 0.60 material on harvest can treat $h = 5$ and $h = 1$ as the same question.

Why this is a pedagogical case, not a modeling wave: the *estimand* changed with h; the *act* moved materially; the honest API is to require a user or data-driven bandwidth policy; the dishonest API is to pick $h = 1$ because it is the middle of a table. The supported claim is that a reliability surrogate requires an explicit bandwidth policy. Not a universal h^*. An automatic bandwidth rule would close a genuine gap by minting a false authority. The book teaches refusal instead.

21.5 Hardening moved the numbers and did not reverse the decisions

After GI evidence is ingested, the decision layer is asked the same overlay questions again. On the qualification fixture the answers were:

User policy	Pre-GI	Post-GI	Reading
No preference	unresolved	unresolved	GI did not mint a preference
Risk-neutral	deterministic	deterministic	Confirmed
Reliability-first	fail-closed (no common parameter reliability metric)	DET/EV/robust tie	Equivalence, not a winner
Robust	deterministic	deterministic	Confirmed

Authority throughout is `EXPERIMENT_CONDITIONED_OPTIMIZATION`. GI confirmation does not upgrade SI local claims to global or causal. Simulation does not promote `PRIOR_ONLY`. Common-depth GI **changed numbers**, and the compared policies remained materially tied on that laboratory fixture, **without reversing** these decisions. The pipeline did what a validation layer is for: it moved the decimals, and it left unresolved preference unresolved. A risk-neutral user

who had already picked the deterministic candidate still had that candidate. A user who had supplied no preference still had no preference. Reliability-first, once a common metric existed, produced a tie among DET, EV, and robust — equivalence inside the overlay, not a winner minted because the table had three names. That is the point of ε and unresolved risk: evidence can move without the system impersonating an organization.

The $\varepsilon = 0.60$ that appears in qualification is `synthetic_user_epsilon`. It is a flag on a laboratory tolerance, so that a later reader does not take 0.60 as a recommended harvest materiality. Do not recommend it. Do not copy it into a live staffing rule. Different consequence horizons fail closed rather than silently sharing one ε.

The public construction is the overlay in that order, with the synthetic flag when the tolerance came from a script rather than from an organization. If no risk policy is supplied, do not invent risk-neutral. If ε is omitted, there is no practical equivalence to report. The mathematical frontier of a supplied set is still not a global optimum of an unenumerated universe.

21.6 Refusal, information, and the human remainder

If all signals leave α^* unchanged, EVSI = 0 even if intervals shrink (Chapter 19) (Howard 1966). Experimentation is for when identification is weak *and* the lever's M is large. Challenger promotion is value of information on **model** uncertainty. Bandits are for when the act *is* the experiment — not a synonym for forecasting, and not an inner loop of SI.

There are situations in which the system should refuse to decide, and the refusal *is* the result. No risk mode: `RISK_PREFERENCE_UNRESOLVED`. No ε, and someone asks whether two plans are equivalent: there is no practical equivalence, and materiality is `INSUFFICIENT_INFORMATION` — the left pair in this chapter's statistical-versus-material figure cannot even be classified as "inside ε" until someone writes the sentence. Reliability asked, no h policy: refuse, or require an explicit policy; the bandwidth table above is what happens if a default is invented instead. `PRIOR_ONLY` driving an irreversible act: do not promote authority; Forecast B of Chapter 19 is still Forecast B after a campaign. Unconditional posterior reliability from one θ-row: `UNAVAILABLE` (Chapter 19). No defensible occupancy x: `INSUFFICIENT_EVIDENCE` for a process sentence (Chapter 22). A target outside trusted support: infeasible, not extrapolated.

The machine may estimate J, $\mathcal{F}$, micro tails, and M. The human still owns the rest, and the rest is not a footnote. Someone has to write ε as a business sentence. Someone has to be the person who prefers Sure to Gamble, or the reverse. Someone has to own h if reliability is used, because $h = 5$ versus

$h = 1$ moved the act. Someone has to say whether `PRIOR_ONLY` may drive an irreversible lever. Someone has to decide whether to buy information, whether to promote a challenger, and whether to decide at all. Saying that preference is unresolved, or that the evidence is insufficient, is how the book teaches that remainder rather than inventing a winner.

Crossed $Y_{b,d}$ remains an advanced limitation (Chapter 19): do not claim unconditional posterior reliability; do not block the book; do not silently implement a new primitive in a chapter. Chance constraints as user constraints, and the surrogate h that can impersonate them, are this chapter's object; Chapter 18 already pointed here and must not invent a default bandwidth. Chapter 15 must not call the decision frontier a solver output: local constrained, eval-limited, and best-validated-among-supplied are three sentences, and they are not this overlay. Chapter 20 treats interchange of gradient and expectation; it is not a second overlay tutorial.

The book's philosophy of forecasting is not "better models, better futures." It is: **name the object, name the information, name the uncertainty, name the act, and refuse when any of those is missing.**

21.7 Exercises

C21-E1. Mathematical. Prove $\approx_\varepsilon$ need not be a single indifference if ε differs by objective.

C21-E2. Failure diagnosis. $\varepsilon = 1.96 \times \text{MCSE}$. Why is the *concept* wrong even if you type a number?

C21-E3. Decision reasoning. Two staffing plans 0.2% apart on expected wait. Nobody wrote a tolerance and nobody wrote a risk mode. Can you pick?

C21-E4. Assumption criticism. Bandwidth 5 vs 1 moved J from 48.84 to 37.69. Package default or explicit h?

C21-E5. Conceptual. When should a sharper forecast *not* change the act?

C21-E6. Application. Using the Sure/Gamble table, which act does expected payoff pick? Which act does $\text{CVaR}_{0.05}$ pick? Why is that not a scientific controversy?

C21-E7. Interpretation. A ranking of supplied states is printed. Why must it not be read as a global optimum of the industry?

C21-E8. Method selection. Reliability-first is requested and nobody wrote a target. What do you issue?

C21-E9. Interpretation. GI changed numbers and DI did not reverse. What did ε and unresolved risk just do?

C21-E10. Epistemic restraint. You are asked for "the recommended policy" with `PRIOR_ONLY`, no ε, no risk, no h. List every legal refusal. Do not emit a winner.

CHAPTER 22

Which Structure Is This Problem?

Depends on. Chapters 5–21.

The reader now has two honest worlds. In one, tomorrow does not remember: maps, an efficient mix, a parent that only sees totals (Chapters 5–8). In the other, tomorrow does remember: occupancy is a box that carries, a process proposes a shortlist, paired worlds score it (Chapters 9–19). Preference, when declared, overlays the result (Chapter 21).

The question is not which package. It is: at *this* grain, does the leftover stock change the act you would take?

22.1 English before labels

Ask three questions.

1. Is there a stock the next act can actually see — files still in the building, patients still in the ward — or only a KPI total?
2. If you empty that stock and re-solve, does the preferred consequence move by more than the organization already called noise?
3. Are the maps independent, or do they share a pool Chapter 5 already refused to glue?

If there is no defensible stock, a static mix among maps may still be honest, or you may not have enough to say. If there is a stock and emptying it does not move the act, the parent of Chapter 7 is allowed its ignorance: treat the children as totals. If emptying it *does* move the act, keep the stock and use the process chapters. Shared saturation is not "approximately a process." It is still the wrong maps.

Those three answers are all the routing this book owns. The labels FRONTIER / DYNAMIC / insufficient evidence are just names for them. They are this book's names, not methods in the operations-research literature.

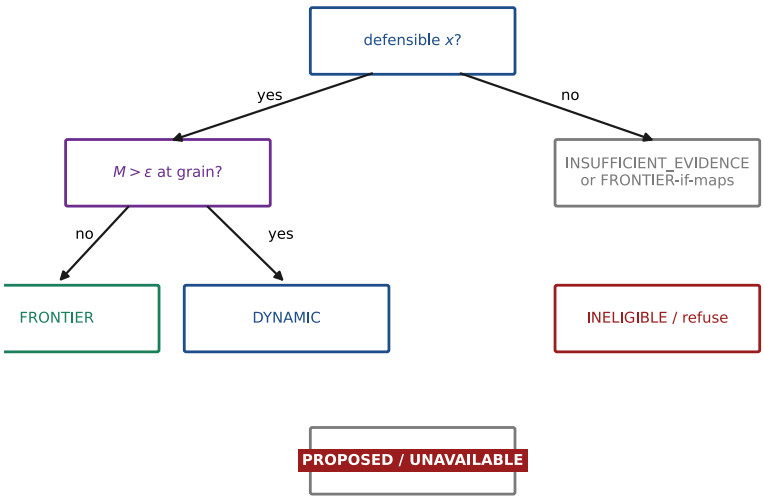


Figure 22.1: The same three questions as a tree. A faded fourth leaf is not a shipped method.

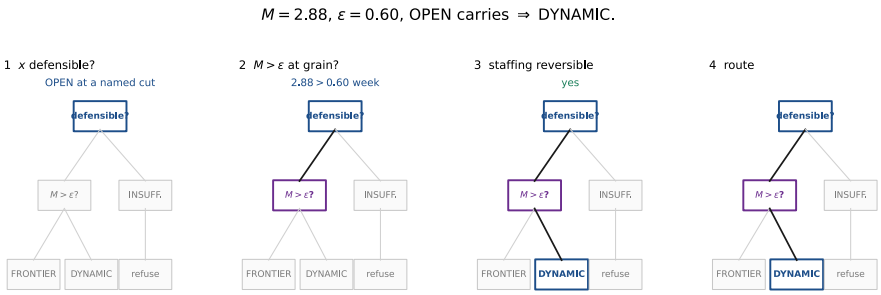


Figure 22.2: Four stills: is the stock defensible; does the gap exceed the tolerance; is the roster reversible; the process leaf lights.

Grain is part of the question. A claims shop, two grains, **new** paired numbers — not Chapter 7's month:

grain	J with leftover kept	J leftover emptied	gap	tolerance 1.0
this week	22.0	18.4	3.6	process: keep the stock
this quarter	61.0	60.4	0.6	maps: dropping the stock is cheap

Same files. Opposite structure. Duration of maturation does not appear in the table.

A child efficient set that *depends on incoming stock*, then Bellman at the parent, would close Chapter 7's pipe (Bellman 1957; Bertsekas 2012). That object is not a shipped method here. If someone asks for it, the honest sentence is that it is unavailable — not a fourth route, and not a function to pretend to export. Approximate DP (Powell 2011) is out of this monograph's qualified scope. Sequence in a calendar widget is not Bellman.

22.2 Exercises

C22-E1. Application. Using the week/quarter table, which structure at which grain?

C22-E2. Method selection. A manager says "we bought a process engine, therefore this is a process problem." Which question did they skip?

C22-E3. Failure diagnosis. Two channels share one impression pool. Is that a process, a static mix, or the shared-well refusal of Chapter 5?

C22-E4. Hostile. Someone asks for a child frontier that depends on last month's leftover OPEN. What do you say, and what do you *not* emit?

CHAPTER 23

Monday Morning

Depends on. Chapters 1–22.

Justified practice is not a slogan. It is what you can still do at 8 a.m. when a roster has to be locked before tonight's waiting room is observed.

A ward must choose tonight's nurse roster. Two acts: staff 12, or staff 14 (two extra nurses on overtime). Overflow occurs if waiting-room occupancy exceeds 16. The board has not yet said whether overflow is a catastrophe or a cost. Finance has not yet said whether two extra FTE are noise.

23.1 The origin

What the ward knows at 8 a.m.: current waiting room 9, identified service rates from a long internal series, the roster still mutable. What it does not know: tonight's arrivals, tonight's census. A dashboard that "forecasts" tonight from tonight's census is leakage (Chapter 2). The statement being asked for is an intervention, staffing, not a regression of historical completions on historical spend. Service rates are identified. An outreach elasticity that has never seen an experiment is not this morning's lever (Chapter 4).

The waiting room is stock, not last night's flow (Chapter 2). Tomorrow remembers: leftover waiting is still there at 7 a.m. unless it completed or left. At the *night* grain the leftover changes the act; at the quarter it may not (Chapter 22).

23.2 Three worlds, two acts

Given the identified rates, three equally likely arrival worlds — Quiet, Typical, Surge — are enough to staff. Occupancy at 7 a.m.:

	Quiet	Typical	Surge	mean	overflow (>16)
staff 12	10	14	22	15.3	Surge only

	Quiet	Typical	Surge	mean	overflow (> 16)
staff 14	8	11	16	11.7	none

These are entity worlds given frozen rates, not a posterior on the rates, not a recession scenario, and not last week's residual (Chapter 19). One ribbon of width "about 12" would have mixed them.

Search can propose the two rosters. Validation, if you have more than one candidate, pairs the worlds (Chapter 18). There is no third roster hiding in the solver.

Mean occupancy favors 14 ($11.7 < 15.3$). Overflow chance favors 14 (0 versus 1/3). That is not yet a decision. Finance cares about the two extra FTE. The board cares about overflow.

23.3 Preference has to speak

Chapter 21's arithmetic, in this ward. Twenty such nights. Treat "Quiet / Typical / Surge" as 8/8/4 nights in twenty for staff 12 (so $4/20 = 0.20$ overflow — close to the one-third toy; use the three-world table as the teaching object and do not launder it into a fourth fixture).

On the three-world table, expected occupancy is not the overflow event. If the organization writes "four extra waiting patients is noise, overflow is not," then $15.3 - 11.7 = 3.6 < 4$ and the *mean* occupancy difference is not material, while overflow is. Staff 14. If it writes "overflow on one night in three is acceptable; two extra FTE are not," staff 12. If it will not write either sentence, stop. That stop is the remainder, not a missing row in the ranking.

Downside occupancy as CVaR on the worst world: staff 12 sees 22, staff 14 sees 16. Downside still picks 14. The disagreement, when it exists, is *who pays for the extra nurses versus who owns overflow* — two honest people, two acts. The machine may compute the table. It may not pick the person.

23.4 After Tuesday's census

Tuesday's count is in. The miss, if any, does not name its cause. Three stories (Chapter 19): the count was mis-keyed; the forecast is persistently high and the slope did not move; the slope moved. Attaching Tuesday's rows to last night's champion issues a new forecast from the same revision. It does not train, and it does not promote. Promotion is a human signature if a challenger is even on the table.

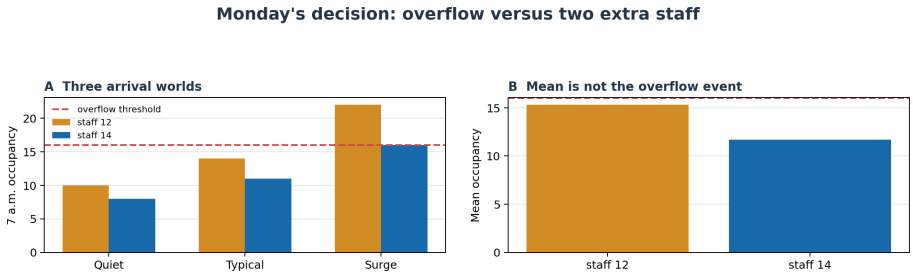


Figure 23.1: Occupancy for staff 12 and staff 14 across Quiet, Typical, and Surge worlds, with the overflow threshold and mean occupancy shown separately.

If every plausible signal would have left tonight's roster unchanged, stop spending on sharpness. The extra model was not for this act.

23.5 Exercises

C23-E1. Application. Using the occupancy table, which act does expected occupancy pick? Which act does "no overflow" pick?

C23-E2. Decision reasoning. The board writes "four extra waiting patients is noise; overflow is not." Which act, and which column of the table did they ignore?

C23-E3. Decision reasoning. Finance writes "overflow on one night in three is acceptable; two extra FTE are not." Which act?

C23-E4. Failure diagnosis. A dashboard titles itself "forecast" and uses tonight's midnight census at 8 a.m. What did it use that $\mathcal{I}_t$ did not contain?

C23-E5. Hostile. Tuesday's occupancy is two heads high. Someone wants to blend a last-error patch into the rates *and* fit a regime dummy. What does Chapter 19 forbid?

C23-E6. Method selection. At the quarter, emptying leftover waiting moves expected occupancy by 0.4, and the organization called 1.0 noise. Do you need the process engine for *that* grain?

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